



Consist Switch

User Manual



Consist Switch User Manual version 2.0.4
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Document scope

Information in this document applies to:

- Train Switch product models equipped with 8 or 13 Ethernet ports
- Consist Switch product models equipped with 8 or 14 Ethernet ports
- Consist Switch product models equipped with 19 or 28 Ethernet ports (see the supplementary information about the firmware of these devices in Unavailable services in 7.2.x firmware series on page 251)

Document History

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1.0.0	First edition.	1.0.0
1.0.1	Improvements and corrections in QoS section and commands reference.	1.0.0
1.0.2	Console connection/configuration information.	1.0.0
1.0.3	More information in QoS section.	1.0.0
1.0.4	Added information on DHCP snooping; corrections.	1.2.0
1.0.5	Minor corrections.	1.2.0
1.1.0	New edition: major review, updated contents, many improvements.	1.5.0
1.2.0	Language improvements. New section for LACP. IGMP, QoS and DHCP section revisited.	1.12.1
1.3.0	Added information on REST interface, 802.1X authentication, interface speed settings commands, many improvements.	2.0.5
1.3.1	Added info on firmware upgrade without losing saved configuration, described new behavior of STP and multicast flooding upon topology changes.	2.4.0
1.3.2	Updated technical specifications.	2.4.0
1.3.3	Added mechanical drawings for base version.	2.4.0
1.3.4	Mechanical and electrical specifications moved to Installation Manual. Updated safety information.	3.0.4
1.4.0	Updated information for new major software release.	3.2.1
1.5.0	Updated information for new major software release.	4.0.0
1.6.0	Updated information for new major software release.	5.0.1
1.7.0	Updated information for new major software release.	5.2.0
1.8.0	Updated information for new major software release.	6.0.0
1.9.0	Updated information for new major software release.	6.2.0
1.9.1	Added NTP client and NTP server sections.	6.2.0
1.9.2	Added SVI interfaces dynamic status section.	6.2.1
1.9.3	Added log on file details and diagnostics subsystem.	6.4.0
1.9.4	Added password security, account locking, banner.	6.4.0
1.9.5	Added DHCP group classless static route (option 121).	6.4.0

1.9.6	Added rate limiting priority queues and profiles.	6.4.0
1.9.7	Command list updated.	6.4.0
1.9.8	Amended diagnostics subsystem commands.	6.4.0
1.9.9	TCDS updated and Multi-coach support added; amended diagnostics subsystem commands; updated credits section.	6.4.0
2.0.0	Updated diagnostics data-sets external documentation reference.	6.4.0
2.0.1	Added annex with the list of services not supported in the 7.0.x firmware series.	6.4.0/7.0.x
2.0.2	Added distributed groups in link aggregation sections; added TTCMP chapter.	6.4.1/7.0.x
2.0.3	Added port mirror setup with 'configure' command; fixed cross reference to Copyright credits section; content common to TBN styled the same way on both manuals; updated command reference.	6.4.2/7.0.x
2.0.4	Improved connected MAC explanation; added error code 8 details and added reference to LED n.4 behavior in TTCMP section; fixed NAT example; updated QoS section; updated command reference.	6.4.2/7.2.x

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Chapter 1 - **Safety instructions**

1.1 Usage

The product must only be used for the intended purpose described in the present manual.

The equipment should be installed only in areas where access is restricted to qualified personnel.

Before installing this equipment, carefully read the instruction manual to get all the necessary technical information.

1.2 Requirements for personnel

The product is an electronic equipment whose installation and setting require a professional knowledge only available to qualified and purposely trained technicians.

Therefore, the operator should refrain from trying to install a Consist Switch by his own if not specifically qualified and trained for this purpose, as this attempt could damage the equipment and/or cause injuries to the operator himself.

In any case, the product should be installed by authorized personnel only and used in accordance with the instructions specified in this manual.

Failure to do so could result in poor product performance, personal injury and/or damage to the train.

In general, the installation and use procedures described in this manual must be followed carefully.

1.3 Electrical safety

Make sure that the power supply is switched off before making any electrical connections or removing the power cord.

During installation, make sure that the power cord has been connected with the correct polarity and that the supply voltage is in the range specified in this manual.

An input voltage exceeding the maximum acceptable value could badly damage the product. Do not open the unit for any reason whilst it is powered.

**WARNING**

THE EQUIPMENT IS DESIGNED TO BE CONNECTED ONLY TO A SAFETY EXTRA-LOW VOLTAGE (SELV) IN ACCORDANCE WITH IEC/EN 60950-1.

The Consist Switch is protected with a non-replaceable internal fuse.

1.4 Grounding

The equipment must be grounded through the screw indicated in the present manual.

1.5 Service and maintenance

As there are no serviceable parts or adjustments, the operator should not attempt to open the equipment. Moreover, any attempt to open the device unit by not authorized personnel would nullify the warranty.

1.6 Overheating

**WARNING**

ATTENTION HOT SURFACES RISK OF BURNING

Make also sure that the environmental conditions (temperature and humidity) are in the range specified in this manual.

To avoid thermal problems, make sure that the product is properly installed so that the heat generated can be dissipated correctly, and that the devices installed around the device does not cause the temperature rise.

1.7 Declaration of conformity

The product is in conformity with the following EC directives:

Directive	Title
2004/108EC	Electromagnetic Compatibility (EMC)

The product is in conformity with the following reference standard:

Standard	Title
EN 50121-3-2:2006	Railway applications - Electromagnetic compatibility - Part 3-2: Rolling stock - Apparatus
EN 50155:2007	Railway applications - Electronic equipment used on rolling stock
EN 61373:1999	Railway applications - Rolling stock equipment - Shock and vibration tests
EN 60950-1:2007	Information technology equipment - Safety - Part 1: General requirements
EN 50124-1:2001+A1	Railway applications Insulation coordination Basic requirements -Clearances and creepage distances for all electrical and electronic equipment
IEEE 802.3at:2009	PoE, Data Terminal Equipment Power via the Media Dependent Interface Enhancements

Chapter 2 - Product description

2.1 Overview

The Consist Switch is an Ethernet managed switch specifically designed for inter-networking applications in rolling-stock environments.

Besides the standard features also found in similar products, it provides facilities for on-board network discovery and configuration which enable fast device deployment/replacement and fine-grained control of operating stations.

The full list of features and product characteristics can be found in the Technical specification chapter.

The product key features are the following:

	13 ports model	14 ports model	19 ports model	28 ports model
Fast Ethernet (FE) ports	8 or 13	8 or 14	4 or 19	12
Gigabit Ethernet (GbE) ports	5	6	11 or 15	11 or 16
10Gigabit Ethernet (10GbE) ports	N/A	N/A	4	5
PoE (optional)	on 8 FE ports	on 8 FE ports	on 8 FE or 4 GbE + 4 FE ports	on 4 GbE + 12 FE ports
Hardware Bypass	up to 2	up to 3	up to 2	up to 2
Internal network logic and data path	Gigabit capable		10 Gigabit capable	

Table 1: Product Specific Features Overview

- Up to 3 couples of ports may be equipped with a Hardware Bypass so that, in case of failure, the electrical continuity on up to 3 lines will be guaranteed.
- Wire speed switching performances.
- USB port for console operations.
- A separate Management connector for Konuendo Total Device Maintenance (KTDM), a technology capable of off-band operations ranging from firmware installation and/or upgrade up to configuration or user data loading or custom system level operations.

- A command line user interface (CLI) for system configuration and management. It can be accessed via system console or a Secure Shell (SSH) network session.
- A REST-based Web Services interface for programmatic remote interaction.
- Per-port DHCP server
- Managed switch standard features: SNMP, RSTP, VLAN, QoS, LACP, IGMP snooping, etc...
- Optional Train Topology and Configuration Management Protocol (TTCMP©) for configuring, monitoring and managing the device.

Please refer to VDS Rail representatives for information about Konuendo Consist Network Concept (KCNC) and TTCMP©.

2.2 At a glance

The next sections present a general overview of the front panel of the Consist Switch. All the connectors are grouped for easier understanding of their capabilities. All the Ethernet connectors are part of the same switch. Differences in names define only some special features.

2.2.1 14 ports Consist Switch

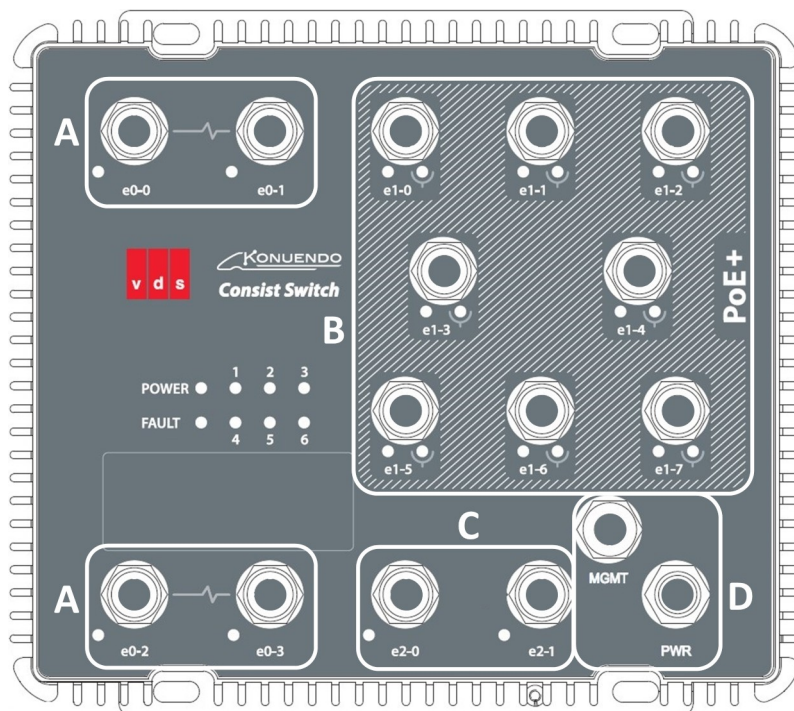


Figure 2.1: 13 ports Consist Switch connectors

Ports labeled *e0-0* and *e0-1* (A) may be connected by a Hardware Bypass, making the device completely transparent to the network in case of a power outage or failure. In such case all the traffic will flow undisturbed across the device. The same applies for

e0-2 and *e0-3* ports (A).

Ports *e0-x* (A) and ports *e2-x* (C) may be Gigabit capable (depending on the model).

Ports from *e1-0* to *e1-7* (B) are Fast Ethernet ports. They may be PoE equipped, depending on the model.

Port *e2-x* (C) are supposed to be up-link connections towards the Backbone network, if present. They are, however, standard switched Ethernet port. They may be connected by a Hardware Bypass.

The *PWR* connector is the one to be used to connect a power supply.

The *MGMT* connector exports the system console and a USB port used for offline maintenance operations (see *Konuendo Total Device Management* at page 207).

NOTE

In the Consist Switch with 13 ports, the *e2-1* (C) is not present and the *MGMT* connector functions are supplied by distinct *CONSOLE* and *MNT-H* ports.

2.2.2 19 ports Consist Switch

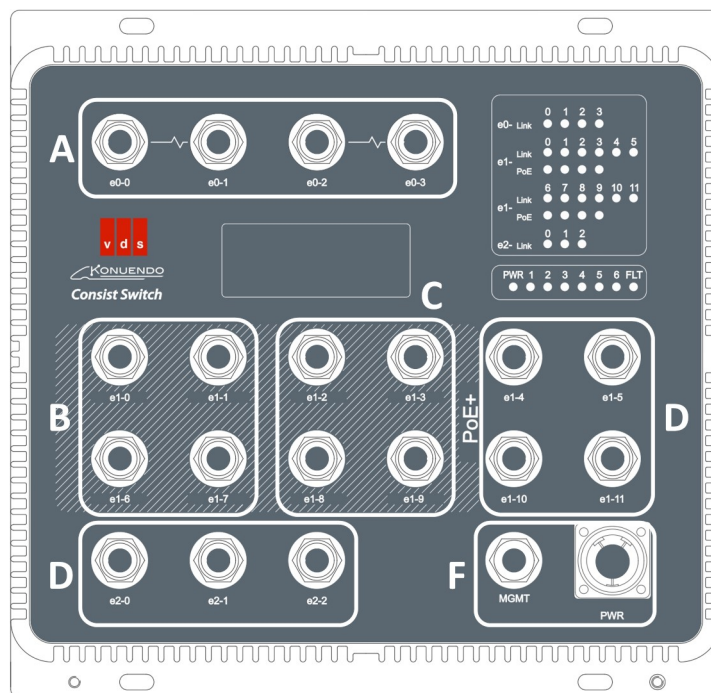


Figure 2.2: 19 ports Consist Switch connectors

Ports labeled *e0-0* and *e0-1* (A) may be connected by a Hardware Bypass, making the device completely transparent to the network in case of a power outage or failure. In such case all the traffic will flow undisturbed across the device. The same applies for *e0-2* and *e0-3* ports (A).

Ports *e0-x* (A) may be Gigabit or 10 Gigabit capable (depending on the model)

Ports *e1-0*, *e1-1*, *e1-6* and *e1-7* (B) are Fast Ethernet ports. They may be PoE equipped, depending on the model.

Ports *e1-2*, *e1-2*, *e1-8* and *e1-9* (C) may be, depending on the model, Gigabit capable and/or PoE equipped ports.

Ports *e1-4*, *e1-5*, *e1-10*, *e1-11* and *e2-x* (D) may be Gigabit capable (depending on the model).

The *PWR* connector is the one to be used to connect a power supply.

The *MGMT* connector exports the system console and a USB port used for offline maintenance operations (see *Konuendo Total Device Management* at page 207).

2.2.3 28 ports Consist Switch

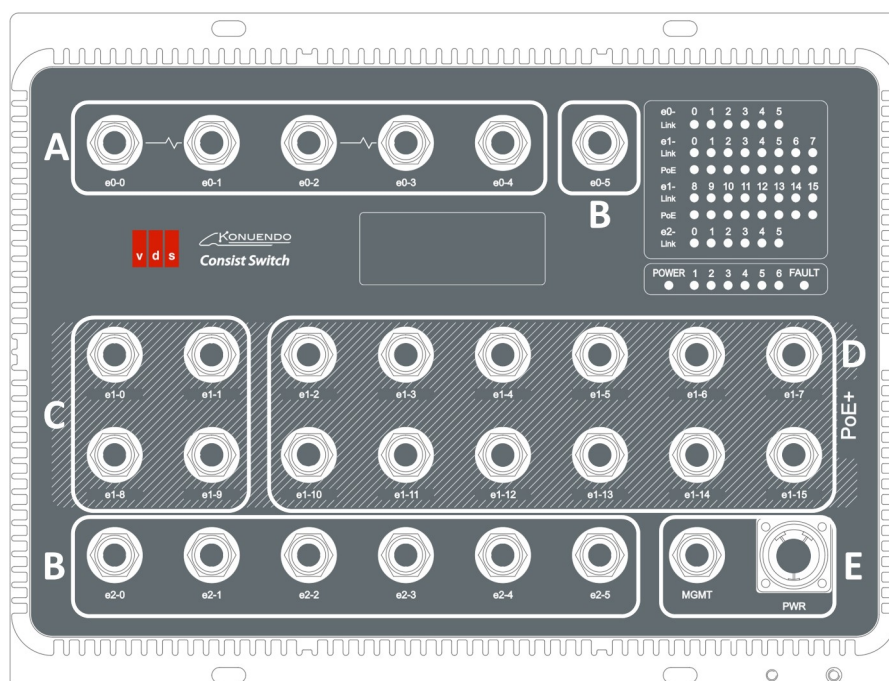


Figure 2.3: 28 ports Consist Switch connectors

Ports labeled *e0-0* and *e0-1* (A) may be connected by a Hardware Bypass, making the device completely transparent to the network in case of a power outage or failure. In such case all the traffic will flow undisturbed across the device. The same applies for *e0-2* and *e0-3* ports (A).

Ports *e0-0* to *e0-4*, (A) are Gigabit or 10 Gigabit capable (depending on the model)

Ports *e0-5* and *e2-x* (B) are Gigabit capable.

Ports *e1-0*, *e1-1*, *e1-8* and *e1-9* (C) are Gigabit capable ports. They may be PoE equipped, depending on the model.

Ports *e1-2* to *e1-7* and *e1-10* to *e1-15* (D) are Fast Ethernet ports. They may be PoE equipped, depending on the model.

The *PWR* connector is the one to be used to connect a power supply.

The *MGMT* connector exports the system console and a USB port used for offline maintenance operations (see *Konuendo Total Device Management* at page 207).

Chapter 3 - Quick start

3.1 From power-on to the prompt

In order to start working with the Consist Switch, you need to:

- ⇒ connect the device to a proper power source, following recommendations and guidelines as specified in the *Safety instructions* section (page 11)
- ⇒ connect the CONSOLE cable to a PC and configure the serial port with the following parameters:

speed:	115200 bps
parity:	none
stop-bits:	1

- ⇒ configure your terminal emulator to support a *standard VT100* terminal type
- ⇒ access device console through the terminal emulator and log into the administration interface using the following credentials:

username:	admin
password:	admin

You're now ready to configure and manage the Consist Switch. Beware of the fact that you now have access to the device with the highest privilege level (administrator), so you are in full control of it.

You can enter commands by typing them at the prompt; for example type:

```
A@SW-Std.03-00-CA>show version
```

and press **Enter**. The program will show information about the running software.

IMPORTANT NOTE

The device factory configuration is suitable for most network environments. Since Consist Switch settings can be critical, in order for the networks it connects to work properly, please read carefully through this manual before making any configuration changes.

Chapter 4 - User interface

The CLI is a simple text-based command-line oriented user interface (also referred to as the *shell*) which provides an effective tool for managing all the product details.

You can access the CLI through a terminal emulation software if the CONSOLE port of the Consist Switch is connected to a PC with the specific USB cable.

The parameters for the connection on the serial line are: 115200 baud, no parity, 8 data bits, 1 stop bit. The terminal emulation standard is ANSI/VT100.

Alternatively, once the switch has been assigned its own IP address, you can access the CLI through an SSH connection.

In both cases, once you are connected, you must log in with the right credentials to operate on the system.

The credential used for logging in defines the activities you are allowed to execute and the commands that you have access to. See the *Users* section on page 75 for more information about supported users and their default passwords.

The system welcomes you with a prompt like the following:

```
** Konuendo Platform Configuration Shell **
**          Railway Switch Version          **
**          Press '?' for Help              **
U@SW-Std.03-00-CA>
```

The prompt reports you the abbreviated user name specified at login time and the current host name assigned to the device (*U* and *SW-Std.03-00-CA* in the example above).

4.1 Shell interaction

At the shell prompt you may issue commands. The commands you can use are determined by the *shell level* you are into.

The shell level establishes a context in which the user is restricted to operate only on certain parts/components of the system (such as a network interface, for example) or to perform only certain operations. Levels are designed both to group together some commands that logically belong to same aspects and to enforce authentication constraints, so that switching to a certain level may require you to have previously logged in using specific credentials. Levels are described in *Table 2: Shell levels* below.

Shell level	Description
-------------	-------------

base	This is the initial level in which you enter when you log in and access the shell. At this level you can get information about the switch status and configuration or move to a further level.
system administration	On this level you can execute commands which pertain to the device administration activities such as installing a new system image, delete the active configuration or reboot the system. You can switch to this level using the <code>sysadmin</code> command. Only the <i>admin</i> user can access this level.
global configuration	On this level you can execute commands which modify the configuration of global settings such as the system host name or the QoS mappings, for example. You can switch to this level using the <code>configure</code> command. Only the <i>admin</i> and the <i>operator</i> users can access this level.
specific configuration	On this level you can execute commands which modify the settings for a specific system element. For example, the maximum MTU size or the IP address of a given network interface. The command to use for moving yourself into this level varies depending on the element under configuration. The same security restrictions which applies to the <i>global configuration</i> level also apply to this level.

Table 2: Shell levels

When on a level other than *base* level, the prompt reports information about current level in square brackets. In *system administration* level:

```
A@SW-Std.03-00-CA>sysadmin
A@SW-Std.03-00-CA[SysAdmin]>
```

While on *global configuration* level:

```
A@SW-Std.03-00-CA>configure
A@SW-Std.03-00-CA[configure]>
```

On *specific configuration* level (in this case the configuration context is network interface *e0-2*):

```
A@SW-Std.03-00-CA>configure interface e0-2
A@SW-Std.03-00-CA[config-if e0-2]>
```

The shell can be at just one level at a time and the `exit` command allows the user to return to the previous level at any time.

From the base level an `exit` terminates the shell and brings you to the login prompt again:

```
O@SW-Std.03-00-CA[configure]>exit
O@SW-Std.03-00-CA>exit
SW-Std.03-00-CA login:
```

The same result can be achieved at any level by pressing **Ctrl-D**. This is equivalent to a `do exit` command (see *Do Form* section below).

When switching to a different level, the prompt will change accordingly:

```
O@SW-Std.03-00-CA>configure interface vlan1
O@SW-Std.03-00-CA[config-if vlan1]>exit
O@SW-Std.03-00-CA[configure]>exit
O@SW-Std.03-00-CA>
```

The shell *specific configuration* level will be useful when you have to enter, for example, several commands for configuring a given network interface:

```
O@SW-Std.03-00-CA>configure interface vlan1
O@SW-Std.03-00-CA[config-if vlan1]>ip address 192.168.1.10 255.255.255.0
O@SW-Std.03-00-CA[config-if vlan1]>enable
O@SW-Std.03-00-CA[config-if vlan1]>mtu 1490
O@SW-Std.03-00-CA[config-if vlan1]>exit
O@SW-Std.03-00-CA[configure]>exit
```

However, it is always possible to issue any command from the base level by inserting the whole command:

```
O@SW-Std.03-00-CA>configure interface vlan1 ip addr 192.168.1.10 255.255.255.0
O@SW-Std.03-00-CA>configure interface vlan1 enable
```

4.2 Do Form

On each level only a subset of commands is allowed. However, from every level, commands from base level are still available provided that the user prefixes them with the word `do`. For example:

```
O@SW-Std.03-00-CA>configure interface e2-0
O@SW-Std.03-00-CA[config-if e2-0]>do show system time
Fri Jul 30 13:50:03 2010
O@SW-Std.03-00-CA[config-if e2-0]>exit
O@SW-Std.03-00-CA[configure]>exit
O@SW-Std.03-00-CA>
```

Simply invoking `show system time` would return an error:

```
O@SW-Std.03-00-CA[config-if e2-0]>show system time
Error in command, param is "show" [wrong]
O@SW-Std.03-00-CA[config-if e2-0]>
```


4.3 No Form

To reverse the action of a configuration command it is sufficient to prefix it with the word *no*. For example, the following are the commands used to enable and disable the *e2-0* network interface:

```
O@SW-Std.03-00-CA>configure interface e2-0 enable
O@SW-Std.03-00-CA>no configure interface e2-0 enable
O@SW-Std.03-00-CA>
```

The so called no-form of commands can be used at any level. The same result of the commands above may be obtained with:

```
O@SW-Std.03-00-CA>configure interface e2-0
O@SW-Std.03-00-CA[config-if e2-0]>enable
O@SW-Std.03-00-CA[config-if e2-0]>no enable
O@SW-Std.03-00-CA[config-if e2-0]>
```

4.4 Getting help

Pressing the '?' key will show suggestions for current input. Depending on the current context, suggestions may consist in a list of commands or a list of parameters. If no suggestion is available, it will simply have no effect.

For example, by pressing '?' at the basic prompt you get:

```
O@SW-Std.03-00-CA>                                (? pressed)
auto-logout          Set the shell autologout feature
clear                Clear object counters
configure            Enter configuration mode
debug               Activate debug messages
exit                 Exit the shell
ping                 Ping a network IP address or a hostname
save                 Save running configuration to non-volatile memory
show                 Show running system information
sysadmin             Enter system administration mode
```

or in the middle of a command:

```
O@SW-Std.03-00-CA>configure interface              (? pressed)
<ifname>          Set interface parameters
O@SW-Std.03-00-CA>configure interface
```

or in the middle of a command word:

```
O@SW-Std.03-00-CA>configure inte                    (? pressed)
interface
O@SW-Std.03-00-CA>configure inte
```

If the words entered so far on the line may constitute a valid command, the '?' key will suggest this with a final <CR> help line:

```
0@SW-Std.03-00-CA>configure interface vlan1      (? pressed)
  description      Interface specific description (128 chars max)
  enable           Enable interface
  exit             Exit from interface configuration level
  fast-link-detection  Configure fast link-down detection mechanism on Gigabit...
  ip               Interface IPv4 configuration commands
  lldp             LLDP protocol permitted on this interface
  mtu              Set interface MTU
  poe              Configure interface PoE mode
  tcDs             Set TCDS interface properties
  vrrp             Configure Virtual Router Redundancy Protocol (VRRP)
  <CR>
```

4.5 Command completion

The 'TAB' key will complete the current token and, when the completion is performed, it will behave exactly like the '?' key, showing the on-line suggestions:

```
0@SW-Std.03-00-CA>configure inte      (TAB pressed)
0@SW-Std.03-00-CA>configure interface (TAB pressed)
  <ifname>          Set interface parameters
```

Chapter 5 - General configuration

This chapter overviews the configuration capabilities of the Consist Switch and describes the commands for setting its basic operating parameters.

5.1 Device configuration

In order to safely manage changes in the operational parameters of the device, its configuration has been distinguished in a *running* and a *startup* configuration, whose characteristics are described in the table below:

Configuration	Description
running	The active configuration which resides in memory and which drives device operation at runtime. It is updated whenever a configuration command is entered by the user. It persists across CLI sessions but is lost if the device is switched off or rebooted.
startup	The configuration which resides on the non volatile memory of the device. It contains the sequence of commands to execute when the device starts up. It persists across reboots or power cycles.

Table 3: Running and startup configurations

The Consist Switch can be configured by entering commands using the CLI (see *User interface* chapter, page 23).

When you enter configuration commands, you always alter the *running* configuration. In order to make your changes effective even at the next system restart, you need to save the *running* configuration to the non volatile memory using the command:

```
O@SW-Std.03-00-CA>save running-config
```

This will make your current configuration the *startup* configuration from the next system restart on.

You can examine the *running* configuration using the command:

```
O@SW-Std.03-00-CA>show running-config
```

while the contents of the *startup* configuration can be examined by means of the command:

```
0@SW-Std.03-00-CA>show startup-config
```

The startup configuration file can be deleted from the non-volatile memory using the command:

```
A@SW-Std.03-00-CA>sysadmin delete startup-config
```

Please note that in order to run this last command, the operator must be logged in as *admin* user.

5.2 External configuration

Normally the configuration details are stored in the device flash, ready to be used at every reboot.

It is possible, however, to have the configuration stored in an external USB key, so that no details are kept inside the Consist Switch. At every boot the configuration is read from the USB storage and applied.

See *External configuration* chapter (page 215) for instructions on how to set up the USB key.

5.3 Configuring IP address

Configuring the Consist Switch with an IP address expands your ability to manage the device and use its features. By default, the device is configured with no IP address assigned. You can assign an IP address to the Consist Switch by configuring the management interface (*vlan1*) as follows.

At *base configuration* level you can assign a fixed IP address to the interface:

```
0@SW-Std.03-00-CA>configure interface vlan1 ip address 192.168.1.1 255.255.255.0
```

In this example the address 192.168.1.1 with a netmask of 255.255.255.0 has been assigned to the device. Alternatively you can configure the interface to use a dynamic address assigned by a DHCP server (if available on your network) with the command:

```
0@SW-Std.03-00-CA>configure interface vlan1 ip address dhcp
```

If the device and the management stations are located on different networks you should also configure a default gateway address. In this case, assuming for example that your gateway IP address is 192.168.1.254, use the following command:

```
0@SW-Std.03-00-CA>configure ip route 0.0.0.0 0.0.0.0 192.168.1.254
```

If you need name resolution you can configure the address of the name server to be used. For example in order to use 192.168.1.10 as name server enter the command:

```
0@SW-Std.03-00-CA>configure ip nameserver 192.168.1.10
```

Remember that the modifications made so far are immediately effective, but they will be lost if you reboot the system without saving the running configuration.

5.4 Configuring device host name

You can assign a host name to the device with the `configure system hostname` command. For example:

```
O@SW-Std.03-00-CA>configure system hostname csw-1
O@csw-1>
```

Note that the prompt has been updated to reflect the change.

5.5 Basic port settings

By default, all the device ports are enabled with speed value and duplex mode automatically negotiated with peers.

An interface (in this example `e0-0`) can be disabled issuing the following command:

```
O@SW-Std.03-00-CA>no configure interface e0-0 enable
```

or re-enabled issuing:

```
O@SW-Std.03-00-CA>configure interface e0-0 enable
```

You can get ports status information using one of the `show interface base` commands. For example the following command will return a compact representation of all ports status:

```
A@SW-Std.03-00-CA>show interface summary

Port types:  L0 Loopback
              FE Fast Ethernet
              GE Gigabit Ethernet
              LAG Link Aggregation Group
              SVI Software Virtual Interface
              IST Interswitch Trunk

Port  Type  Status  Line  Speed  Duplex
-----
lo    L0    enabled up    -      -
e0-0  FE    enabled down   -      -
e0-1  FE    enabled down   -      -
e0-2  FE    enabled up    100    Full
e0-3  FE    enabled down   -      -
e1-0  FE    enabled up    100    Full
e1-1  FE    enabled down   -      -
e1-2  FE    enabled down   -      -
e1-3  FE    enabled down   -      -
e1-4  FE    enabled down   -      -
e1-5  FE    enabled down   -      -
e1-6  FE    enabled down   -      -
e1-7  FE    enabled up    100    Full
e2-0  FE    enabled up    100    Full
grp2  LAG    enabled up    -      -
vlan1 SVI    enabled up    -      -
```

while the following command will display more detailed information about a given port:

```
0@SW-Std.03-00-CA>show interface e0-0
Interface e0-0 is enabled, line protocol is up
  Interface index is 3
  Hardware is switched Fast Ethernet (switched)
  Link AutoNegotiation is Enabled
  Speed: 100 Mb/s Duplex: Full MDI: Auto
  Flow Control is Disabled
  Hardware address is a0:59:3a:03:00:d6
  Default COS: 0  COS override: Disabled
  QOS trusting: both
  Queuing strategy: mq
  UP BROADCAST RUNNING MULTICAST  MTU: 1500
  RX packets:3245071  TX packets:4038734
```

5.6 Configuring Port Speed and Duplex

Each available switched-port's speed and duplex settings are by default configured in Auto Negotiation mode; this way it's up to the underlying port Ethernet-hardware to negotiate proper link's parameters with the other peer during linkUp routines.

Anyway a properly authenticated user (*operator/admin*) can manually override such settings from the command shell; you can force the speed value for the interface:

```
0@SW-Std.03-00-CA>configure interface e0-1 speed 10
```

or the duplex mode:

```
0@SW-Std.03-00-CA>configure interface e0-1 duplex half
```

Please note that, if you choose to manually set the speed or duplex to a fixed value, the involved interface will exit Auto Negotiation mode as a whole: in other words if you decide to specify a fixed NON-negotiated value for the speed setting also the duplex-mode setting will exit Auto Negotiation state and vice-versa; in such situations the default settings, automatically enforced, are 100Mbps/Full-Duplex.

IMPORTANT NOTE about Auto Negotiation

Parallel detection is used when a device that is capable of auto-negotiation is connected to one that is not. This happens if the other device does not support auto-negotiation or auto-negotiation is administratively disabled. In this condition, the device that is capable of auto-negotiation can determine and match speed with the other device. This procedure cannot determine the presence of full duplex, so half duplex is always assumed.

5.7 Configuring MDI/MDI-X

Each available switched-port's is by default configured to automatically detect if the connection would require a crossover cable and automatically chooses the MDI or MDI-X configuration to properly match the other end of the link.

Anyway a properly authenticated user (*operator/admin*) can manually override such settings from the command shell, forcing the interface to MDI or MDI-X mode:

```
0@SW-Std.03-00-CA[config-if e0-1]>mdi-mode auto
0@SW-Std.03-00-CA[config-if e0-1]>mdi-mode mdi
0@SW-Std.03-00-CA[config-if e0-1]>mdi-mode mdix
```

5.8 Configuring Flow Control

Ethernet flow control is a mechanism for temporarily stopping the transmission of data in case of a sending station is transmitting data faster than the other end of the link can accept it. . Consist Switch implements flow control by means of the pause frame mechanism as defined by the IEEE 802.3x standard.

Flow control is disabled by default, but a properly authenticated user (operator/admin) can enable it on a per-port basis using the following command:

```
0@SW-Std.03-00-CA[config-if e0-1]>flow-control enable
```

5.9 Fast Link Detection

5.9.1 Overview

VDS Rail introduced this mechanism to speed up link-down events detection to overcome some limitations imposed by the IEEE Ethernet standard on Gigabit ports; as a consequence, this mechanism, described summarily in the following, is thought to be used under certain particular conditions and has some drawbacks and limitations to be aware of.

Basically the problem lies in the long timeouts stated by the IEEE Gigabyte Ethernet Standard and implemented by the related silicon-based state machines; these timeouts could lead to a link-down event detection timeout of up to approximately 700 ms.

In order to overcome this limitation VDS Rail networking products use alternative interrupts' sources (namely carrier-false) as markers and precursors of possible link-down events in order to promptly detect a link-down event in as low as 5 ms.

However, since the above mentioned IEEE state machines' timeouts have generally their good reasons to exist, **this solution has its drawbacks**: it can lead to false positives under some circumstances, for example false link-down signaling: when such false positives arise the link will be promptly reverted back to UP state, thus causing anyway some sort of service disruption. Anyway in the vast majority of cases this mechanism serves well its purpose to speed up link-down detection on Gigabit ports without any collateral bad effect.

False link-down events can be triggered in fact by various factors:

- bad cabling;
- bad connectors or junctions;
- badly plugged connectors or junctions;

- high stresses on the cable itself during operations.

In order to cope with these, occasional, false positives the Fast Link Detection mechanism:

- exposes statistical data about carrier-false events to help troubleshooting;
- implements a surge protection mechanism that automatically disables the carrier-false interrupts emission and handling when a certain (non configurable) threshold is exceeded; this avoids the chance that a burst of carrier-false events triggered by a faulty cable could keep a link bouncing up and down indefinitely;
- offers a way to completely disable the Fast Link Detection mechanism as a whole on a per-port basis through a standard shell command.

Please note that when the Fast Link Detection mechanism has been disabled, explicitly or through the means of the surge protection safety mechanism, the link-down detection reverts to the IEEE standard longer timeouts.

5.9.2 Basic Operations and Configuration

It is possible to check the current configuration and state of the Fast Link Detection mechanism on a Gigabit interface with a simple `show interface` command:

```
A@SW-Std.03-00-CA>show interface e0-0  
Interface e0-0 is enabled, line protocol is down  
    Interface index is 3  
    Hardware is Gigabit Ethernet (switched)  
    Link AutoNegotiation is Enabled  
    Speed: Auto Duplex: Auto MDI: Auto  
    Flow Control is Disabled  
Fast Link Detection is enabled <<<<<<<<<<<<<<<<<<<<<<<<  
    Hardware address is a0:59:3a:80:09:00  
    Default COS: 0   COS override: Disabled  
    QOS trusting: both  
    Queuing strategy: mq  
    UP BROADCAST MULTICAST MTU: 1500  
    RX packets:0 TX packets:0
```

Mechanism is enabled by default on Gigabit interfaces and such configuration is NOT explicitly reported in running-configuration: only the *no-form* is reported for such a configuration command.

Using the same command it's also possible to check if the mechanism has been auto-disabled by the surge protection mechanism:

Moreover, using a `show interface <intf> details` command it is possible to gather statistical data about the carrier-false events detected on the system:

Additionally, the **surge protection** safety mechanism *can be reset explicitly* simply issuing *disable/enable* commands on the interface; please note, however, that such operation will not reset also the interface statistical counters:

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[illegible]

As above mentioned it is also possible to **disable the Fast Link Detection** mechanism as a whole on a per-port base using the proper configuration command: `no configure interface <intf> fast-link-detection enable`

[illegible]

This way you can also reset the statistical data about carrier-false events:

[illegible]

Chapter 6 - Redundancy

6.1 Spanning Tree Protocol

The Consist Switch supports the Rapid Spanning Tree Protocol (RSTP) as specified into the IEEE-802.1D-2004 standard. This ensures a loop-free topology also in a deployed network design containing multiple redundant links between switches.

6.1.1 RSTP Overview

A detailed description of the RSTP protocol is out of the scope of this manual, but we will recap a few fundamental concepts in RSTP terminology.

In an Ethernet-based network any kind of physical topology must be reduced to a simply, fully connected, symmetric and loop-free logical topology in order to work; this is done by the RSTP process through the dynamic construction and maintenance of a *spanning-tree* logical topology, rooted at one of the bridges, belonging to such interconnected LAN, which has been previously elected by protocol to be the *Root Bridge*.

Bridge Protocol Data Units (BPDUs) are the means by which the participating switches exchange the information needed to calculate the shared logical topology.

Once enabled, the protocol is able to attain its objectives of maintaining a fully connected, simple, symmetric and loop-free topology, without the need of being configured. In any moment network management can properly choose the desired final "spanned" topology by means of the protocol's configuration parameters.

IEEE-802.1D-2004 RSTP is an evolution of the old IEEE-802.1D-1998 Spanning Tree Protocol and it is backward fully compatible with any older switch already present in the deployed network.

In IEEE-802.1D-2004 RSTP terminology each port of the switch can be in one of the following logical states regarding its learning and forwarding states:

- **Discarding**: a port in this state does not forward normal frames nor update the forwarding databases with freshly learned MAC source addresses. This state is named **BLK** by the shell and in this manual.
- **Learning**: a port in this state does not forward normal frames but is actively updating the forwarding databases with freshly learned MAC source addresses. This state is named **LRN** by the shell and in this manual.
- **Forwarding**: a port in this state actively forwards normal frames and is also actively updating the forwarding databases with freshly learned MAC source addresses. This state is named **FWD** by the shell and in this manual.

Moreover, in contrast with the old STP algorithm, in RSTP each port has also a *port*

role in the topology; the standard defines the following roles for a port managed by RSTP:

- **Root:** the port represents the selected path to an *upstream* switch which is on the least cost way to the current *Root Bridge*. This role is named **ROOT** by the shell and in this manual.
- **Alternate:** the port represents an alternative path to an *upstream* switch which is on the way to the current *Root Bridge*, but with a higher cost as determined by the spanning-tree algorithm using the currently chosen configuration parameters; the port can be used in case of failure of the main **ROOT** port. This role is named **ALTR** by the shell and in this manual.
- **Designated:** the port represents a path which connects the switch to a *downstream* switch providing connectivity toward the leaves of the spanning-tree active topology. This role is named **DESG** by the shell and in this manual.
- **Edge:** the port has been determined, automatically or by the means of configuration parameters, to be a connection to some sort of end-device, and so not actively participating to the determination of the active topology. This role is named **EDGE** by the shell and in this manual.
- **Backup:** this is an alternative path toward the leaves of spanning-tree. It can be used as a substitute of a **DESG** port, but this role can be seen only when this port is attached to the same LAN segment as the **DESG** port, and this is possible only when it is attached to a repeater/hub or back to another port on the same switch. This role is named **BCKP** by the shell and in this manual.

6.1.2 Displaying RSTP status

At any time you can display the status of the RSTP protocol process issuing the command `show spanning-tree` from the *base level*:

```
0@SW-Std.03-00-CA>show spanning-tree

Selected redundancy protocol RSTP is running.

Root bridge : 32768/a0:59:3a:03:00:d2
Root port: e1-5
Root path cost: 200000
Hello Time 1 sec Max Age 10 sec Forward Delay 15 sec

Bridge ID: 53248/a0:59:3a:03:00:cc
Hello Time 1 sec Max Age 10 sec Forward Delay 15 sec
TxHoldCount 6
```

name	id	role	state	cost	port_mode
e2-0	128.13	ALTR	BLK	200000	rstp,p2p,autoEdge
e1-5	128.10	ROOT	FWD	200000	rstp,p2p,autoEdge
e1-4	128.9	EDGE	FWD	200000	rstp,p2p,autoEdge
e1-2	128.7	EDGE	FWD	200000	rstp,p2p,autoEdge

In the upper part of the above output you can find data regarding the currently determined *Root Bridge*, its priority and MAC address, the local **ROOT** port chosen, the root path cost for the path leading to such bridge and the timers values advertised by the *Root Bridge* itself. Immediately beneath such data, but before ports data, you can

find configuration information regarding the switch itself on which you have just issued the command. At the end you have a summary reporting RSTP data for each port currently enabled on the system and considered by the running RSTP; here you can find listed for each port:

- **id:** its priority
- **role:** its current role in the topology
- **state:** its forwarding state
- **cost:** its associated port-cost
- **port_mode:** a set of flags indicating some specific configuration parameters:
 - **rstp/stp:** this flag specifies if this port is running in RSTP mode, or if it has reverted to old STP mode since it has received an old STP Configuration BPDU frame on this port and so it has determined to be connected to a bridge running old STP
 - **autoEdge/adminEdgeOn/adminEdgeOff:** this flag reports the edge mode set for the port: when *autoEdge* is displayed the edge mode of this port will be automatically/dynamically determined depending on received BPDU frames. If instead you have administratively configured the port's edge mode by means of the peer-interface spanning-tree edge mode commands, here you will see the actual administrative edge mode status (*adminEdgeOn* or *adminEdgeOff*).
 - **p2p:** this flag states that the port is a point-to-point full-duplex link capable of RSTP hand-shaking.

6.1.3 Configuring RSTP parameters

You can actively influence the finally calculated active topology by properly modifying the spanning-tree protocol standard configuration parameters. These parameters can be divided into two distinct sets:

- the set of spanning-tree global configuration parameters, available at the *global configuration* level – sub-menu `spanning-tree`, which holds spanning-tree parameters that pertain to the switch as a whole, like the *BridgePriority*, various timers and the running status of the protocol itself, as you can see in the following on-line help screenshot:

```
0@SW-Std.03-00-CA>configure spanning-tree           (? pressed)
enable          Enable redundancy protocol process
forward-delay   Change Forward Delay timer value(seconds)
hello-time      Change Hello Time timer value(seconds)
max-age         Change Max Age timer value(seconds)
mode            Select redundancy protocol mode to use.
priority        Change Bridge Priority value
tx-hold-count   Change TX Hold Count value. Range is 0-10. Set 0 for
infinite.
```

- the set of the per-interface spanning-tree configuration parameters, available at the interface-specific configuration level – sub-menu `spanning-tree`, which

holds spanning-tree parameters that relate to that specified interface, like *PortPriority* and *PortPathCost*, as you can see in the on-line help:

```
0@SW-Std.03-00-CA>configure interface e0-2 spanning-tree (? pressed)
edge          Set interface spanning-tree edge property to auto detect
enable        Enable redundancy protocol on this port
port-cost     Set interface spanning-tree port-cost property value
port-priority  Set interface spanning-tree port-priority property value
```

By factory default, in the absence of any specific configuration the device will power up with the RSTP process running on each available interface of the device using the default parameters.

Parameter	Value
STP Mode	rapid-stp
HelloTime	2 sec
ForwardDelay	15 secs
MaxAge	20 secs
TxHoldCount	6
BridgePriority	32768
PortPathCost	200000
PortPriority	128
Port Edge Mode	auto

Table 4: RSTP default parameters

When these parameters are set to their default values, they are not listed in the *running* configuration.

The admitted ranges for the above RSTP parameters are in compliance with the values recommended by the standard and the system will complain with an informational error message when trying to set unacceptable values. For example:

```
0@SW-Std.03-00-CA>configure spanning-tree priority 24577
Value 24577 NOT ALLOWED for Bridge Priority param.
Must be in range <0-61440> with 4096 increments.
```

or

```
0@SW-Std.03-00-CA>configure spanning-tree forward-delay 4
Value 4 for ForwardDelay param is NOT consistent with actual MaxAge param value:
20
This condition should be met: 2 x (ForwardDelay - 1) >= MaxAge
```

6.1.4 Port Edge Mode

In a typical deployment not all the enabled ports in the device are used to interconnect various LAN segments through inter-switch links, but most likely many of these ports will be connected to some sort of end device; such ports will never be able to generate any kind of topology loop, so that they can be safely transitioned to the

forwarding state without additional delay, and at the same time any link state transition happening on one of these ports should not trigger any topology change event to be sent throughout the network, in order to avoid the resulting unnecessary flushing of forwarding databases and the subsequent unneeded broadcasts and unknown unicast flooding. This kind of ports are usually called *edge ports*; please be aware that BPDUs are equally transmitted also on interfaces operationally or administratively determined to be in edge mode.

In order to cope with this situation, RSTP allows the user to set the edge mode of a port, as you can see in the on-line help:

```
0@SW-Std.03-00-CA>configure interface e2-0 spanning-tree edge      (? pressed)
  auto      Set interface spanning-tree edge property to auto detect
  off       Set interface spanning-tree edge property to administratively off
  on        Set interface spanning-tree edge property to administratively on
```

Each port can be set in one the following modes:

- **on**: the network operator has explicitly configured this port as an edge port connecting an end-device. This port will fast transition to forwarding state when enabled, and it won't be source of any topology change upon link-status events.

ATTENTION

Please note that such a port, administratively forced to be in edge mode, will anyway *immediately revert* to NON-edge mode when a BPDU is received on that interface. Moreover, keep in mind that *temporary loops* may arise on a port forced to be considered edge and subsequently attached to a bridge-device, until the first BPDU is received.

- **off**: the network operator has explicitly configured this port as an inter-switch link connecting two devices running a spanning algorithm (and so capable of generating forwarding loops troubles).
- **auto**: the RSTP process will determine autonomously whether this port is used for inter-switch connection or connects directly to an end-device; an *auto* edge port is operationally considered to be an edge port (used for end-device direct connection) when no BPDUs are received on that port for a time equal to *MigrationTime*, which is a fixed non configurable timer recommended by the standard to be set at 3 secs. This is the default edge configuration for all ports.

6.1.5 Selecting Running Spanning Tree Protocol Version

The renewed IEEE-802.1D-2004 RSTP redundancy protocol standard is designed to be backwards-compatible with the old IEEE-802.1D-1998 STP specification.

This aim is currently attained basically in two ways :

- *per-port STP compatibility mode*: whenever an old style STP *config BPDU* frame is heard on a port, that port is immediately rolled back to run in old STP

compatibility mode, since an old bridge not supporting new RSTP fast sync handshakes is now supposed to live on that port's segment. This per-port condition is automatically detected and this behavior cannot be further configured by user in any manner.

- *bridge STP compatibility mode*: the RSTP protocol was also natively designed to run in old STP compatibility mode on the switch as a whole; this behavior is user-configurable. As you can see from the following on-line help excerpt:

```
operatorSW-Std>configure spanning-tree mode          (? pressed)
  rapid-stp          Select Rapid-STP mode as redundancy protocol
  stp                Select STP compatibility mode as redundancy protocol
```

Using the above commands a user can explicitly decide that this switch will run the redundancy protocol in old STP compatibility mode. Note that when reverting an already running RSTP instance to old STP compatibility mode, the protocol stack will be re-initialized and so all the involved ports (usually all the available ones) will have their RSTP protocol stack restarted, so that such ports will be re-started from the Disabled/BLK state and then they'll be slowly (as stated in old STP) transitioned to their LRN/FWD states (as needed) as you can see in the following:

```
0@SW-Std.03-00-CA>configure spanning-tree mode stp
0@SW-Std.03-00-CA>show spanning-tree

Selected redundancy protocol STP is running.

Root bridge : 32768/a0:59:3a:03:00:d2
Root port: e1-5
Root path cost: 200000
Hello Time 1 sec  Max Age 10 sec  Forward Delay 15 sec

Bridge ID: 53248/a0:59:3a:03:00:cc
Hello Time 1 sec  Max Age 10 sec  Forward Delay 15 sec
TxHoldCount 6

name      id      role    state    cost      port_mode
=====
e1-5     128.10  ROOT    BLK      200000    stp,p2p,autoEdge
e1-2     128.7   DESG    BLK      200000    stp,p2p,autoEdge

.....
"...in the meanwhile a cable was connected to port e0-3..."
.....

0@SW-Std.03-00-CA>show spanning-tree

Selected redundancy protocol STP is running.

Root bridge : 32768/a0:59:3a:03:00:d2
Root port: e1-5
Root path cost: 200000
Hello Time 1 sec  Max Age 10 sec  Forward Delay 15 sec

Bridge ID: 53248/a0:59:3a:03:00:cc
Hello Time 1 sec  Max Age 10 sec  Forward Delay 15 sec
TxHoldCount 6

name      id      role    state    cost      port_mode
=====
e1-5     128.10  ROOT    FWD      200000    stp,p2p,autoEdge
e1-2     128.7   DESG    FWD      200000    stp,p2p,autoEdge
e0-3     128.4   DESG    LRN      200000    stp,p2p,autoEdge
```

Please note that when running the whole RSTP instance in old STP compatibility

mode you will lose also the per-port capability to automatically determine the EDGE mode of a port: in other words, when running in old STP mode the default per-port *autoEdge* mode will result in a port always being detected as non-edge designated and so it will be slowly transitioned to FWD state as required. In the case that we absolutely need that port being recognized as EDGE, it will be necessary to explicitly state its edge condition using a proper configuration command:

```
0@SW-Std.03-00-CA>configure interface e1-2 spanning-tree edge on
0@SW-Std.03-00-CA>show spanning-tree
```

Selected redundancy protocol STP is running.

Root bridge : 32768/a0:59:3a:03:00:d2
 Root port: e1-5
 Root path cost: 200000
 Hello Time 1 sec Max Age 10 sec Forward Delay 15 sec

Bridge ID: 53248/a0:59:3a:03:00:cc
 Hello Time 1 sec Max Age 10 sec Forward Delay 15 sec
 TxHoldCount 6

name	id	role	state	cost	port_mode
e1-5	128.10	ROOT	FWD	200000	stp,p2p,autoEdge
e1-2	128.7	EDGE	FWD	200000	stp,p2p,adminEdgeOn
e0-3	128.4	DESG	FWD	200000	stp,p2p,autoEdge

6.1.6 Disabling RSTP Process

As already said, in the absence of any initial configuration the spanning-tree process is enabled by default and runs with a default set of parameters: anyway you can stop the RSTP process as a whole at any time by issuing from the *base* level:

```
0@SW-Std.03-00-CA>no configure spanning-tree enable
```

which causes the spanning-tree process to be stopped immediately as you can see in the output of the `show spanning-tree` command:

```
0@SW-Std.03-00-CA>show spanning-tree
```

Selected redundancy protocol RSTP is NOT running.

Bridge ID: 32768/a0:59:3a:03:00:cc
 Hello Time 1 sec Max Age 10 sec Forward Delay 30 sec
 TxHoldCount 6

Alternatively you can instead stop the RSTP protocol only on a single port issuing from the *interface* level:

```
0@SW-Std.03-00-CA>no configure interface e0-2 spanning-tree enable
```

This causes the spanning-tree protocol to be stopped immediately but only on the specified port, which is then transitioned to NotManaged (Forwarding) state and, from now on, excluded from the running spanning-tree instance; as a consequence BPDUs are no more transmitted from such an interface.

IMPORTANT NOTE

Please pay attention to the fact that explicitly stopping the spanning-tree algorithm as above specified will have the effect to immediately transition all the involved ports to the FORWARDING state, and so, if physical loops are present in the topology, this will inevitably lead to *broadcast storms* and subsequent network services disruption until the physical loop is interrupted or the RSTP process is re-enabled.

6.2 Hardware bypass

Depending on the model, the Consist Switch may be equipped with up to 3 *Hardware Bypasses*.

On selected couples of Ethernet ports, the *Hardware Bypass* is an optional utility that, in case of failure, guarantees the electrical continuity between the two ports, so that the switch is virtually transparent on the network line.

In models where this feature is installed, the *Hardware Bypass* is normally activated (the 2 or 3 ports are short-cut) until the switch is fully started. Only at this point the ports involved in the *Hardware Bypass* are connected to the internal switch circuitry and the switch may begin to manage network packets normally.

The *Hardware Bypass* advantages depend strongly on the topology used. While in several occasions its use is a true benefit, in other architectures its usefulness may be not so relevant.

6.3 Link Aggregation

As an additional redundancy option, the device supports the creation of up to 6 or 7 (depending on the model) different virtual interfaces, each one composed by the aggregation of up to 6 single links. Once created, each of these *group* virtual interfaces is seen by the system as an additional regular switched interface (with only small usage limitations).

By means of such *group* interfaces, the user can attain:

- an increased aggregated throughput, configurable upon user's needs in linear increments, through the dynamic addition and removal of links to the group interface;
- a further level of redundancy to use in addition or in alternative to the classical spanning-tree protocols and hardware bypass facilities.

6.3.1 Overview

There are some characteristics to keep in mind at groups creation time:

- up to 7 virtual group interfaces can be created, named *grp0* to *grp6*¹ (such interfaces are named **Groups**);
- up to 6 single switched-interfaces (no groups nor routed interfaces) can be aggregated into one group (such component-interfaces are named **Group-Members**);
- each switched interface available on the device can be part of **one and only one** group at a time;
- the set of the group-members should be composed of homogeneously configured interfaces (if this is not the case a “Warning Message” will be issued on the console):
 - all links must be configured with the same speed;
 - all links must be configured in Full Duplex mode (since Link Aggregation is supposed to be used only on point-to-point links);

IMPORTANT NOTE

It's advisable **not** to leave the group-members interfaces in **Auto Negotiation** mode but to explicitly specify the members speed and Full-duplex setting in a homogeneous way; this will avoid possible troubles originating from interactions with *Parallel Detection* mechanism of the standard Auto Negotiation feature; an Auto Negotiation capable peer **cannot** correctly determine duplex link-state when connected to a peer NOT capable or NOT configured for Auto Negotiation, and in such cases the Auto Negotiation enabled peer will fall back to *Half Duplex* mode ('*Parallel Detection*'): as a consequence such link could be then excluded from Link Aggregation pool.

- a similar and compatible corresponding group interface must be configured on the other switch involved in group definition;
- only two switches can be connected through an aggregated link, which, in other words, means that every member belonging to the same group **must** be part of a connection between the same two switches (no multi-point aggregations).

Once created, the behavior of a group can be summarized as follows:

- as long as at least one of the group-member links is up, the whole group interface is up;
- when the last group-member link goes down the whole group goes down as well;
- the traffic load is automatically balanced among all the currently available (link up) group-members on the basis of the Ethernet MAC-addresses of the end-points of the data flows.

¹*grp5* on the models where the number of possible groups is 6

6.3.2 Link Aggregation Control Protocol (LACP)

The Link Aggregation Control Protocol (LACP) is a standard protocol that provides a method to control the bundling of group members to form the group interface.

When LACP is enabled the switch transmit periodical packets (LACPDU) down all group members ports informing the partner about member and group properties.

If on the other side LACP is also used, and the member properties matches, the port is *bounded* to it's group interface and the traffic can flow between the two devices. If group member properties does not match the port is set to *unbundled* and removed from the set of composing group members. In any case an *unbundled* port starts immediately acting again as a normal switched port and is also put back into the owning spanning-tree instance (if any running), so that it can be then used by the system as a normal port until an LACP agreement with the partner is reached again: at that point the freshly bundled port will be immediately removed from the spanning-tree instance and inserted back into the set of composing group members. This dynamic interaction between LACP and RSTP eases LACP based group configuration phase: you can manually configure LACP groups across a live network without the need to access each single network device via console, since the traffic flow is minimally disrupted while LACP negotiates links.

The implementation of the LACP protocol is compliant with the IEEE 802.3ad/802.1ax standard making the device inter-operable with equipment from other vendors that also comply with the standard.

Advantages of LACP over static configuration

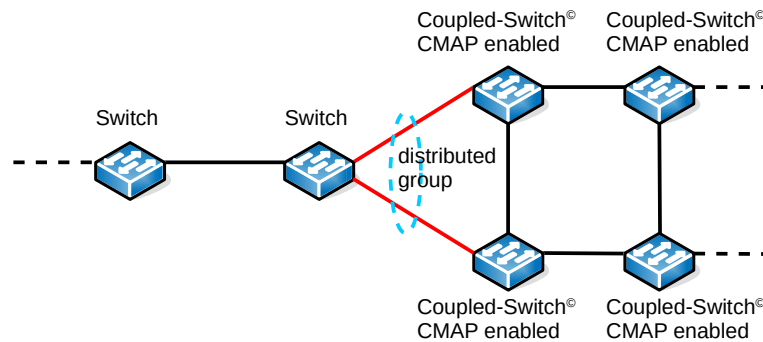
Using LACP instead of a static configuration gives a better detection of link failures and cable or configuration mismatching, avoiding to use failed links which would result in data packet loss.

With a static configuration, for example, a mismatch configuration where the number of group members is different on the two devices can't be detected (as soon as the links are up) and could prevent some traffic to flow correctly between the two devices.

Moreover, as already stated above, using LACP to configure link aggregation enables the user to manually configure groups across a live network minimally disrupting network connections while asynchronously configuring groups across participating peers.

6.3.3 Distributed Groups

Distributed groups is the kind of link aggregation that must be used for connecting a switch to a dual-homed link aggregation managed by the CMAP protocol (Coupled-Switch® backbone ports). See the following topology diagram and refer to Coupled-Switch© mode from page 167 for further information and details.



IMPORTANT NOTE

Do NOT use distributed groups to connect to dual-homed static or LACP groups since these kind of link aggregations are NOT managed by the CMAP protocol. Use distributed groups only and exclusively for connecting to Coupled-Switch® backbone lines with CMAP protocol enabled.

6.3.4 Creating Group Virtual Interfaces

The procedure for creating an aggregated group interface can be summarized as follows:

- **pre-verify** that the above specified constraints are enforced;
- **ensure spanning-tree protocol is enabled** using the usual `show spanning-tree` command: RSTP is needed only if the links belonging to the two participating switches are already connected each other at group creation time, or if LACP aggregation protocol is used; this avoids creation of temporarily loops until the aggregation is properly defined on both ends. You could anyway disable RSTP afterward as a whole or on a group interface after the link aggregation has been fully configured.

IMPORTANT NOTE

When creating a new group interface it is instead mandatory that RSTP would NOT be disabled (on a per-port basis) on any of the candidate group members; it could have been surely disabled as a whole on the switch BUT not on the single candidate link.

- **create empty group interface**

At first, a new **empty** virtual group interface should be created; this can be done simply issuing the command `configure interface <ifname>` where the `<ifname>` parameter must be chosen among the allowed names for group virtual interface (which are: `grp0`, `grp1`, `grp2`, `grp3`, `grp4`, `grp5` and, where supported, `grp6`) and it must not be an already existent interface.

```
0@SW-Std.03-00-CC>configure interface grp0
0@SW-Std.03-00-CC[config-if grp0]>
```

As a result an empty new group interface is created and the command-shell moves to the *specific configuration* level of the new interface; a new interface line appears in the running-configuration as you can see:

```
0@SW-Std.03-00-CC[config-if grp0]>do show running-config
....
interface grp0
....
```

- ***select the aggregation protocol***

By default, the aggregation protocol associated to the newly created group interfaces is set to static.

If you want to enable the LACP protocol negotiation you should execute the following command:

```
group mode lacp [ fast | slow ]
```

The `fast` and `slow` options will request the partner to send protocol LACPDU's respectively at fast (every 1 sec) or slow (every 30 sec) rate. The default mode is `fast`.

If you want to create a *distributed* group you should execute the following command:

```
group mode distributed
```

You can get back to the default aggregation protocol by issuing the `no` form of the command or by issuing the command:

```
group mode static
```

- ***specify group-members***

The next step needed after creation of the interface is to populate it with up to 6 group-members using the command:

```
group members <ifname> [ <ifname> ... ]
```

Such interface level command is available only for *group* interfaces and causes the provided space-separated list of interfaces to become members of the aggregation: the transition will happen immediately in *static* and *distributed* mode, or will be delayed till LACP successful negotiation, when `lacp` mode was selected (more details in the next sections).

```
0@SW-Std.03-00-CC[config-if grp0]>group members e0-0 e0-2
0@SW-Std.03-00-CC[config-if grp0]>do show running-configuration
....
interface grp0
  group members e0-0 e0-2
....
```

- ***enable group*** interface as a whole

As a last step in bringing into life our aggregated link, the group interface

should be enabled with the usual command:

```
0@SW-Std.03-00-CC[config-if grp0]>enable
```

A few words on what has been going on in the system up to now:

- a new group has been created;
- an aggregation protocol has been chosen: *static*, *LACP* or *distributed* mode;
- composing group members have been defined; this step ends up with different effects on the system depending on the previously chosen aggregation protocol:
 - ***static* or *distributed* mode:** upon definition such group-members interfaces have been immediately excluded from the running spanning-tree instance and such ports have been transitioned to the *blocking* state; this is needed to avoid loops, during the asynchronous phase of the definition of aggregations on the two participating switches. At the same time the freshly created virtual interface *grp0* has been added to the running spanning-tree instance, although still in a down state, since the group interface itself is still disabled. This way you can experience traffic disruption across the group link until the link aggregation is not properly and symmetrically configured also on the other side of the group: to avoid this, you should have carefully configured RSTP parameters in advance to keep available an alternate data path to the aggregation peer. (In other words, just created groups should NOT become ROOT/FWD until the other end has been fully configured, if you're in static mode and you want to maintain connectivity to other peer).
 - ***lacp* mode:** until LACP has successfully negotiated a defined group member for bundling, the involved link continues to act regularly as a normal switched interface and to belong to the owning spanning tree instance; this way no loops arise and the network connections flowing through the involved interfaces are not disrupted. As soon as a link is successfully bundled by LACP negotiation, it is then immediately excluded from the owning spanning tree instance, transitioned into blocked state, and then inserted into the owning group, which has already also begun participating as a whole into any running RSTP instance (if group already enabled). This behavior minimally impacts flowing network connections so that you can safely configure groups asynchronously across involved peers using *KMsh* command shell via a remote ssh connection passing through the group aggregation links you're configuring.
- the group has been finally enabled and now behaves as a regular switched-interface in place of the participating members as you can see also from the status of the spanning-tree:

```

0@SW-Std.03-00-CC>configure interface grp0
0@SW-Std.03-00-CC[config-if grp0]>do show spanning-tree

Selected redundancy protocol RSTP is running.

Root bridge : 32768/a0:59:3a:03:00:cc
Root port: This is the R00T Bridge.
Root path cost: 0
Hello Time 1 sec Max Age 10 sec Forward Delay 15 sec

Bridge ID: 32768/a0:59:3a:03:00:cc
Hello Time 1 sec Max Age 10 sec Forward Delay 15 sec
TxHoldCount 6

name      id      role    state    cost      port_mode
=====
grp0      128.14  DESG    FWD      200000    rstp,p2p,autoEdge
e1-4      128.9   DESG    FWD      200000    rstp,p2p,autoEdge
e1-1      128.6   EDGE    FWD      200000    rstp,p2p,autoEdge

```

As already said, this group interface roughly acts as a regular switched interface and you can operate on it as you would on a standard one, apart from some limitations described in the following lines and promptly reminded by the command-shell upon operation.

When using LACP, as soon as a member link drops out of aggregation, such link dynamically goes back into the running RSTP instance and starts acting as normal switched interface.

As an example, we can change the per-interface spanning-tree port-cost and edge parameters on the group in the usual way, whereas the attempt to change the VLAN membership fails because this is not allowed on a group interface (as we will see later on):

```

0@SW-Std.03-00-CC>configure interface grp0
0@SW-Std.03-00-CC[config-if grp0]>spanning-tree edge off
0@SW-Std.03-00-CC[config-if grp0]>spanning-tree port-priority 96
0@SW-Std.03-00-CC[config-if grp0]>do show spanning-tree

Selected redundancy protocol RSTP is running.

Root bridge : 32768/a0:59:3a:03:00:cc
Root port: This is the R00T Bridge.
Root path cost: 0
Hello Time 1 sec Max Age 10 sec Forward Delay 15 sec

Bridge ID: 32768/a0:59:3a:03:00:cc
Hello Time 1 sec Max Age 10 sec Forward Delay 15 sec
TxHoldCount 6

name      id      role    state    cost      port_mode
=====
grp0      96.14   DESG    FWD      200000    rstp,p2p,adminEdgeOff
e1-4      128.9   DESG    FWD      200000    rstp,p2p,autoEdge
e1-1      128.6   EDGE    FWD      200000    rstp,p2p,autoEdge

0@SW-Std.03-00-CC[config-if grp0]>switchport mode trunk
Cannot change vlan's properties of a group device as a whole. Act on each single
slave component instead.

```

Since the manual creation of groups is an asymmetrical operation, performed on one of the two switches at time, the following must be kept in mind:

IMPORTANT NOTE

When using static or distributed mode aggregation protocol, while loops are avoided during group creation, communication across the group can be disrupted and cannot be guaranteed until the aggregated channel has been properly defined on both participating ends; traffic disruption is instead minimally impaired when using LACP mode aggregation protocol.

Please note that, if RSTP is enabled and running on a group interface, no matter what the aggregation protocol is, and if you want to force the RSTP process to preferably use the more bandwidth-capable aggregated link you just created, **you must explicitly lower** the link spanning-tree port-cost for that group interface using the per-interface *spanning-tree port-cost* command:

```
0@SW-Std.03-00-CC>configure interface grp0 spanning-tree port-cost 2000
```

6.3.5 Getting information about group interfaces

You can get either detailed or summarized information about the current status of one or all group interfaces and their members by issuing

```
show link-aggregation [interface <grp>]
```

or

```
show link-aggregation summary
```

Following some examples of command usage:

```
A@sw-Std.03-00-CC>show link-aggregation
Interface grp0
  Aggregation protocol is Distributed

  Status: enabled/up

  Port  Status  Bndl  Actor key  Partner key  Primary
  -----
  e0-2   Up       Yes   --         --          No
  e0-0   Up       Yes   --         --          Yes

Interface grp1
  Aggregation protocol is Static

  Status: enabled/up

  Port  Status  Bndl  Actor key  Partner key  Primary
  -----
  e0-3   Up      --    --         --          No
  e0-1   Up      --    --         --          Yes

Interface grp2
  Aggregation protocol is LACP
  System ID: 32768/A0:59:3A:03:00:CE  Requesting fast LACPSUs
  Partner ID: 32768/00:80:63:BB:5F:18  Requesting fast LACPDUs

  Status: enabled/up

  Port  Status  Bndl  Actor key  Partner key  Primary
  -----
  e1-7   Up      Yes   0x09      0x22        No
  e1-5   Up      Yes   0x09      0x22        Yes
```

Note that if using LACP protocol *System* and *Partner* information together with member *bundling status* and *keys* are reported.

Alternatively you can get a summary report by issuing the command:

```
A@sw-Std.03-00-CC>show link-aggregation summary

  Port flags: (U) - up   (b) - boundled   (*) - primary port

  LAG   Protocol   Status      Ports
  -----
  grp0  Distributed  enabled/up  e0-2(Ub) e0-0(Ub*)
  grp1  Static       enabled/up  e0-3(U)  e0-1(U*)
  grp2  LACP        enabled/up  e1-7(Ub) e1-5(Ub*)
```

6.3.6 Managing Group Virtual Interfaces

Once the group interface is created, you can still dynamically vary its group-members composition using the same command used to populate it at creation time, specifying the updated list of wanted group-members:

```
group members <ifname> [ <ifname> ... ]
```

So that you can add or remove further members to/from an existent group:

```

0@SW-Std.03-00-CC>configure interface grp0
0@SW-Std.03-00-CC[config-if grp0]>do show running-config
...
interface grp0
  group members e0-0 e0-2
  ...
...
0@SW-Std.03-00-CC[config-if grp0]>group members e0-0 e0-2 e1-4
0@SW-Std.03-00-CC[config-if grp0]>do show running-config
...
interface grp0
  group members e0-0 e0-2 e1-4
  ...

```

As stated above, any new freshly-added/bundled link member will be excluded from the running spanning-tree instance as a single link, and on the other side any composing member link removed/unbundled from the group will be re-inserted into the running spanning-tree instance as a single entity.

Just keep in mind that *you cannot completely empty* the group of all its members; in that case the system warns you to destroy the whole group instead.

In addition to the changes to the group composition, you can also modify common properties as you would do with a normal switched-interface, but some limitations arise depending on the fact that you are trying to change some properties on a group interface or on a normal switched-interface which is currently participating in a group as a member.

Such limitations can be summarized as follows:

Interface Command	Allowed on Group	Allowed on Group-member	Notes
speed	No	No	-
duplex	No	No	-
dhcp-server	No	No	-
ip dhcp-option82	Yes	No	Anyway the dhcp-option82 payload will reference the originating group-member and not the enclosing group.
ip dhcp-snooping	Yes	No	-
lldp	No	Yes	-
qos	No	Yes	Must be set consistently across group members by user.
spanning-tree	Yes	No	Can be set on group members too, but only if they're UNBOUND

Interface Command	Allowed on Group	Allowed on Group-member	Notes
			slaves of an LACP group interface.
<code>switchport</code>	No	Yes	Must be set consistently across group members by user.
<code>enable</code>	Yes	Yes	-
<code>dot1x</code>	No	No	-

As additional notes about qos and switchport commands:

- **qos:** it is up to the user to set the quality of service parameters across the participating group members in a consistent way;
- **switchport:** it is up to the user to consistently define the VLAN membership properties across the participating group-members using the switchport command; this means that, in order to put a group into a specified VLAN or to define a group as an interlink trunk port (a inter-switch link across which all defined VLANs can flow), the user has to manually set the switchport property on all the composing group-members.

6.3.7 Destroying Group Virtual Interfaces

Whenever you do not need anymore an aggregated link you can destroy it, and all of its composing members will be released from the group and will act again as normal switched interfaces participating to any running spanning-tree instance as single links.

To destroy a group as a whole simply issue the command:

```
** Konuendo Platform Configuration Shell **
**      Railway Switch Version      **
**      Press '?' for Help          **
0@SW-Std.03-00-CC>no configure interface grp0
```

You can observe that the composing members, now released, have immediately been included again in the running spanning-tree instance.

```
A@SW-Std.03-00-CC>show spanning-tree
```

```
Selected redundancy protocol RSTP is running.
```

```
Root bridge : 32768/a0:59:3a:03:00:cc
```

```
Root port: This is the ROOT Bridge.
```

```
Root path cost: 0
```

```
Hello Time 1 sec Max Age 10 sec Forward Delay 15 sec
```

```
Bridge ID: 32768/a0:59:3a:03:00:cc
```

```
Hello Time 1 sec Max Age 10 sec Forward Delay 15 sec
```

```
TxHoldCount 6
```

name	id	role	state	cost	port_mode
e1-4	128.9	DESG	FWD	200000	rstp,p2p,autoEdge
e1-1	128.6	EDGE	FWD	200000	rstp,p2p,autoEdge
e0-2	128.3	DESG	FWD	200000	rstp,p2p,autoEdge

In any case when destroying, modifying or creating groups always remember that, when such operations are done manually some anomalies in data flow and reported spanning-tree ports state could happen, until both ends of the group are not properly set in a consistent and symmetric way; the system anyway is designed to reduce to a minimum data flow disruption and loop creation upon group definition.

Chapter 7 - VLANs

The Consist Switch supports the definition of IEEE-802.1Q Virtual LANs (VLAN) as a means of broadcast-domain separation.

7.1 VLANs overview and terminology

As specified in IEEE-802.1Q standard, using VLAN abstraction you can associate a group of ports of the switch, typically belonging to the same IP sub-network, to a specific VLAN ID (VID), thus restricting the burden of traffic related to such sub-network to that closed subset of ports, where the only devices really interested in that traffic allegedly reside.

If you define a VLAN which crosses the physical device switch boundary and extends across multiple different switches, you need a mean to maintain such VLANs separation when your frames travel across the inter-switch links: IEEE-802.1Q standard maintains such separation by *tagging* frames egressing from an inter-switch link (a *trunk* from now on).

A VLAN is uniquely identified across switches by the means of its VID, which is a 12-bit integer number between 1 and 4094 chosen by the user when the VLAN has been defined (integers 0 and 4095 are reserved by the standard).

You must be aware, when referring to a port of the switch as a VLAN member, that each available port can be configured to be in one of the following two, mutually exclusive, modes regarding VLAN configuration:

- **access-port:** a port configured in access mode forwards only *untagged* frames belonging to the associate VLAN ID. This means that each frame which comes in an access-port must be *untagged* (or will be discarded), and will then be associated to the VLAN domain which contains such port. On the other side each frame that egresses such port surely belongs to the port's VLAN and will be transmitted *untagged*. This kind of port is generally used to connect end-devices and not as an inter-switch link, but this is not always the case.
- **trunk-port:** a port configured in trunk mode is generally used to forward traffic belonging to multiple VLANs across two switches since it handles both *tagged* and *untagged* frames. Traffic egressing a trunk-port will be automatically tagged with an 802.1Q tag header stating its VLAN membership by means of its VID number. As a special case one VLAN per-trunk will egress untagged; we will call this the *trunk-native* VLAN.

When you create a trunk-port all the already existent VLANs will be forwarded across that trunk and, if not otherwise specified during trunk creation, the untagged trunk-native VLAN is always the default VLAN with VID 1. Moreover, if not otherwise specified, a trunk-port will forward frames belonging to every possible VLAN defined and to be defined on the system; in case, instead, it had been required to selectively

inhibit some VLANs to cross over a certain inter-switch link you should use the pruning mechanism to explicitly allow or deny a well-defined set of VIDs to flow across such trunk-port.

7.2 Default Native VLAN 1

In the absence of any configuration, each port of the switch is configured as an access-port belonging to the *default native* VLAN, which is the VLAN with VID 1. Moreover, when you remove a port from a previously created VLAN, the removed port will become part again of the *default native* VLAN 1. This is also, if not otherwise specified, the trunk-native VLAN for each and every trunk you will create on the switch.

VLAN 1 is the only VLAN which has also a *routed interface* (named *vlan1*) associated with it, which at the end, if properly configured, is the interface through which you can manage the switch remotely.

Note that if you remove all switch ports from the default native VLAN you will lose the capability to reach the device remotely via *vlan1* configured interface, since there will be no more underlying physical switched interfaces backing the *vlan1* routed one.

You cannot destroy or rename the default native VLAN 1: you can only use its ports to define other VLANs using the commands illustrated further in this manual.

7.3 VLAN management

In order to take a look at the existent VLANs, from the *base* level issue the command `show vlans`:

```
0@SW-Std.03-00-CA>show vlans
VLAN Name                               AccessPorts          TrunkPorts
-----
0001 default_vlan                       e0-0 e0-1 e0-2 e0-3
                                         e1-0 e1-1 e1-2 e1-3
                                         e1-4 e1-5 e1-6 e1-7
                                         e2-0
```

The `show vlans` command displays currently existent VLANs and ports membership: the *TrunkPorts* column lists which ports on the system is considered to be an inter-switch link and consequently will transmit *tagged* frames, except for the ones belonging to the trunk for which they are native, which will be transmitted *untagged*.

It's also possible to require information about only a subset of existing VLANs using the command `show vlans <xx[,yy][,ww-zz]>`:

```

0@SW-Std.03-00-CA>show vlans 1,22-60
VLAN      Name                               AccessPorts          TrunkPorts
=====
0001      default_vlan                      e0-1 e1-0 e1-1 e1-2  e0-0 e0-2 e1-3
                                e1-6 e1-7 e2-0
0022      V0022                            e0-3 e1-4            e0-0 e0-2 e1-3
0050      V0050                            e1-5                e0-0 e0-2

```

In case, instead, it's required to gather more detailed information about the currently defined trunk-ports, you can use the command `show interface trunks`:

```

0@SW-Std.03-00-CA>show interface trunks
Trunk  Line  Native  Tag  Prune Set
-----
e0-0   down  0001    No   allow 1-4096
e0-2   down  0022    No   allow 1-4096
e1-3   down  0050    No   deny 30-70

```

which, among other things, displays for each defined *trunk*:

- **Native:** the defined *native-vlan* for this *trunk*
- **Tag:** the tagging state of the above mentioned *native-vlan*
- **Prune Set:** the set of allowed or denied VLANs on this *trunk*

7.4 Creating VLANs

The most straightforward way to create a new VLAN is to explicitly name its VID using `vlan` command. It is possible, if needed, to associate a literal description using the same command:

```

0@SW-Std.03-00-CA>configure vlan 10 name test_vlan
0@SW-Std.03-00-CA>show vlans
VLAN Name                               AccessPorts          TrunkPorts
=====
0001  default_vlan                      e0-0 e0-1 e0-2 e0-3  e1-0 e1-1 e1-2 e1-3
                                e1-4 e1-5 e1-6 e1-7  e2-0
0010  test_vlan

```

In this way you have just created a new VLAN with VID 10 named *test_vlan*; note that since the only things which matters in VLAN management in the end is VID number, you can safely omit to name the VLAN: in such a case the system will automatically generate a VLAN name based on its VID in the form V0010.

7.5 Managing VLANs ports membership

After having defined a new VLAN with the desired VID as showed above, you still have to explicitly associate the desired ports to the VLAN itself using the *interface configuration level* command `switchport`; the following will set up a couple ports in

access and trunk mode:

```
0@SW-Std.03-00-CA>configure
0@SW-Std.03-00-CA[configure]>interface e1-4 switchport mode access vlan 10
0@SW-Std.03-00-CA[configure]>interface e2-0 switchport mode trunk native vlan 10
0@SW-Std.03-00-CA[configure]>interf e1-5 switchp mode trunk prune allow 10,15-25
0@SW-Std.03-00-CA[configure]>do show vlans
```

VLAN	Name	AccessPorts	TrunkPorts
0001	default_vlan	e0-0 e0-1 e0-2 e0-3 e1-0 e1-1 e1-2 e1-3 e1-5 e1-6 e1-7	e2-0
0010	test_vlan	e1-4	e1-5 e2-0

Please note that the starting tokens of the fourth command has been shortened to fit the line width of this manual; but they are correctly accepted by the menu since there is no prefix ambiguity with any other valid menu token.

As you can see by the above `show vlans` command output, we have just associated to VLAN 10, the port `e1-4` as an access-port and the port `e2-0` as a trunk *native* for VLAN 10: this means that from now on each and every VLAN (not pruned) which will be created will be forwarded on trunk `e2-0`, taking care to transmit every egressing frame *tagged* except for the frames belonging to the user-defined trunk native VLAN 10, which are instead transmitted *untagged*.

Moreover, we defined also interface `e1-5` as trunk-port, and we associated a pruning interval so that the only vlans allowed to be forwarded on trunk `e1-5` will be VLAN 10 and the VLANs in the range 15-25 inclusive.

The prune set for a trunk can be defined as:

- the set of allowed VLANs:

```
0@SW-Std.03-00-CA>conf interf e1-5 switchp mode trunk prune allow 10,15-25
```

- the set of denied VLANs:

```
0@SW-Std.03-00-CA>conf interf e1-5 switchp mode trunk prune deny 10,15-25
```

System default is `allow 1-4094`, which means that no pruning is enforced on every newly created trunk, unless explicitly requested by configuration.

IMPORTANT NOTE

Setting **non homogeneous pruning sets** among the defined trunks can lead to inter-switch **connectivity troubles** on the pruned set of VLANs, because the standard RSTP process currently implemented on the switch simply ignores the VLANs existence (and their pruned state) when it has to determine the active topology, and eliminating possibly physical loops.

Using `switchport` command you are free to move ports directly between already defined or still non-existent VLANs, both in access and in trunk mode. Using the *no-form* of `switchport` command, you can remove the port from a VLAN and put it back to the default native VLAN 1. If you do not specify the VLAN membership while using command `switchport`, the default VLAN with VID 1 is assumed, thus the following

commands will have both the same effect:

```
0@SW-Std.03-00-CA>no configure interface e1-4 switchport mode access vlan 10
0@SW-Std.03-00-CA>configure interface e1-4 switchport mode access
```

An alternative way to create a new VLAN in the system is to simply define a port mode using `switchport` command with a non-existent VID; in this way the new VLAN will be created implicitly; this works for both *access* or *trunk* port modes.

Take a look at the following command sequence issued in the above scenario:

```
0@SW-Std.03-00-CA>configure interface e1-7 switchport mode access vlan 22
VLAN 22 did NOT exist. Created vlan 'V0022'.

0@SW-Std.03-00-CA>configure interface e0-3 switchport mode trunk native vlan 66
Required trunk-native VLAN 66 did NOT exist. Created vlan 'V0066'.

0@SW-Std.03-00-CA>show vlans
```

VLAN Name	AccessPorts	TrunkPorts
0001 default_vlan	e0-0 e0-1 e0-2 e1-0 e1-1 e1-2 e1-3 e1-5 e1-6	e0-3 e2-0
0010 test_vlan	e1-4	e0-3 e1-5 e2-0
0022 V0022	e1-7	e0-3 e1-5 e2-0
0066 V0066		e0-3 e2-0

We have indirectly created two new VLANs, simply setting a port in access or trunk mode and naming a still not existent VID; note that in such a case you cannot specify a name for the VLAN so the default name is generated for you by the system upon creation. However, you can change such name using an explicit `vlan` command:

```
0@SW-Std.03-00-CA>configure vlan 66 name demoring
0@SW-Std.03-00-CA>show vlans
```

VLAN Name	AccessPorts	TrunksPorts
...		
0066 demoring		e0-3 e2-0
...		

When you define a new VLAN, traffic tagged as belonging to that VLAN will be forwarded on each switch port configured in trunk mode, for which the new VLAN is not pruned, and on each switch port configured in access mode for that VLAN. Thus, defining a VLAN without explicitly assigning access ports makes sense, since all incoming traffic tagged as belonging to a VLAN which has not been defined on the switch would be otherwise discarded.

7.6 Destroying VLANs

Although you can remove all ports from a VLAN by putting them back to default VLAN 1 using the `switchport` command, an already existent VLAN is not destroyed simply because it has no more access or native trunks associated with it. Thus, it

remains forwarded through all existent trunks, for which it's not pruned, until explicitly removed from the system.

Let us suppose this scenario:

```
0@SW-Std.03-00-CA>show vlans
```

VLAN	Name	AccessPorts	TrunkPorts
0001	default_vlan	e0-0 e0-1 e0-2 e1-0 e1-1 e1-2 e1-3 e1-5	e0-3 e2-0
0010	test_vlan	e1-4 e1-6	e0-3 e1-5 e2-0
0022	V0022	e1-7	e0-3 e1-5 e2-0

You can explicitly remove a whole VLAN using the *no-form* of `vlan` command. In this case if the VLAN was not empty yet, the system could tackle this condition in two different ways:

- when no trunk has been defined as *native* for this VLAN, it proceeds to remove the whole VLAN, by removing in advance all the access ports defined for it, and then removing the VLAN itself from the list of existent VLAN forwarded by any existent trunk on system:

```
0@SW-Std.03-00-CA>no configure vlan 10
Deleting Vlan 10.
Removing also all its access ports: e1-4 e1-6
```

- if a trunk has been previously defined as *native* for this VLAN, the system refuses to take further action: in such a case you first manually remove the native trunk using explicitly the `switchport` command:

```
0@SW-Std.03-00-CA>no configure vlan 22
Cannot delete Vlan 22.
Trunk e2-0 marked as native for this vlan.
```

In this case you can remove such VLAN in the following way:

```
0@SW-Std.03-00-CA>no configure interface e2-0 switchport mode trunk
0@SW-Std.03-00-CA>no configure vlan 22
Deleting Vlan 22.
Removing also all its access ports: e1-7
```

Chapter 8 - Quality of Service (QoS)

8.1 QoS on Ethernet packets

Quality of Service allows you to prioritize packets so that time sensitive data can be transferred smoothly and with minimal delay over your network.

Priorities are assigned to packets as they enter the switch by inspecting their QoS fields.

Packet QoS requirements can be defined at MAC level (layer 2 of the OSI reference model) or inside the layer 3 protocol header.

Note that layer 2 QoS fields are lost as soon as the packet is forwarded by a layer 3 device (router), while layer 3 QoS fields are maintained.

8.2 Layer 2 QoS

The Consist Switch supports layer 2 QoS as defined by the IEEE 802.1p specification. This is an extension of the IEEE 802.1Q (VLAN tagging) standard and works in tandem with it.

802.1Q standard specifies a tag added to the Ethernet header in order to carry VLAN information. This tag has two parts: the VLAN ID (12-bits) and a Class of Service (CoS) value (3-bits).

The CoS bits provide space for 8 different levels of service as listed below:

CoS	Traffic characteristics
0 (lowest)	Best effort
1	Background
2	Spare
3	Excellent effort
4	Controlled load
5	"Video" < 100 ms latency
6	"Voice" < 10 ms latency
7 (highest)	Network control

Table 5: Class of Service (CoS) levels

Because of the use of 802.1Q tagging, layer 2 QoS should be used on trunk ports only.

Normal access ports, in fact, discard tagged frames.

On access ports you can associate incoming packets with a user defined CoS value which will be used inside the 802.1Q tag if the packet is transmitted tagged.

8.3 Layer 3 QoS

The Consist Switch supports layer 3 QoS for IP packets as defined in RFC 2474/2475 Differentiated Services Architecture (DiffServ).

DiffServ uses the 6-bit Differentiated Services Code Point (DSCP) field in the standard IP header for packet classification purposes. This field replaces the 3-bit *IP precedence* field originally used to classify and prioritize types of traffic.

QoS treatment is defined as *Per-Hop Behavior* (PHB) and is determined by DSCP values.

Up to 64 different DSCP values can be defined. However, most networks use only the following PHBs:

- **Default PHB:** associated with DSCP 0, typically best effort characteristics.
- **Expedited Forwarding (EF) PHB:** characteristics of low delay, low packet loss and low jitter, suitable for voice, video and other real-time traffic.
- **Assured forwarding (AF) PHB:** it defines 4 different levels of forwarding assurances for IP packets *AF classes*. Each class has 3 different levels of drop precedence. The combination of classes and drop precedence yields 12 separate DSCP encoding from AF11 through AF43. Should congestion occur between classes, the traffic in the higher class is given priority.
- **Class selector (CS) PHB:** it maps the TOS precedence bits into a DiffServ class. So there are 8 DSCP encoding from CS0 to CS7.

8.4 Priority levels

Priorities are assigned to packets as they enter the switch by inspecting the layer 2 or layer 3 QoS fields. The priority value is then used when the packet has to be transmitted from the output port. Each port has 8 independent output queues, one for each supported priority. Packets are transmitted in the order received for any given priority. Output queues are numbered from 1 to 8, with queue 8 having the highest priority and queue 1 the lowest.

The order the frames are transmitted is controlled by the scheduling mechanism associated to the port. Two different schedulers are available:

- **weighted:** an 33, 25, 17, 12, 6, 3, 2, 1 weighing scheme is applied to the 8 priorities. This prevents low priority traffic from being starved by high priority traffic.
- **strict priority:** all top priority frames are transmitted until that priority queue

is empty. Then the next lower priority queue is served. Using this approach, the packets with the lowest priority are transmitted only when there are no other higher priority packets to send. In some cases a high rate of packets with high priority can prevent packets with lower priority from being sent.

NOTE

On Consist Switch equipped with firmware version 6.x the number of queues is limited to 4 (this is shown clearly by the CLI menu help messages) and the weighing scheme is 8, 4, 2, 1.

By default, the *weighed* mechanism is used, but you can change this setting and define how many queues should be served using the strict priority mechanism, choosing a number of queues from 1 up to 6 or all queues.

For example, from interface-specific configuration level, the command:

```
A@SW-Std.03-00-CA>configure interface e0-1 qos prio-queues 1
```

specifies that queue 1 should be served using the strict priority mechanism while the other queues will continue to use the weighed mechanism.

8.5 Priority assignment to incoming packets

As mentioned before, the priority level assigned can be based on the 802.1p CoS field contained inside 802.1Q tagged frames or on the DiffServ Code Point (DSCP) field contained in the IP packets header.

You can choose how priority is assigned on a per interface basis. For example the command:

```
A@SW-Std.03-00-CA>configure interface e0-1 qos trust cos
```

defines that the initial priority should be assigned basing on the 802.1p CoS field value, while the command:

```
A@SW-Std.03-00-CA>configure interface e0-1 qos trust dscp
```

forces the use of the IP DSCP code point value as the priority selector.

By default, ports are configured to trust both fields with precedence given to 802.1p CoS field (`qos trust both`).

Note that IP DiffServ is a layer 3 mechanism for carrying QoS information. It is preserved while packets flow across a switched network and can't be changed by layer 2 switches.

802.1p, instead, is a layer 2 mechanism, it is contained only in 802.1Q tagged packets and can be respected or forced by layer 2 switches.

When an untagged frame enters a port configured to trust the CoS field, it will be assigned the port default CoS value. If the frame is already tagged, the CoS value will remain unchanged.

You can override the CoS assignment of incoming packets forcing the default CoS value of the port to be assigned to all packets. The command:

```
A@SW-Std.03-00-CA>configure interface e0-1 qos default cos 2 override
```

sets the default CoS value for the port *e0-1* to 2, forcing this value on all packets that enter the port.

Remember that the 802.1p CoS value is maintained only if the packet is forwarded tagged out from a trunk port.

8.6 Priority mapping

A priority map is used to associate CoS and DSCP values to one of the eight available priority levels, that is the 8 egress queues.

You can inspect the current mapping using the command:

```
A@SW-Std.03-00-CA>show qos prio-map
Queue   CoS      DSCP
-----
 1       0       cs0
 2       1     cs1,af11,af12,af13
 3       2     cs2,af21,af22,af23
 4       3     cs3,af31,af32,af33
 5       4     cs4,af41,af42,af43
 6       5     cs5,ef
 7       6     cs6
 8       7     cs7
-----
```

On Consist Switch equipped with firmware version 6.x the mapping is slightly different since the number of available priority queues is limited to 4:

```
A@SW-Std.03-00-CA>show qos prio-map
Queue   CoS      DSCP
-----
 1       1,2     cs0,cs1,af11,af12,af13
 2       0,3     cs2,af21,af22,af23,cs3,af31,af32,af33
 3       4,5     cs4,af41,af42,af43,cs5,ef
 4       6,7     cs6,cs7
-----
```

If required you can change this mapping for the CoS and DSCP values respectively. For example the command:

```
A@SW-Std.03-00-CA>configure qos cos-map 1 0,1,2,3
```

assigns a priority level of 1 to the first four 802.1p CoS values.

You can also remap the DSCP code points. The command:

```
A@SW-Std.03-00-CA>configure qos dscp-map 2 af41,af42
```

maps the code points *Assure Forwarding 41* and *Assure Forwarding 42* to the priority queue 2.

While on Consist Switch equipped with firmware version 6.x the DSCP code points can be referred to exclusively using their string aliases, from the firmware version 7.2.1 the DSCP code points can be referred to either using their string aliases or using their hex

values. The following table reports the available aliases and their associated values:

Code point alias	Hex	Bitmask			
CS0	00	000000	AF32	28	011100
CS1	01	001000	AF33	30	011110
AF11	10	001010	CS4	32	100000
AF12	12	001100	AF41	34	100010
AF13	14	001110	AF42	36	100100
CS2	16	010000	AF43	38	100110
AF21	18	010010	CS5	40	101000
AF22	20	010100	EF	46	101110
AF23	22	010110	CS6	48	110000
C3	24	011000	CS7	56	111000
AF31	26	011010			

8.7 Rate limiting

Under some circumstances it may be required to limit the network traffic entering or leaving a switch port in order to guarantee the planned overall level of QoS.

For example, if you have associated a port to the highest priority level and you are using a strict prio-queue for serving that priority level, you might want to be sure that, if the end device connected to the port begin transmitting packets in uncontrolled way due to, for example, some failure on the end device itself, this will not have any effect on the whole network.

In this case, if you know that the end device should not produce more that 2 Mbps of data, you could limit the maximum rate accepted by the port to that value.

Another requirement could be connected with the level of broadcast traffic. Too many packets of this type can be problematic and it can result in potentially severe network congestion.

All these issues can be easily resolved by using the rate limit feature.

The interface command

```
switchport rate-limit inbound|outbound|broadcast <value> [K|M|G]
```

can be used to define the maximum value of traffic rate accepted or emitted by the selected port. The rate can be expressed in kbps, Mbps (by default) and Gbps².

For example, the following command limit to 5 Mbps the traffic entering port *e0-2*:

```
A@SW-Std.03-00-CA>configure interface e0-2 switchport rate-limit inbound 5 M
```

As the incoming traffic exceeds this limit, the switch begins to drop packets entering port *e0-2*.

The following command limit to 1 Mbps the amount of broadcast traffic accepted by port *e1-4* reducing the impact of what is commonly called a *broadcast storm*:

² In the firmware series version 6.x the Gbps unit is not available.


```
A@SW-Std.03-00-CA>configure interface e0-2 switchport rate-limit broadcast 1 M
```

Outbound traffic values can be also specified in PPS (Packets Per Second):

```
A@SW-Std.03-00-CA>configure interf e0-2 switchport rate-limit outbound 10000 PPS
```

These commands can be used together on the same interface.

Broadcast limit is a subset of the overall incoming traffic and can be associated with a lower value.

NOTES

- Rate limit settings with *inbound direction* and *broadcast direction* are mutually incompatible.
- The rate limiting is meaningful for *physical interfaces only*; related commands cannot be issued on link aggregations (group interfaces).

8.7.1 Priority queues rate limiting

Rate limiting settings can also be applied to the priority queues defined in the Priority mapping section.

The interface command can be used to define the maximum value of queued traffic rate accepted by the selected port:

```
[no] configure interface NAME switchport rate-limit inbound VALUE UNIT prio QUEUE
```

where *QUEUE* represent one of the eight priority queues (a value in the range $[1..8]^3$).

For example, the following command limits to 5 Mbps the traffic with priority mapped to queue 1 entering port *e0-2*:

```
A@SW-Std.03-00-CA>configure interface e0-2
A@SW-Std.03-00-CA[config-if e0-2]>switchport rate-limit inbound 5 M prio 1
```

NOTE

A priority queue rate limit setting with inbound direction is incompatible with a not priority based rate limit setting with inbound direction.

8.7.2 Profiles

Rate limit settings can be group together in a profile and applied to multiple interfaces.

The following command will create a profile (or remove it in the no-form):

```
[no] configure rate-limit profile NAME
```

- 3 On Consist Switch equipped with firmware version 6.x the number of queues is limited to 4 and this is shown by the CLI menu help messages.

Multiple settings can be defined in a profile as follows:

```
A@SW-Std.03-00-CA>configure rate-limit profile mylimits inbound 1 M prio 4
A@SW-Std.03-00-CA>configure rate-limit profile mylimits inbound 2 M prio 3
A@SW-Std.03-00-CA>configure rate-limit profile mylimits outbound 7 M
```

In the example the profile has been given the name `mylimits`.

A profile of rate limit settings is applied to a switch port interface by the command:

```
[no] configure interface INTERFACE switchport rate-limit profile NAME
```

As an example, the `mylimits` profile is applied to interface `e1-0`:

```
A@SW-Std.03-00-CA>configure interface e1-0
A@SW-Std.03-00-CA[config-if e1-0]>switchport rate-limit profile mylimits
```

Any change to a profile will apply to all the ports the profile has been applied to.

8.7.3 Committed Burst Size

The Consist Switch implements the rate limiting feature using the leaky bucket scheme. The behavior of the bucket for the inbound traffic can be finely tuned up by specifying the minimum size of a data burst that the bucket should be able to accommodate.

The actually applied value is shown together with the other details of any physical interface. The following is the output for an interface with no limits applied:

```
A@SW-Std.03-00-CA>show interface e1-1 details
...
Rate Limiting
-----
outbound unlimited
inbound unlimited (burst n/a)
```

The following is the output for an interface limiting both the outbound and broadcast traffic:

```
A@SW-Std.03-00-CA>show interface e2-0 details
...
Rate Limiting
-----
outbound kbps:34000
inbound kbps:4032 (burst bytes:135299)
```

This feature and its details described in the next sub-sections is not available on Consist Switch equipped with firmware series version 6.x.

8.7.3.1 *User supplied Committed Burst Size*

A Committed Burst Size value, in Bytes, can be specified by the user as optional parameter to the commands described above, per interface:

```
configure interface NAME
switchport rate-limit inbound|broadcast VALUE K|M|G burst BURST [prio QUEUE]
```

or per profile:

```
configure rate-limit profile NAME
    inbound|broadcast VALUE K|M|G burst BURST [prio QUEUE]
```

The supplied value (*BURST*) overrides the default one that sometimes can result in slightly inaccurate rates (especially for TCP traffic).

8.7.3.2 Automatic optimal Committed Burst Size

An optimal Committed Burst Size value can be automatically computed by the device. This facility is enabled by the use of the *auto* token instead of a numeric value, as follows:

```
configure interface NAME
    switchport rate-limit inbound|broadcast VALUE K|M|G burst auto [prio QUEUE]
```

or per profile:

```
configure rate-limit profile NAME
    inbound|broadcast VALUE K|M|G burst auto [prio QUEUE]
```

IMPORTANT NOTE

The automatic computation of the optimal burst size can last up to some seconds. The use of this feature can significantly increase the boot time that can last up to some minutes (in the worst case). A suggested good practice is to use this feature once, then read the computed value and use it in the final deployed configuration in place of the *auto* token.

Chapter 9 - Time

System date and time can be shown with the command `show system time`. The output will be similar to the following:

```
O@SW-Std.03-00-CA>show system time
Fri 13 May 2011 14:52:40 [UTC]
```

System clock is not backed up by an internal battery so the system time and date must be re-assigned on every boot or synchronized with external sources by means of one of the supported time protocols.

Components used and system design ensure that time keeping is the most accurate possible, but if you want to keep many devices synchronized with greater accuracy you should rely on a time synchronization service. The system can be configured to use one of such services (see *Auto-synchronization* section below).

To obtain information about the current clock configuration, you can use the `show clock` command. The output will be similar to the following:

```
O@SW-Std.03-00-CA>show clock
Current time: Fri 13 May 2011 14:51:27
Time protocol: none
Time server(s):
Timezone: Universal Coordinated Time
```

9.1 Configuration

In order to set the current time you need to be logged in as *admin*. First you need to set the timezone with the command

```
configure system timezone ZONE COUNTRY
```

For example:

```
A@SW-Std.03-00-CA>configure system timezone Europe Rome
```

You can obtain a list of valid values for *ZONE* and *COUNTRY* by pressing '?' while typing the command. This setting will be saved as soon as a `save running-config` command is issued.

In order to manually set the time issue the command

```
sysadmin set time DD MM YYYY HH mm SS
```

The meaning of the parameters is the following:

DD	day of the month
MM	month
YYYY	year
HH	hours
mm	minutes
SS	seconds

For example:

```
A@SW-Std.03-00-CA>sysadmin set time 02 08 2017 10 08 15
```

9.2 Auto-synchronization

The system clock can be configured for automatic synchronization with one or more time sources, by means of TIME or NTP protocols.

The protocol to be used for automatic synchronization can be set with the command:

```
configure system clock source PROTOCOL ADDRESS [ ADDRESS ]
```

where valid values for *PROTOCOL* are `ntp` and `timep`, and *ADDRESS* is the IP address of a reachable network resource. You can specify up to two server addresses.

Once the synchronization service has been configured, the device will synchronize regularly with time sources.

9.3 NTP client

If the system clock is configured to be automatically synchronized with NTP (as NTP protocol client), the user can obtain information about the status of the protocol issuing the `show ntp status` command. The output will be similar to the following:

```
A@SW-Std.80-31-60>show ntp status
Reference ID      : AC100164 (172.16.1.100)
Stratum          : 6
Ref time (UTC)   : Tue Apr 02 07:17:37 2019
System time      : 0.000000000 seconds fast of NTP time
Last offset      : -102256784.000000000 seconds
RMS offset       : 102256784.000000000 seconds
Frequency        : 0.000 ppm slow
Residual freq    : -156.536 ppm
Skew             : 1000000.000 ppm
Root delay       : 0.013945554 seconds
Root dispersion  : 2.441490173 seconds
Update interval  : 0.0 seconds
Leap status      : Normal
```

By default the NTP client polls the time source (server) every 16 seconds. This interval can be made longer or shorter using the command `configure ntp client polling-time TIME` where *TIME* is a value (in seconds) in the range 1-8192. Once set, this configuration becomes effective after a device reboot.

NOTE

The *polling-time* parameter is rounded up to an actual value that is the closest power of two (1, 2, 4, 8, 16, ...). For example, any value in the range 7-12 corresponds to an actual value of 8 seconds; lower values will be mapped to 4 (or less), greater ones will be mapped to 16 (or more).

9.4 NTP server

The Consist Switch can act as time source (NTP server) for other devices in the network. This functionality is not enabled by default and can be toggled with the command `[no] configure ntp server enable`.

Chapter 10 - Security

The Consist Switch lives in a heterogeneous network environment, so it is mandatory to adopt all the options to guarantee the right security level.

10.1 Users

The access to the system CLI is protected by an authentication phase during which a user must supply the right credentials to gain access to the system.

In the Consist Switch there are 3 predefined users, reflecting different roles for professionals interacting with the device.

The operations those users are allowed to perform on the switch are exactly defined by their role. These roles are:

- **user:** this role is entitled only to watch at the device conditions without any chance to modify its behavior. This is the safest role.
- **operator:** this role gains the possibility to access the *global* or *specific* configuration level. This is the normal role used to configure the switch.
- **admin:** this role has total control over the device. It grants all the privileges and, because of this, it allows to use the *system administration* level commands. These commands deal with the most critical aspects of the device such as firmware upgrades, etc...

The initial password for those users are set equal to the user name itself.

The passwords may be changed by a configuration level command, however restricted to the *admin* role. For the user *operator*, for example, the following command must be used:

```
A@SW-Std.03-00-CA>configure system operator password SecretPassword
```

This command is recorded in the configuration file with the password encrypted.

If, for any reason, passwords are lost, they may be changed back to their factory values with a low level system operation detailed in *Password Recovery*, page 205.

NOTE

There is a group of commands that, for consistency sake, can only be issued by the *admin* role whilst defined outside the *system administration* level.

10.1.1 Enforcing password security

The password security strength can be enforced enabling the *password complexity check* feature.

If the check is enabled then any new password has to contain at least one lowercase and one uppercase letter, one number and one special character; furthermore it has to be at least 8 characters long.

This feature is turned off by default and can be activated with the following command:

```
configure system password complexity enable
```

10.1.2 Account locking

A user account can be locked for a given amount of time after a given number of failed login attempts. Account locking is off by default and can be enabled with the following command:

```
configure system login failed-attempts <1-100> [lock-timeout <1-1000000>]
```

The lock timeout part is optional and has a default value of 180 seconds.

NOTE

Account locking policy can not be used together with RADIUS Server authentication and it will be ignored in case RADIUS authentication is configured.

10.2 CLI Users RADIUS Authentication

The previously described CLI authentication phase can also be completed leveraging an external Remote Authentication Dial In User Service (RADIUS) authentication server.

In order to use such RADIUS based authentication for local users (user/operator/admin) you'll have at first to define a proper set of RADIUS parameters, and then to associate such radius based authentication-method to each desired user; some specific privileges' restrictions may apply when performing such configurations.

10.2.1 Configuring RADIUS Authentication

As already said, in order to use RADIUS authentication you must at first define the parameters needed to identify and successfully communicate with the chosen RADIUS authentication server (you must have *admin* full administration privileges to define primary and backup radius-servers and theirs options):


```
A@SW-Std.03-00-CA>configure system radius-server primary 172.16.16.111 secret
A@SW-Std.03-00-CA>configure system radius-server backup 172.16.16.112 secret2
A@SW-Std.03-00-CA>configure system radius-server backup options auth_port=1814
```

where possible radius options are showed in the following help excerpt:

```
A@SW-Std.03-00-CA>configure system radius-server primary options <optionlist>
configure and enable primary radius server options:
optionlist is opt1=val1[,opt2=val2[,opt3=val3...]]
where optx are: auth_port, acct_port, retry_cnt, timeout, src_ipaddr.
```

After having configured a valid server we can choose what kind of authentication-method to use for each defined CLI local user:

- Forcing local authentication only (default) for user *user*

```
A@SW-Std.03-00-CA>configure system authentication user local
```

- Forcing radius authentication only for user *user*

```
A@SW-Std.03-00-CA>configure system authentication user radius
```

- Allowing both RADIUS and local authentication for user *user*; in this last case, at first, user authentication is attempted using the defined external RADIUS server(s), and then system falls back to usual local authentication-method:

```
A@SW-Std.03-00-CA>configure system authentication user radius local
```

In the above examples we issued all the needed shell commands as user *admin*, but keep in mind that when dealing with the choice of the authentication-method the following privileges' restriction applies:

- user *user* cannot change either his/her own authentication-method, since have no access to configure level commands
- user *operator* can change authentication-method for himself/herself and for the lower-privileged user *user*, but cannot change authentication-method for higher-privileged user *admin*
- user *admin* can change authentication-method for any user

10.3 802.1X Port Authentication

The device supports port-based authentication through standard protocol 802.1X-2004; the Consist Switch embeds an 802.1X PAE Authentication server, that, after being properly configured and enabled, can be used to filter clients' access to physical ports depending on the outcome of 802.1X authentication exchanges. The device is not able to act as a 802.1X PAE Supplicant.

10.3.1 802.1X: Overview and terminology

Some terminology and concepts used to explain 802.1X configuration and operation:

- **PAE - Port Access Entity:** a logical entity associated to a physical LAN port, that

supports one or both of the 802.1X roles of Supplicant and/or Authenticator.

- **Supplicant:** it's the name used for the external client trying to get access to the network via 802.1X port authentication schemes. Usually it's a software layer embedded into the client's operating system. The Consist Switch does not support this role as a PAE, so we'll use this term to refer only to an external client attempting access.
- **Authentication Server:** it's the ultimate entity which decides if the supplicant-provided credentials were good, and access is to be allowed or not. In our scenario it's an external RADIUS Authentication/Accounting server.
- **EAP (Extensible Authentication Protocol):** it's the inner protocol used to securely carry authentication information between supplicant and authentication server.
- **EAPOL (EAP Over Lan):** this is the outer layer 2 protocol carrying all the authentication exchanges between the supplicant and the authenticator. It's the only type of traffic allowed on the 802.1X controlled port till authentication is successfully completed.
- **RADIUS:** this is the outer layer 3 protocol used by the authenticator to relay all the inner EAP authentication exchanges, received from the supplicant via EAPOL frames, to the RADIUS Server.
- **Authenticator PAE:** it's the intermediate system (this device) which talks directly with the Supplicants and Authentication Server and it's in charge of:
 - effectively inhibiting access and blocking all traffic on Unauthorized ports except authentication related frames' exchanges during the initial 802.1X authentication phase.
 - relaying EAP requests from the Supplicant to the Authentication Server properly unpacking such requests from containing EAPOL, and repacking them into well formed RADIUS packets, well-known to the final authentication system.
 - relaying EAP responses from Authentication Server to the Supplicant properly unpacking such responses from containing RADIUS packets, and repacking them into well formed EAPOL frames, well-known to the final Supplicant entity.
 - granting access to successfully authenticated supplicants.
 - maintaining/updating access-accounting information with the specified accounting server (where configured).

Additionally, from the perspective of the 802.1X Authenticator PAE a port can be configured in the one of following states:

- **force-authorized:** it's the default state in which a port is brought-up when 802.1X is not configured on that port. All traffic flows are allowed and no filtering is enforced on such a port; 802.1X does not care about any EAPOL frame possibly traveling across this port. The port is in fact not managed at all by 802.1X Authenticator. A port can be forced into this state at any time

through CLI commands.

- **force-unauthorized**: when in this state all traffic flows are instead inhibited and any EAPOL frame is likewise ignored by the 802.1X Authenticator PAE; also in this case the port is practically not managed by 802.1X. A port can be forced into this state at any time through CLI commands.
- **auto**: when a port is set to this state, the 802.1X Authenticator PAE starts managing such port and, if 802.1X is properly configured, the Authenticator starts also relaying any authentication exchange from the Supplicant to the AuthenticationServer. When a port is in this state only EAPOL authentication frames are allowed to flow through the port, till Supplicant credentials are successfully verified. A port can be configured in this state at any time through CLI commands.

Moreover, we should consider that a port controlled by 802.1X could be connected not only directly to a supplicant end-device, but also to an intermediate device like a switch/hub or an access-point; in this particular configuration we could end up having multiple per-port end-devices trying to get access to the LAN. To address such needs an 802.1X controlled port can be configured also to act in two different modes:

- **host-mode single**: in this mode *each and every* supplicant hitting the port MUST be authenticated before being granted access to the LAN.
- **host-mode multi**: in this mode instead *only the first* supplicant is in charge to perform a successful 802.1X authentication: once this first client, called hereinafter a **sentinel**, is authenticated, the port is virtually opened to any other client which shows up after, even if these subsequent clients won't be able to authenticate themselves (sort of **guest nodes**); but in this case, as soon as *the sentinel* fails re-authentication and disappears all the possibly unauthorized guest nodes are immediately forced out, while any other regularly authenticated supplicant (other than the sentinel) continue to hold the granted access.

Finally, when a proper and complete 802.1X configuration is in place, a controlled port can transition also to one of the following states:

- **authorized single**: this port was set in host-mode *single* and at least a supplicant has already been authenticated.
- **authorized_multi**: this port was set in host-mode *multi* and at least a supplicant has been authenticated (the *sentinel*). The access is then allowed to any following client on this port without the need of any additional authentication for those clients; this is true as long as the first authenticated client, the sentinel, remains authenticated.

10.3.2 802.1X Basic Configuration

Briefly, a basic and functional 802.1X Authenticator PAE configuration must fulfill at least the following areas:

- **define** an *aaa-domain* which holds the parameters needed to reach an active

and configured RADIUS Authentication Server from the Consist Switch hosting the 802.1X Authenticator.

- **associate** such *aaa-domain* to the port(s) on which you want to enable 802.1X access control.
- **enable** dot1x on such port(s)

Defining an *aaa-domain* for the back-end RADIUS Server simply means associating a *domain name* to a group of parameters needed to effectively contact and communicate with an external back-end RADIUS Authorization/Accounting Server(s).

Every radius *aaa-domain* can contain up to four server identity comprising:

- 2 Authentication Servers: primary and backup
- 2 Accounting Servers: primary and backup

The minimum required *aaa-domain* configuration must include at least one authentication server in order to be used for 802.1X access control. An example of basic configuration involving two distinct authentication and accounting servers follows:

```
A@SW-Std.03-00-CA>configure aaa-domain
A@SW-Std.03-00-CA[aaa-domain]>radius myDomainName auth-serv primary address
                                     172.16.16.10 secret mySecret
A@SW-Std.03-00-CA[aaa-domain]>radius myDomainName acct-serv primary address
                                     172.16.16.100 secret mySecret2
```

The chosen domain name must be one single word not exceed 32 characters in length.

At this point we can continue carrying on our 802.1X configuration on the desired ports.

Say you want to enable 802.1X access control on port *e0-3* using the above specified RADIUS Authentication and Accounting Server parameters, you have simply to:

- **associate** the desired *aaa-domain* to the port:

```
A@SW-Std.03-00-CA>configure interface e0-3 dot1x aaa-domain myDomain
```

- **enable** 802.1X PAE Authenticator functionalities on such port:

```
A@SW-Std.03-00-CA>configure interface e0-3 dot1x port-control auto
```

From now on such port will be managed by 802.1X using the specified external RADIUS Server; you can see progress of an authentication exchanges enabling *dot1x debug*:

```
A@SW-Std.03-00-CA>debug dot1x
...
... after a supplicant successfully authenticated on port
...
dot1x |EVENT [e0-3] -- link-status up
dot1x |[e0-3] -- Registered supplicant 000000000343 as sentinel.
dot1x |EVENT [e0-3] -- AUTH-COMPLETE 000000000343 172.16.16.10
```

NOTE

When defining an **aaa-domain** holding the parameters for the external radius server, keep in mind that *end-to-end full connectivity* should exist between the Authenticator PAE and such remote RADIUS Authentication Server for all this to work: this means, obviously, that must exist a valid route to get to such remote machine, but *also*, less obvious, that the 802.1X traffic flow must be routed through a switch physical port which belongs to the **native Vlan 1** (one), since that's the only vlan currently associated to the switch routed-interface *vlan1*.

10.3.3 802.1X Status and Debug

We can enable dot1x debug messages issuing:

```
A@SW-Std.03-00-CA>debug dot1x
```

An example output could be:

```
dot1x |EVENT [e0-3] -- link-status up
dot1x |[e0-3] -- Registered supplicant 000000000343 as sentinel.
dot1x |EVENT [e0-3] -- AUTH-COMPLETE 000000000343 172.16.16.10
```

which shows that:

- link status changed for a dot1x controlled port
- a supplicant is marked as sentinel being the first authenticated supplicant on such port
- basic authentication data is briefly reported as:

```
dot1x |EVENT [e0-3] -- AUTH-COMPLETE 000000000343 172.16.16.10
      ^^^      ^^^      ^^^      ^^^
      |        |        |        |
      {port}  {type event} {mac-address} {used AuthServ IP addr}
```

Possible *authentication events* reported in debug messages are:

- **AUTH-COMPLETE:** a new unknown supplicant has successfully authenticated.
- **AUTH-CONFIRMED:** an already authenticated supplicant has successfully renewed its authenticated state: this could be due to re-authentication attempts triggered both side (by Supplicant or by Authenticator)
- **AUTH-FAILED:** a new or known supplicant has failed first authentication or re-authentication. This incident will be reported into log file too.

We can obtain data about 802.1X port states and authenticated supplicants also on demand using the proper *show dot1x* informational command:

```
A@SW-Std.03-00-CA>show dot1x supplicants
```

Ifname	dot1XState	HostMode
e0-1	force-unauthorized	single
e1-2	auto	multi
e1-4	auto	multi
e1-5	force-unauthorized	single
e1-6	auto	multi
e2-0	authorized_single	single

Supplicant	Vlan	LastAuth	ActiveAuthServ
000000000343 *	1	Fri 1 Jan 2010 00:30:15 [UTC]	172.16.16.100

Issuing the above command you'll see that for each port, in a state other than *force-unauthorized*, are reported:

- its **dot1xState**:
 - force-unauthorized
 - auto
 - authorized_single
 - authorized_multi
- its **HostMode**:
 - single
 - multi
- a table containing any *currently authenticated supplicant*, listing for each of these:
 - its mac-address
 - an * **mark** indicating that such supplicant is the chosen sentinel for such port
 - the assigned VLAN (for the client and for the port)
 - the last time authentication has successfully completed
 - the currently active (last used) Authentication Server IP address

10.3.4 802.1X Advanced Configuration

Apart the basic 802.1X configuration we've seen so far, it's possible to fine tune also the following other *dot1x* related parameters accessible from the dot1x per-interface menu level:

- **eapol-version**: this is needed for being able to choose between version 2 (default) and version 1 EAPOL frame format; needed to address some specific issues with some old supplicant's client software not properly supporting new version frames.

```
A@SW-Std.03-00-CA>configure interface e0-0 dot1x eapol-version 1
```

- **host-mode:** this can be used to select between single (default) and multi authentication host-mode (as previously described).

```
A@SW-Std.03-00-CA>configure interface e0-0 dot1x host-mode multi
```

- **reauth-timer:** this is used to specify the per-port re-authentication interval to use; it's the amount in seconds between Authenticator triggered re-authentication attempts: when set to 0 (zero) it effectively disables re-authentications attempts as a whole. Default is 0 (zero): disabled.

```
A@SW-Std.03-00-CA>configure interface e0-0 dot1x reauth-timer 120
```

Additionally, it's possible also to instantly trigger a *forced* re-authentication on a chosen port; such action can be triggered on **all** supplicants currently authenticated on that port or on a single specific supplicant using its MAC address:

```
A@SW-Std.03-00-CA>configure interface e0-0 dot1x reauthenticate station all
dot1x |EVENT [e2-0] -- AUTH-CONFIRMED 00000000343 172.16.16.10
```

or

```
A@SW-Std.03-00-CA>configure interface e2-0 dot1x reauth station 00:00:00:00:03:43
dot1x |EVENT [e2-0] -- AUTH-CONFIRMED 00000000343 172.16.16.10
```

10.3.5 802.1X – Maintenance Access Via Local Users Back-end

Standard 802.1X port security solutions are generally cumbersome to configure and maintain: as we have seen, we should configure a few redundant Radius server in addition to the required switch local configurations.

Even though this is an acceptable scenario on a large scale deployment with multiple Radius authority configured to serve a multitude of supplicant end-devices, when, instead, the typical use case would be to simply allow access to the network for **only maintenance reasons** to authorized personnel, such complex Radius back-end configuration is particularly demanding.

In order to address these needs VDS Rail provides a special 802.1X based port-security solution that **does not depend on the configuration of an additional authentication server**, nor external neither internal.

After having enabled 802.1X support on the selected ports in the usual way, the conceived solution permits to easily associate such ports to a special '*aaa-domain*' named **local-db**: using such *aaa-domain* as back-end, port access will be granted to any 802.1X Supplicant that uses valid **local users credentials** (*admin / operator / user*); such binding will remain effective no matter what kind of users authentication was chosen (local vs radius), and across any possible change of authentication tokens through different means (password change both via *KMsh* and *TTCMP Network Designer*). In other words if you have access to the management functionalities of the switch you will have access also to any 802.1X secured port configured to use the **local-db** special *aaa-domain*.

The only limitation of such mechanism is that the 802.1X Supplicant **MUST** be

configured to use **EAP-TTLS/PAP** as EAP authentication method: in this way a TLS tunnel will be established during phase1, while in the following phase2 the user credentials will travel in clear inside such encrypted channel using simple PAP. This double mechanism requires the 802.1X Supplicant client to configure not only the desired *user/password credentials* BUT also to provide a valid VDS provided CA certificate: such certificate will be effectively used by the client to authenticate the server (VDS switch) and to bring up the outer TLS tunnel.

In the following you can see how such a *configuration* is made, and how a supplicant is successfully registered using the '**local-db**':

```
A@SW-Std.03-00-CA>debug dot1x
A@SW-Std.03-00-CA>configure interface e1-7
A@SW-Std.03-00-CA[config-if e1-7]>dot1x port-control auto
dot1x |EVENT [e1-7] -- port-control auto
A@SW-Std.03-00-CA[config-if e1-7]>dot1x aaa-domain local-db
dot1x |EVENT [e1-7] -- link-status down
dot1x |EVENT [e1-7] -- link-status up
dot1x |[e1-7] -- Registered supplicant 1c7508e6ae29 as sentinel.
dot1x |EVENT [e1-7] -- AUTH-COMPLETE 1c7508e6ae29 local-db
A@SW-Std.03-00-CA[config-if e1-7]>do show dot1x supplicants
```

Ifname	dot1XState	HostMode
e1-7	authorized_single	single
	Supplicant	Vlan
	1c7508e6ae29	* 1
	LastAuth	Fri 1 Jan 2010 02:34:01 [UTC]
	ActiveAuthServ	local-db
e2-0	auto	single

It's anyway anytime possible to revert such a port to a standard authentication back-end server in the usual manner; in the hypothesis that an *aaa-domain* named **testradius** has been configured in advance:

```
A@SW-Std.03-00-CA>debug dot1x
A@SW-Std.03-00-CA[config-if e1-7]>dot1x aaa-domain testradius
dot1x |EVENT [e1-7] -- link-status down
dot1x |EVENT [e1-7] -- link-status up
dot1x |EVENT [e1-7] -- link-status up
dot1x |[e1-7] -- Registered supplicant 1c7508e6ae29 as sentinel.
dot1x |EVENT [e1-7] -- AUTH-COMPLETE 1c7508e6ae29 172.16.16.10
A@SW-Std.03-00-CA[config-if e1-7]>do show dot1x supplicants
```

Ifname	dot1XState	HostMode
e1-7	authorized_single	single
	Supplicant	Vlan
	1c7508e6ae29	* 1
	LastAuth	Fri 1 Jan 2010 02:38:53 [UTC]
	ActiveAuthServ	172.16.16.10
e2-0	auto	single

```
dot1x |EVENT [e1-7] -- AUTH-COMPLETE 1c7508e6ae29 172.16.16.10
```

Please **note** that even though anytime you change the chosen *aaa-domain* to **local-db** or viceversa, the 802.1X port state will be *toggled up and down* and re-authentication itself will be triggered, the exact behavior in terms of timings depends very much indeed, on the chosen 802.1X client Supplicant software: for this reason it is possible to

obtain far more predictable results when performing such a configuration change after having unplugged the related port, in order to be sure that the client re-authenticates in a small time as cable is plugged in again next time. This is not an issue anyway in a standard boot sequence, where a well defined configuration is in place way before the port itself is enabled.

A typical **802.1X Supplicant configuration** using EAP-TTLS/PAP as authentication is depicted in the following figure (from a Ubuntu Linux system with NetworkManager); note that the user, after having selected the proper external and internal methods (Tunneled TLS and PAP), provides not only the proper credentials (in this case the standard default) but also a VDS certificate used for the TLS tunnel establishment.

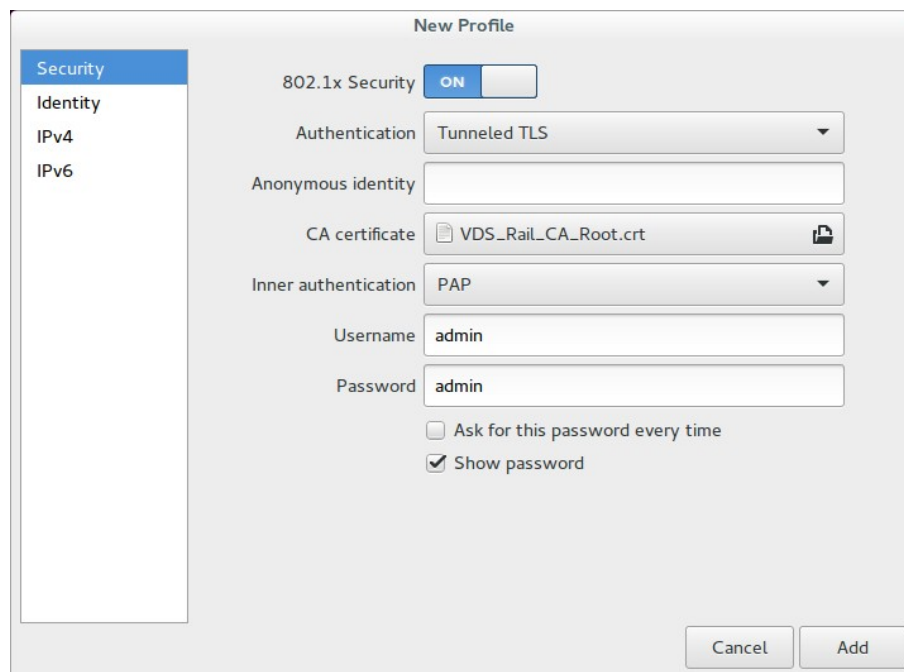


Figure 10.1: Example of 802.1X Supplicant configuration

10.4 Communication services

It is possible to manage the Consist Switch in different ways.

While the serial console is always available, other services that permit such communications may be enabled or not.

In detail, the communication services are:

- **SSH**: a secure protocol, usually enabled
- **telnet**: a less secure way to interact with the device, usually disabled
- **HTTP-based REST-like web services**: a communication paradigm best suited to be used from the applications, based on the HTTP protocol.

- **HTTPS-based REST-like web services:** same as the above, based on HTTPS instead, so much stronger on the security aspects.
- **L2 console:** The only network mechanism providing a connectivity with the Consist Switch without an IP address, using uniquely the device MAC address.

Although some of the listed services are enabled by default at boot time, they may be switched on or off with some specific configuration commands:

```
A@SW-Std.03-00-CA>configure service ssh-server enable
```

NOTE

These commands require an *admin* role to be executed.

10.5 Login banner

A customized banner can be displayed to the user before any interactive login. This feature can be enabled and the message customized by means of the command:

```
configure system login banner "<customized message>"
```

Double quotes are mandatory. New lines can be inserted by typing `\\n`, which will appear in the command line as a `\n`.

This is an example

```
A@SW-Std.03-00-CA>configure system login banner "UNAUTHORIZED ACCESS TO THIS
DEVICE IS PROHIBITED\n"
A@SW-Std.03-00-CA>exit
# ssh admin@10.1.2.3
UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED
Password:
** Konuendo Platform Configuration Shell **
**      Railway Switch Version      **
**      Press '?' for Help          **
A@SW-Std.03-00-CA>
```

Chapter 11 - DHCP services

The Dynamic Host Configuration Protocol (DHCP) is a protocol defined in RFC 2131 to pass configuration information to hosts on an IP network.

This chapter describes the DHCP services available on the Consist Switch: per-port DHCP server, DHCP snooping, DHCP option 82 and DHCP trusted interfaces.

11.1 Per-port DHCP server

The Consist Switch is equipped with an integrated DHCP server which eases the setup of the network environment by handling the automatic assignment of an IP address to DHCP clients connected to a specific port of the Consist Switch.

The IP addresses offered to clients are configured on a per-port basis and do not require any specific information about the clients itself, such as, for example, its MAC address, unless multiple clients should be connected to the same port. This way the client device can be replaced without changing the configuration of the Consist Switch.

The configuration parameters offered to the connected devices are distinguished in two types:

- *port-specific* parameters, valid for a given port only; they can be defined in the *specific configuration* level of the port;
- *shared* parameters, valid for many ports and collected in named groups called *DHCP options groups*; they can be defined in the *global configuration* level.

At port level, the DHCP server is automatically enabled if at least one of its parameter is set in the configuration.

At global level, for the DHCP requests being handled by the server, the latter must be enabled using the following command from *global configuration* level:

```
[ no ] configure ip dhcp-server enable
```

The *no-form* of the command will disable the service globally, and thus also on each port.

11.1.1 Single-client configuration

If a single end-device will be connected to the port, a unique IP address is required.

To offer a specific address to whichever client performs a request on a given port, use the command:

```
[ no ] configure interface NAME dhcp-server client-address ADDRESS [ NETMASK ]
```

The *NAME* parameter specifies the port that the server is being configured on. The

NETMASK parameter is optional: if not present, the natural mask of the IP class the address belongs to is used. The *no-form* of the command will remove such configuration.

In the following example we set the IP address and the network mask that will be offered to a client connected on *e0-1* port:

```
A@SW-Std.03-00-CA>configure interface e0-1 dhcp-server client-address
                               10.14.1.1 255.255.255.0
```

11.1.2 Multi-client configuration

For more complex requirements, where the Consist Switch should be able to support more than one single device per port, two different types of configurations are available: *pool*-based and *selector*-based.

11.1.2.1 Pool-based

In *pool*-based configurations, the addresses which will be offered to clients are defined as a range with the following command:

```
[ no ] configure interface NAME dhcp-server client-address-pool START END
```

Where *START* and *END* are initial and final IPv4 addresses of the range from which the server will choose the one to offer. The *END* address must be inferior than the *START* address, or the command will fail.

For example the following configuration:

```
A@SW-Std.03-00-CA>configure interface e0-1dhcp-server client-address-pool
                               10.1.1.1 10.1.1.10
```

will make the server offer clients the first available address in the range from 10.1.1.1 up to 10.1.1.10 (included).

11.1.2.2 Selector-based

For a finer control of address allocation, *selector*-based configurations can be defined to make the server offer different addresses depending on specific information found in the DHCP messages received.

There are four types of selectors available, and each one qualifies clients as follows:

- **hostname:** consider option 12 (Hostname) in the DHCP message;
- **client-id:** consider option 61 (Client-identifier) in the DHCP message;
- **vci:** consider option 60 (Vendor Class ID) found in the DHCP message;
- **vlan:** consider the 802.1q (VLAN) tag found in the packet carrying the DHCP message; this selector is meaningful only if the port which the server is active on is configured as a *VLAN trunk* port.

Each configuration line which includes one of the above selectors is also called a *selector-based rule*.

The general form of a command which defines a selector-based rule is the following:

```
[ no ] configure interface NAME dhcp-server client-address  
                                ADDRESS [ NETMASK ] SELECTOR VALUE
```

where *SELECTOR* is one of *hostname*, *client-id*, *vci* or *vlan*, and *VALUE* depends on the selector type as follows:

- *hostname*: a string specifying the expected value of option 12;
- *client-id*: the type of the option 61 value (*ascii* or *hex*) and a string specifying the expected value;
- *vci*: a string specifying the expected value of option 60;
- *vlan*: an integer between 1 and 4094 which specifying the VLAN ID.

It is possible to define any number of selector-based rules, but it is not possible to mix rules based on different selector types for each port.

If a DHCP discovery/request message does not match any of the defined rules, no answer is provided.

A rule based on *client-id* selector describes also how the selector's value should be considered when comparing it with the actual option's value:

- *ascii*: the selector's value must be specified as an ASCII string; the *type* field (first byte) in the option's value is assumed to be 0, and the remaining bytes are compared to the selector's value;
- *hex*: the selector's value must be specified as a sequence of bytes expressed in double-digit hexadecimal form; the option's value is then compared byte-by-byte to the selector's value, including the *type* field (first byte).

For more information on the meaning of option 61 value consult RFC 2132.

In the following example we configure the server to offer two different addresses to clients which announce themselves with *host-1* and *host-2* values as option 12 (*hostname* selector):

```
A@SW-Std.03-00-CA>configure interface e0-1 dhcp-server client-address  
                                10.16.1.1 255.255.255.0 hostname host-1  
A@SW-Std.03-00-CA>configure interface e0-1 dhcp-server client-address  
                                10.16.1.2 255.255.255.0 hostname host-2
```

Same as the above example, but using *vlan* selector:

```
A@SW-Std.03-00-CA>configure interface e0-1 dhcp-server client-address  
                                10.16.1.3 255.255.255.0 vlan 3  
A@SW-Std.03-00-CA>configure interface e0-1 dhcp-server client-address  
                                10.16.1.4 255.255.255.0 vlan 4
```

Same as the above example, but using *client-id* selector:

```
A@SW-Std.03-00-CA>configure interface e0-1 dhcp-server client-address  
                                10.16.1.5 255.255.255.0 client-id ascii host-1  
A@SW-Std.03-00-CA>configure interface e0-1 dhcp-server client-address  
                                10.16.1.6 255.255.255.0 client-id hex 0103040b01a088
```

11.1.2.3 *Fallback address*

There are cases where some of the devices logically connected to a given port and needing to obtain their IP address via DHCP are unknown or considered so by purpose.

For known devices it is possible to define which specific address will be offered them by means of one or more selector-based rules. For unknown devices it is however required that they somehow obtain an address, possibly undefined *a priori*.

To let unknown clients obtain a valid address, the per-port DHCP server provides the concept of *fallback address*.

A *fallback address* is an address offered when none of the defined selector-based rules match the request.

Fallback addresses can be obtained in two, mutually exclusive ways:

- picking the first available address from a defined *fallback pool*, or
- *forwarding* the DHCP message on the network and letting another server, if any, handle it.

To define the *fallback pool* for a given port use the command:

```
[ no ] configure interface NAME dhcp-server fallback address-pool START END
```

while:

```
[ no ] configure interface NAME dhcp-server fallback forward-mode
```

makes the server to forward to the network any unknown request coming from a given port.

With the above command, the message will be forwarded out from all the ports of the Consist Switch, except the one it came from.

11.1.3 Configurable options and fields

The following options and fields can be set in messages *sent* by the server of the Consist Switch through a specific port.

11.1.3.1 *Hostname (option 12)*

The configuration of option 12 (Hostname) is compatible only with single-client configurations or with multi-client configurations using *vlan* selector rules. In all other cases this option cannot be configured.

To configure option 12 use the command:

```
[ no ] configure interface NAME dhcp-server client-name HOSTNAME [ vlan VID ]
```

The *no-form* of the command will remove such an option.

Please note that the only supported selector for this option is *vlan*. If the port is configured to serve multiple clients, the address configuration must be based on the *vlan* selector only.

In the following example the host name *ip-cam1* will be included in any DHCP offer:

```
A@SW-Std.03-00-CA>configure interface e0-1 dhcp-server client-name ip-cam1
```

The following host names will be included in DHCP offers depending on the VLAN tag found in received packets:

```
A@SW-Std.03-00-CA>configure interface e0-1 dhcp-server client-name host-2 vlan 2
A@SW-Std.03-00-CA>configure interface e0-1 dhcp-server client-name host-3 vlan 3
```

11.1.3.2 Boot file name

It is possible to set the DHCP messages Boot file name field (file) value with the command:

```
[ no ] configure interface NAME dhcp-server client-boot-file
                                         FILENAME [ SELECTOR VALUE ]
```

where the optional *SELECTOR* parameter can be one of *client-id* or *hostname* selectors. The *client-id* selector supports only ASCII values.

The *no-form* of the command will set this field to “no value”.

Please note that the above command will not add the option 67 (Bootfile name) but instead set a field in the DHCP message.

11.1.3.3 Provisioning URL (option 66)

It is common practice in the industry to use DHCP option 66 (*TFTP server name*) to inform a client of a known network location from which it can fetch data for its purposes (a common example is application-specific configuration data). This is why option 66 in VDS Rail devices is called *Provisioning URL*.

To set the value of option 66 use the command:

```
[ no ] configure interface NAME dhcp-server client-provisioning URL
```

The *no-form* of the command will remove such an option.

11.1.4 Groups of options

A number of DHCP options can be shared among different ports by defining an *options group* and then associating such a group with the desired ports. The options defined in the group will then be included in all DHCP replies transmitted out from those ports.

Using groups it is possible to avoid configuring the same options for many ports where the DHCP service is enabled.

A group is defined by the following command:

```
[ no ] configure ip dhcp-server group NAME
```

The *no-form* of the command will remove the group definition.

For example:

```
A@SW-Std.03-00-CA>configure ip dhcp-server group ip-cameras
A@SW-Std.03-00-CA[config-dhcp-g ip_cameras]>
```

The above command creates a group named *ip-cameras* and then enters the *specific configuration* level for this group. Notice that the prompt has changed accordingly with the new configuration context.

Once you have defined the group by its name, you can specify the options it includes as in the following examples:

```
A@SW-Std.03-00-CA[config-dhcp-g ip-cameras]>default-router 192.168.1.254
A@SW-Std.03-00-CA[config-dhcp-g ip-cameras]>dns-server 192.168.1.10
A@SW-Std.03-00-CA[config-dhcp-g ip-cameras]>netmask 255.255.255.0
A@SW-Std.03-00-CA[config-dhcp-g ip-cameras]>time-server 192.168.1.10
```

When a port has been associated with a group and at the same time its DHCP server configuration includes the definition of a network mask, the latter definition takes precedence over the one from the group, if any.

Once the configuration of the group is terminated, the `exit` command will cause the shell to return to the *global configuration* level.

Once defined, the group can be associated with one or more ports with the following command:

```
[ no ] configure interface NAME dhcp-server client-group GROUP
```

The *no-form* of the command will remove the group association.

For example, from the *specific configuration* level for interface *e0-1*:

```
A@SW-Std.03-00-CA>configure interface e0-1 dhcp-server client-group ip-cameras
```

11.1.4.1 Subnet Mask (option 1)

To add option 1 in a group definition use the command:

```
[ no ] configure ip dhcp-server group NAME netmask ADDRESS
```

The *no-form* of the command will remove the option from the group.

11.1.4.2 Router (option 3)

To add option 3 in a group definition use the command:

```
[ no ] configure ip dhcp-server group NAME default-router ADDRESS
```

The *no-form* of the command will remove the option from the group.

11.1.4.3 Time server (option 4)

To add option 4 in a group definition use the command:

```
[ no ] configure ip dhcp-server group NAME time-server ADDRESS
```

At most one address can be specified. The *no-form* of the command will remove the option from the group.

11.1.4.4 Domain Name Server (option 6)

To add option 6 in a group definition use the command:

```
[ no ] configure ip dhcp-server group NAME dns-server ADDRESS
```

At most one address can be specified. The *no-form* of the command will remove the option from the group.

11.1.4.5 Lease time (option 51)

By default the lease period is equal to 1 hour.

Since there is no permanent storage of lease information, the latter will be lost when the Consist Switch reboots.

To add option 51 in a group definition use the command:

```
[ no ] configure ip dhcp-server group NAME lease-time MINUTES
```

Where *MINUTES* is a value between 1 and 1440 (1 day). The *no-form* of the command will remove the option from the group.

11.1.4.6 Server identifier (option 54)

As stated in RFC 2131:

«To accommodate potentially incomplete network connectivity, a server MUST choose an address as a 'server identifier' that, to the best of the server's knowledge, is reachable from the client.»

It is possible to specify a given IP address as option value within the definition of a group with the command:

```
[ no ] configure ip dhcp-server group NAME server-id ADDRESS
```

The *no-form* of the command will remove the option from the group.

If not explicitly set, the server will use the IP address of the management interface (*vlan1*), if defined, as the option's value in its messages.

Note that this is a "virtual" IP address because it does not need to be configured on any Consist Switch interface in order to be used. The server will respond to all messages targeted to this address.

11.1.4.7 Classless Static Route (option 121)

To add a static route, option 121, in a DHCP group definition use the command

```
[ no ] configure ip dhcp-server group NAME static-route NETWORK MASK ADDRESS
```

The *no-form* of the command will set such field to "no value".

For each group only a single static route definition is currently supported.

11.1.4.8 *Next server address field*

To configure the value of the *next server address* (siaddr) field in DHCP messages use the command:

```
[ no ] configure ip dhcp-server group NAME next-server ADDRESS
```

The *no-form* of the command will set such field to “no value”.

11.1.4.9 *Server host name field*

To configure the value of the *server host name* (sname) field in DHCP messages use the command:

```
[ no ] configure ip dhcp-server group NAME sname ( ADDRESS | HOSTNAME )
```

The *no-form* of the command will set such field to “no value”.

11.2 DHCP snooping

When enabled, DHCP snooping forces all DHCP messages entering the Consist Switch to be processed before being forwarded.

DHCP snooping performs three major tasks:

- it keeps a record of which IP addresses are currently allocated to hosts downstream of the device's ports;
- it allows the dynamic insertion/removal of DHCP option 82 (Relay Agent Information) into/from DHCP packets that flow through the device;
- it allows a fine control on the DHCP protocol handshaking by checking that ports that are receiving a DHCP response from a server are trusted by the system.

To globally enable or disable the DHCP snooping service use the following command when in *global configuration* level:

```
[ no ] configure ip dhcp-snooping enable
```

DHCP snooping is enabled by default on all Consist Switch devices.

To enable or disable the service on a specific port, use the following command:

```
[ no ] configure interface NAME ip dhcp-snooping enable
```

When globally enabled, the service is implicitly enabled on all device's ports.

NOTE

Disabling DHCP snooping will also disable insertion of option 82 (Relay Agent Information) in DHCP messages.

11.2.1 Dropping messages

By default DHCP messages are accepted on all Consist Switch ports. The DHCP snooping service can be configured to accept or drop DHCP messages on a given port, to get finer control over the protocol activity.

11.2.1.1 *Trusted ports*

The *trust configuration* of a port specifies if DHCP *replies* transmitted from a server to a client can enter that port.

To change the trust configuration of a port use the command:

```
[ no ] configure interface NAME ip dhcp-snooping trusted-port
```

When none of the ports of the Consist Switch has an explicit trust configuration, all ports are considered as trusted, and thus all DHCP replies are accepted on all ports.

When at least one port has an explicit trust configuration, all other ports are implicitly considered untrusted, unless configured otherwise.

NOTE

Once DHCP trusting has been enabled, all the ports toward the *authorized* DHCP server must be configured as trusted.

11.2.1.2 *Always Drop*

To unconditionally drop any DHCP message entering from a specific port use the command:

```
[ no ] configure interface NAME ip dhcp-snooping drop-packets
```

The *no-form* of the command will make the port accept any DHCP message again.

11.3 Relay Agent Information (option 82)

DHCP option 82 (Relay Agent Information) is an extension of the DHCP protocol and is defined in RFC 3046.

Option 82 can be used to include information about clients which can be considered by a server for choosing an address to offer. An example of such information is the port number and the MAC address of the Consist Switch from where a request is coming.

Option 82 must be inserted in a DHCP message by a switch at the edge of the network, because only edge switches can uniquely identify the device(s) from which the message originated.

Although option 82 is named *DHCP Relay Agent Information Option*, the device that inserts the option into the message does not need to act as a DHCP relay.

Thanks to its layer 2 switching capabilities the Consist Switch can insert option 82 information, provided that the DHCP snooping service is enabled.

To enable or disable option 82 insertion use the command:

```
[ no ] configure ip dhcp-option82 enable
```

Insertion is enabled by default on all Consist Switch devices. Both *circuit-id* and *remote-id* sub-options of option 82 are also enabled by default as follows:

- *circuit-id* value is the name of the Consist Switch interface from where the DHCP request is entering;
- *remote-id* value is the MAC address of the Consist Switch.

To configure which information has to be used as *circuit-id* value use the command:

```
[ no ] configure ip dhcp-option82 circuit-id ( if-name | if-num )
```

To configure which information has to be used as *remote-id* value use the command:

```
[ no ] configure ip dhcp-option82 remote-id ( hostname | mac-addr | ip-addr )
```

In both cases, using the *no form* of the command will disable the insertion of the given sub-option.

Insertion of option 82 can also be enabled or disabled on a per-port basis with the command:

```
[ no ] configure interface NAME ip dhcp-option82 enable
```

When globally enabled, it is implicitly enabled on all device's ports.

11.4 Services state information

It is possible to obtain information about the current state of DHCP services with the command:

```
show ip dhcp services
```

An example output is the following:

```
dhcp server service is disabled
dhcp snooping service is enabled
dhcp trusting is disabled
dhcp option82 insertion is enabled

dhcp option82 circuit-id suboption is enabled
    circuit id is incoming interface name (if-name)
dhcp option82 remote-id suboption is enabled
    remote id is switch mac address
```

Current address bindings known by the Consist Switch can be obtained with the command:

```
show ip dhcp bindings
```

An example output is the following:

MacAddress	IpAddress	Lease(sec)	Type	VLAN	Port
00:1e:68:9d:5a:46	10.2.2.109	18	dhcp snooping	2	e1-2
00:1e:68:9d:5a:53	10.2.2.120	18	dhcp snooping	2	e1-3
00:1e:54:9a:5a:74	10.3.2.10	240	port server	3	e1-6
00:1e:65:51:4e:18	10.3.1.11	240	port server	3	e1-7

The *Type* column reports whether the lease has been discovered by the DHCP snooping service or if it has been provided by the per-port DHCP server.

To clear all active bindings (leases), globally or for a specific interface, use the command:

```
clear dhcp bindings [ interface NAME ]
```

Chapter 12 - Layer 3 services

Although the Consist Switch is mainly a layer 2 ethernet device, starting from firmware release 5.2.0 some layer 3 functionalities have been added to the system. This chapter describes these new capabilities and their configuration.

In a switched network VLANs are used to split the physical network into different collision domains and layer 3 sub-nets. Devices within a VLAN can communicate with one another without requiring routing. On the other hand devices in different VLANs require a routing device to communicate with one another.

In networks with L2-only switches routing is performed by separate L3 devices but, with the introduction of these new L3 capabilities, a Consist Switch incorporates these routing abilities inside the switch itself.

The functionalities of the L3 management interface (VLAN1) have been extended introducing the possibility to dynamically create multiple L3 interfaces servicing the relevant VLANs. These interfaces are named "Software Vlan Interfaces (SVI)".

12.1 SVI interfaces

A SVI interface is a virtual L3 interface that provides Layer 3 processing for packets to or from all switch ports associated with a specific VLAN.

These interfaces are named *vlanX* where **X** is the **ID** of the **associated VLAN**. This means that the standard *vlan1* interface, is now a SVI interface servicing all the ports that are member of VLAN 1 (initially all the ports).

SVI interfaces are automatically created while entering their interface configuration mode, using the following command:

```
configure interface vlan[1÷4094]
```

The interface cannot be created unless the associated VLAN itself is created beforehand.

There is a one-to-one mapping between VLAN and SVI number, thus only a single SVI can be mapped to a specific VLAN.

Once created, the IPv4 address assigned to the interface can be added or removed using the interface command:

```
[no] configure interface vlanX ip address <address> [netmask]
```

The interface is initially disabled; it can be enabled using the command:

```
configure interface vlanX enable
```

An existing SVI interface can be destroyed using the no form of the configuration command like in the following example:

```
no configure interface vlanX
```

The interface is destroyed also if enabled and configured. In this case all the associated configuration information will be lost.

By default the line protocol status of a SVI interface remains always up no matter if there isn't any active physical port associated with the specific VLAN. This behavior can be changed by issuing the command:

```
configure svi-interfaces status dynamic
```

After this the link status of any SVI interface will reflect the availability of at list one physical port, either access or trunk port, associated with the specific VLAN.

12.2 Enable IPv4 routing

The Consist Switch implements both unicast and multicast IPv4 routing.

While unicast routing can be **static** or it can be dynamically handled through the standard **RIP** and **OSPF** routing protocols, multicast routing can only be handled statically.

By default routing is disabled; it can be enabled or disabled using the global command:

```
[no] configure ip routing enable
```

12.2.1 Configure static unicast routes

Static routes can be added or removed using the following configuration command:

```
[no] ip route <destination-ip> <netmask> <gateway-ip> | <if-name>
```

where:

- `destination-ip` and `netmask` identify the destination network
- `gateway-ip` is the next-hop IP address toward the destination network
- `if-name` is the output interface and it can be used with or without the gateway IP address.

For example the following command:

```
A@VDS-TSW.05-20-CA>configure ip route 192.168.10.0 255.255.255.0 172.16.10.1
```

defines that the next node (hop) on the path to the 192.168.1.0/24 network is at the address 172.16.10.1.

A “*default route*” defines the next hop address for all the destinations that are not contained in the active route table. It can be configured by issuing the following command:

```
A@VDS-TSW.05-20-CA>configure ip route 0.0.0.0 0.0.0.0 192.168.10.1
```

The current routing table can be viewed using the `show ip route` command as in the following example:

```
A@VDS-TSW.05-20-CA>show ip route
Codes: C - connected, S - static, D - dynamic

D> 192.168.20.0/24 via 192.168.1.1 dev vlan1 metric 20
C> 192.168.1.0/24 is directly connected to vlan1
C> 172.16.10.0/24 is directly connected to vlan2
D> 192.168.10.0/24 via 192.168.1.1 dev vlan1 metric 20
D> 192.168.40.0/24 via 192.168.1.1 dev vlan1 metric 20
D> default via 192.168.1.254 dev vlan1 metric 1
```

12.2.2 Configuring RIP protocol

RIP is a relatively old but still commonly used routing protocol. It is defined by the RFC1058.

RIP uses broadcast UDP data packets to exchange routing information. By default routing information updates are sent every 30 seconds. If a router does not receive an update from another router for 180 seconds or more, it removes from its routing table all the routes announced by the “dead” router.

Both Version 1 and Version 2 are supported with version 2 supporting variable-length subnet masks (VLSMs).

By default version 2 is enabled.

Use the following commands to enable RIP:

```
A@VDS-TSW.05-20-CA>configure router rip
```

This will enable a RIP routing process and moves you in router configuration mode.

```
A@VDS-TSW.05-20-CA>configure router rip network ip-address [netmask]
```

this selectively enables RIP on all the interfaces whose IP address falls within the defined network. Multiple network statements can be defined for the different networks configured on the various interfaces.

Note that this statement does not define the networks announced by the RIP protocol but it only identifies the interfaces where the protocol is enabled.

In case where multicast traffic is not forwarded by the underlying LAN, neighbor RIP routers can be statically configured using the command:

```
A@VDS-TSW.05-20-CA>configure router rip neighbor ip-address
```

In this case periodical updates will be transmitted directly (as unicast) to the configured neighbor.

12.2.3 Configuring OSPF protocol

OSPF version 2, defined by the RFC 1253, is a routing protocol designed specifically for IP networks. Differently from RIP, periodically broadcasting the entire routing table, OSPF transmits only the “differences” (deltas?) in case of any change.

OSPF is a complex routing protocol well suited for complex IP networks topologies. Since the network topologies commonly found on modern trains does not have such a high level of complexity, only a subset of the OSPF features has been exposed.

With the aim of reducing the size of the routing table, IPv4 networks can be organized into separate entities named “*areas*” identified by a number. Among all the areas, *area 0* plays a special role. This area is called “*backbone area*” and it must always be present; all the other areas must be somehow directly connected with area 0.

Given that on board IP networks are quite simple, most of the times only the backbone area is used.

OSPF process is enabled using the following command:

```
A@VDS-TSW.05-20-CA>configure router ospf
```

Like for RIP protocol, OSPF routing can be enabled on specific interface basis using the command:

```
A@VDS-TSW.05-20-CA>configure router ospf network ip-address netmask area-id
```

12.3 Configuring multicast routing

The Consist Switch supports also IPv4 multicast routing. Multicast forwarding rules must be statically defined on each multicast router since dynamic multicast routing protocols like PIM or DVRMP are currently unsupported.

Multicast routes can be added using the following configuration command:

```
configure ip mroute <inport> <src-ip[/n]> <mc-grp[/m]> <outport[,outport[...]]>
```

where:

- `inport` defines the ingress interface of the multicast stream to be forwarded;
- `src-ip[/n]` is the source IP of the stream; 0.0.0.0 can be used for matching any address or an optional netmask can be specified for matching multiple sources;
- `mc-grp[/m]` is the IPv4 multicast group address of the stream to be forwarded; an optional netmask can be specified for matching multiple groups;
- `outport(s)` is a comma-separated list of ports where the stream should be forwarded.

For example the following command:

```
A@VDS-TSW.05-20-CA>configure ip mroute vlan1 0.0.0.0 239.10.10.1 vlan2,vlan3
```

defines that the multicast stream to group 239.10.10.1 entering from interface *vlan1* and originated from any source should be forwarded to interfaces *vlan2* and *vlan3*.

The active multicast routing table can be retrieved using the `show ip mroutes [active]` command as in the following example:

```
A@VDS-TSW.05-20-CA>show ip mroutes
```

Active multicast routes				
InIf	Src IP	Group	Pkts	OutIfs
vlan1	0.0.0.0/0	239.194.1.0/24	25623	vlan3
vlan2	0.0.0.0/0	239.192.0.2/32	564	vlan1,vlan3
vlan3	0.0.0.0/0	239.193.0.0/32	657623	vlan1,vlan2

Differently from source specific routes, which are always present, on the multicast routing table the entries for any-source routes are dynamically created as some source begins transmitting. The complete list of the configured multicast routes can be shown using the `show ip mroutes configured` command as in the following example:

```
A@VDS-TSW.05-20-CA>show ip mroutes configured
```

Configured multicast routes				
InIf	Src IP	Group	Tags	OutIfs
vlan1	0.0.0.0/0	239.192.0.1/32	CFG.	vlan2,vlan3
vlan1	0.0.0.0/0	239.194.1.0/24	CFG.	vlan3
vlan2	0.0.0.0/0	239.192.0.2/32	CFG.	vlan1,vlan3
vlan2	0.0.0.0/0	239.194.2.0/24	CFG.	vlan3
vlan3	0.0.0.0/0	239.193.0.0/32	CFG.	vlan1,vlan2

The **Tags** column has a special meaning on other products; in the Consist Switch it will always contain the value `CFG.`, that means that the route has been set in the device configuration.

12.3.1 Multicast routing and IGMP snooping

If IGMP snooping is enabled (see Multicast on page 133), a Consist Switch configured for multicast routing must be also configured with the following command:

```
ip igmp snooping flood-to-cpu
```

This command enables the flooding on known (joined) groups to CPU in order to be eventually routed if a route exist for the specific group.

If not present routing for a specific multicast group might stop working as soon as an End Device joins the same group on any port.

12.4 Router redundancy

The routing service might become a “mission critical” service as soon as the consist network is splitted into separate VLANs and devices connected to different VLANs needs to exchange data continuously. In such case, a redundant configuration with two “routers” configured for working together as a master/backup pair can be deployed, avoiding a single point of failure in network communications.

In this kind of configurations the standard **Virtual Router Redundancy Protocol (VRRP)** is used to achieve redundancy for both unicast and multicast traffics.

Both **VRRP version 2** and **VRRP version 3** are supported; for version 3 only the IPv4 protocol is currently supported.

VRRP enables routers to share a Virtual IP address referred as the gateway by all the hosts on the protected network. The router with the highest priority acts as master for this Virtual IP address while the other one is the backup in case of failure of the master. In order to detect master failures, a keep-alive packet between the two routers is exchanged at a configurable frequency. If the backup router does not hear any advertisement from the master for about three times the advertise period, it becomes the active master and takes ownership of the virtual IP address.

One of the bigger advantages of version 3 over version 2 is the possibility to support keep-alive periods of tens of milliseconds instead of seconds allowing a quicker fault detection.

VRRP can be configured and enabled from the interface configuration menu level.

VRRP activation mandatory parameters are the virtual router identifier (VRID) and the virtual IP address (VIP). All the other ones are optional and, if not defined, they will be used with their default values.

VRRP can be quickly enabled on an interface by issuing the following commands:

```
A@VDS-TSW.05-20-CA>configure interface vlan2 vrrp vrid 1
A@VDS-TSW.05-20-CA>configure interface vlan2 vrrp ip 192.168.100.1
```

You can get the current VRRP subsystem status with:

```
A@VDS-TSW.05-20-CA>show vrrp
Interface vlan2 - state MASTER - instance id undefined
Port status is Enabled/Up
Running VRRP version 2
Virtual router ID 1
Virtual IP address is 192.168.1.100
Priority is 100
Advertisement interval is 1.000 sec
Master router is 192.168.1.2 (local), priority is 100
Master Advertisement interval is 1.000 sec
Master Down interval is 3.609 sec
```

The output, in this example, shows that this device is the active VRRP master and so it owns the virtual IP address 192.168.100.1.

Besides *vrid* and *vip*, there are some other options that can be configured. The following table lists the configurable options with their default values.

Option	Command	Range	Default	Note
VRRP version				
	<code>vrrp version <version></code>	2-3	2	
Router priority				
	<code>vrrp priority <value></code>	1-254	100	The device with higher priority will become the master router.
Time interval between VRRP advertisements transmitted by the active master				
	<code>vrrp advertise <value></code>	1-255	1	The current master is considered dead and a
	<code>vrrp advertise msec <value></code>	20-999	N/A	

new election take place if 3 consecutive advertisement messages are missed.

VRRP can be enabled on more than one routed interfaces. By default the VRRP protocol operates independently on each interface on which it has been configured and enabled. This means that the status of VRRP can be master for one interface and backup for another interface on a device at the same time.

Depending on the network topology, there are situations where you might need to group together different interfaces driven by VRRP so that they share a common VRRP status. For example, this could be the case of a pair of redundant routers connecting other consist networks through a backbone line.

You can group together different instances by associating each interface to a common *VRRP instance*; this is accomplished by setting the *VRRP instance identifier* option on each interface with the command:

```
configure interface <if-name> vrrp instance-id <1-4>
```

When one of the interfaces transitions to the backup state, all the other interfaces associated to the same *VRRP instance* also change their status to backup.

12.5 NAT

NAT (Network Address Translation) is a mechanism used to modify IP address information as packets move across a routing device with the purpose of mapping one IP address space to another. The source IP address, the destination IP address or both may be translated.

If there is not a one-to-one relationship between original and translated address, a stateful translation table is dynamically built in order to allow responses back to the original IP address.

Three different types of NAT are supported:

- source NAT (snat)
- destination NAT (dnat)
- railway NAT (r-nat)

Source NAT

The source IP address is replaced. It is commonly used when the receiver system cannot reply to the real IP address of the sender mainly because of some routing constraints.

Destination NAT

The destination IP address is replaced. It is commonly used to publish a service on a different IP network. In case of a TCP connection the translation table is dynamically updated in order to allow replies in the opposite direction.

Railway NAT

A whole IP network is translated to a different IP network. Both traffic directions are automatically handled for source/destination address replacement. Since the host portion of the address is unchanged, source and destination networks must have the same size.

12.5.1 Configuring NAT

NAT configuration is a two steps activity.

First of all you must define an “ip nat rule” describing the transformation to apply (type of NAT, original and translated addresses, input and output ports and so on). Then you must enable the rule.

A new IP NAT rule can be defined using the following configuration commands:

```
configure ip nat <rule-name>
configure ip nat <rule-name> description <"text">
configure ip nat <rule-name> type <dnat|snat|r-nat>
configure ip nat <rule-name> match <source-ip> <netmask>
configure ip nat <rule-name> in <if-name>
configure ip nat <rule-name> out <if-name>
configure ip nat <rule-name> translate-to <ip>
```

Where:

- **type** is the type of transformation to apply
- **match** is the original IP to change (this is a source or destination depending on NAT type)
- **in** is the incoming port
- **out** is the outgoing port
- **translate-to** is the new value to assign to the matched addresses

Once defined, the NAT rule can be enabled or disabled using the command:

```
[no] configure ip nat <rule-name> enable
```

where *name* is the rule label assigned in the previous step.

In the following example a VCU connected to the consist network is “published” over the backbone using an IP address routed over the backbone.

```
A@VDS-TSW>configure ip nat vcu-backbone
A@VDS-TSW[config-nat vcu-backbone]>description "Publish VCU over the back-bone"
A@VDS-TSW[config-nat vcu-backbone]>type dnat
A@VDS-TSW[config-nat vcu-backbone]>match 192.168.20.10 255.255.255.255
A@VDS-TSW[config-nat vcu-backbone]>in vlan1
A@VDS-TSW[config-nat vcu-backbone]>out vlan2
A@VDS-TSW[config-nat vcu-backbone]>translate-to 10.2.0.10
A@VDS-TSW[config-nat vcu-backbone]>exit
A@VDS-TSW[configure]>ip nat vcu-backbone enable
```

Depending on the VCU application type, translation might be also required for the client source address.

The following rule enables this requirement:

```
A@VDS-TSW>configure ip nat vcu-client
A@VDS-TSW[config-nat vcu-client]>description "vcu client priv. address transl."
A@VDS-TSW[config-nat vcu-client]>type snat
A@VDS-TSW[config-nat vcu-client]>match 10.3.2.0 255.255.255.0
A@VDS-TSW[config-nat vcu-client]>in vlan2
A@VDS-TSW[config-nat vcu-client]>out vlan1
A@VDS-TSW[config-nat vcu-client]>translate-to 192.168.20.1
A@VDS-TSW[config-nat vcu-client]>exit
A@VDS-TSW[configure]>ip nat vcu-client enable
```

Note that the destination address should be changed for the Consist Switch connecting the VCU to the backbone while source IP address should be changed for the Consist Switch connecting client network to the backbone.

Since this double NAT is commonly required on modern on-board topologies where a two level architecture, consist/backbone, is used, the *r-nat* type can become really useful.

In this case the entire consist network is “published” over the backbone and both source and destination mappings are automatically implemented.

In the following example a consist network with IP address 10.0.1.0/24 is published on the backbone using the train wide address 192.168.10.0/24.

```
A@VDS-TSW.05-20-CA>configure ip nat consist_10
A@VDS-TSW.05-20-CA[config-nat consist_10]>type r-nat
A@VDS-TSW.05-20-CA[config-nat consist_10]>match 10.0.1.0 255.255.255.0
A@VDS-TSW.05-20-CA[config-nat consist_10]>in vlan2
A@VDS-TSW.05-20-CA[config-nat consist_10]>out vlan1
A@VDS-TSW.05-20-CA[config-nat consist_10]>translate-to 192.168.10.0
```

This NAT type leaves the host number unchanged while the network part is changed accordingly to the rule.

So a packet exiting from the consist LAN and routed over the train backbone will have its source address changed from network 10.0.1.0/24 to network 192.168.10.0/24 while, at the same time, packets coming from the backbone with a destination address on the 192.168.10.0/24 network are translated to 10.0.1.0/24 and routed to the consist LAN.

12.5.2 Showing NAT status

The current NAT status is analyzed by the `show ip nat` command, as in the following example:

```
A@VDS-TSW.05-20-CA>show ip nat

NAT active rules: 1

Name          Type Match          Ingr. if  Egr. if  Translate-to  Packets  Bytes
-----
TEST RULE     r-nat 172.26.101.0/24  vlan5     vlan6     192.168.101.0  0        0
```

12.6 Traffic Filtering

By default Consist Switch does not enforce any policy in order to limit or block traffic routed through different interfaces. The **Traffic Filter** gives the user the possibility to define which IPv4 traffic is allowed to:

- flow between two different VLANs
- reach the device through a virtual interface

The **Traffic Filter** capability relies on the basic concepts of **filter** and **rule**.

Filter

A sorted list of rules, identified by a unique name, that the device can store and enable. At most one filter can be active at any time on a device.

Rule

The definition of a policy for a certain traffic type, based on a combination of the following items:

- Source IP address
- Destination IP address
- Protocol (any/UDP/TCP/ICMP)
- Source port number
- Destination port number
- Ingress L3 interface
- Egress L3 interface
- Policy (allow/deny)

The rules of a filter are sorted according to a sequence number.

The rule items describe how the packet header should look like to match the rule and what to do with the packet. If the packet matches the rule, the rule policy is applied and the packet is not checked against any other rule. If the rule does not match, then the next one in the filter is consulted. Finally, if there are no more rules to consult, then the default policy is applied. Unless otherwise specified (by an "allow all" last rule), this policy is to deny all the traffic.

The order in which the rules are evaluated matters.

Traffic and rules can be grouped into two families: **forward traffic/rules** and **local traffic/rules**.

Forward traffic

Packets that traverse the device and whose final destination is another device

Forward rules

Rules that apply to forward traffic

Local traffic

Packets that reach the device and whose final destination is the device itself

Local rules

Rules that apply to local traffic

In case the desired default policy is “allow all”, a specific “allow all” rule is required for each family at the bottom of the filter (see examples in the following sections).

12.6.1 Configuring Traffic Filter

Traffic Filter configuration is a three steps activity:

1. filter creation
2. rules definition
3. filter enabling/disabling

First of all an “IP filter” must be created as a container for the rules.

The filter is initially empty and needs a name to be referenced to in the future. The default policy of an empty filter is “accept all”.

Filter creation is performed by the following configuration command:

```
configure ip filter <name>
```

Where **name** is the name of the filter (with a max length of 20 characters).

This command causes the creation of the filter (if not already existent); it also changes the menu context to the `configure ip filter` level, as shown in the following example:

```
A@VDS-TSW.05-20-CA>configure ip filter FILTER
A@VDS-TSW.05-20-CA[config-filter FILTER]>
```

To quit this menu context just type `exit`. The filter remains defined even if empty.

To destroy a filter, use the “no form”:

```
no configure ip filter <name>
```

The filter, if existent, will be disabled (if enabled) and deleted as well as all the contained rules.

The second step is the definition of the rules. The following rule forms are accepted:

```
<seq-no> accept|deny ip|icmp-ping <src> <in-if> <dst> <out-if>
<seq-no> accept|deny tcp|udp <src> <sport> <in-if> <dst> <dport> <out-if>
<seq-no> accept|deny ip|icmp-ping <src> <in-if> local
<seq-no> accept|deny tcp|udp <src> <sport> <in-if> local <dport>
```

Where:

- **seq-no** is the rule sequence number; if a sequence number is reused the corresponding rule is replaced
- **accept** is the policy allowing a matching packet to be forwarded or delivered

to the device

- **deny** is the policy blocking (drop) a matching packet
- **ip** means that no matching is performed on the packet protocol (**icmp-ping**, **tcp** and **udp**, instead, are protocol specific)
- **icmp-ping** means that the rule will match ICMP (Internet Control Message Protocol) “echo request” and “echo reply” packets
- **tcp** means that the rule will match TCP (Transmission Control Protocol) packets
- **udp** means that the rule will match UDP (User Datagram Protocol) packets
- **src** is the source IPv4 device or network address; the special keyword **any** is accepted here to match packets regardless of the source IP address; in order to enter a network address the `x.y.z.t/n` notation must be used
- **sport** is the source TCP/UDP port; the special keyword **any** is accepted here to match packets regardless of the source port
- **in-if** is the ingress interface; the special keyword **any** is accepted here to match packets regardless of the ingress interface
- **dst** is the destination IPv4 device or network address; the special keyword **any** is accepted here to match packets regardless of the destination IP address; in order to enter a network address the `x.y.z.t/n` notation must be used
- **dport** is the destination TCP/UDP port; the special keyword **any** is accepted here to match packets regardless of the destination port
- **out-if** is the egress interface; the special keyword **any** is accepted here to match packets regardless of the egress interface
- **local** means that the packet is a local packet (final destination is the device itself); usually such packets are the “management traffic”

To delete a rule it is sufficient to use the sequence number together with the “no form”:

```
no configure ip filter <name> <seq-no>
```

Finally, a filter can be enabled or disabled issuing the following command:

```
[no] configure ip filter <name> enable
```

where **name** is the name of the filter defined at the first step.

NOTES

Enabling a filter causes the automatic disabling of any previously enabled filter.

Disabling the “enabled filter”, causes the immediate restoring of the “allow all” policy for both forward and local traffic.

Disabling a “not enabled” filter has no consequences at all.

Empty filters implement the “accept all” default policy for both forward and local traffic families. The policy of a traffic family is turned to “deny all” as soon as a rule is added to the filter for that family. For example, if the rule `10 accept ip 192.168.1.1 any local` is added to an empty filter, then the policy for the forward traffic still remains “accept all” while the policy for the local traffic turns to “deny all” (of course except for the traffic matched by the added rule).

12.6.2 Showing Traffic Filter status

The current Traffic Filter status is analyzed by the `show ip filter` command, as in the following example:

```
A@VDS-TSW.05-20-CA>show ip filter

Filter: enabled

Filter name: TEST_3

FORWARD: established 0 packets (0 bytes) - accept 0 packets (0 bytes)
Seq-Num Policy Protocol Source Address Sport Ingr. if Dest. Address Dport Egr. if Packets Bytes
-----
11 deny icmp-ping any e2-0.3 any any any 0 0
91 accept ip any any any any any 0 0

LOCAL: established 31 packets (2378 bytes) - accept 0 packets (0 bytes)
Seq-Num Policy Protocol Source Address Sport Ingr. if Dport Packets Bytes
-----
10 deny icmp-ping any e2-0.3 0 0
90 accept ip any any 5 804
```

12.6.3 Examples

Sequence numbers assignment is arbitrary and affects only the rule sorting.

Deny all ip traffic between VLAN1 and VLAN2

```
A@VDS-TSW.05-20-CA>configure ip filter EXAMP_1
A@VDS-TSW.05-20-CA[config-filter EXAMP_1]>10 deny ip any vlan1 any vlan2
A@VDS-TSW.05-20-CA[config-filter EXAMP_1]>15 deny ip any vlan2 any vlan1
A@VDS-TSW.05-20-CA[config-filter EXAMP_1]>90 accept ip any any any any
A@VDS-TSW.05-20-CA[config-filter EXAMP_1]>95 accept ip any any local
A@VDS-TSW.05-20-CA[config-filter EXAMP_1]>enable
A@VDS-TSW.05-20-CA[config-filter EXAMP_1]>exit
```

Rules 10 and 15 matches the traffic to deny, following the **in-if** and **out-if** items (notice the word **any** used for **src** and **dst**). Rules 90 and 95 match and allow all the remaining traffic not blocked by 10 and 15.

Allow telnet and ssh from host 1.2.3.4 only

```
A@VDS-TSW.05-20-CA>configure ip filter EXAMP_2
A@VDS-TSW.05-20-CA[config-filter EXAMP_2]>10 accept tcp 1.2.3.4 any any local 22
A@VDS-TSW.05-20-CA[config-filter EXAMP_2]>15 accept tcp 1.2.3.4 any any local 23
A@VDS-TSW.05-20-CA[config-filter EXAMP_2]>99 accept ip any any any any
A@VDS-TSW.05-20-CA[config-filter EXAMP_2]>enable
A@VDS-TSW.05-20-CA[config-filter EXAMP_2]>exit
```

This filter is intended to block “incoming” connections that use protocols other than telnet or ssh; for this reason affects only local traffic.

Telnet traffic is destined to TCP port 23 (rule 15) while ssh is to TCP port 22 (rule 10). All the other local traffic is blocked by the default policy.

Rule 99 allows all the forward traffic.

Allow SNMP traffic to 1.2.3.4 from vlan2 to vlan1

```
A@VDS-TSW.05-20-CA>configure ip filter EXAMP_3
A@VDS-TSW.05-20-CA[config-filter EXAMP_3]>10 accept udp any any vlan2
1.2.3.4 161 vlan1
A@VDS-TSW.05-20-CA[config-filter EXAMP_3]>15 accept udp any any vlan2
1.2.3.4 162 vlan1
A@VDS-TSW.05-20-CA[config-filter EXAMP_3]>enable
A@VDS-TSW.05-20-CA[config-filter EXAMP_3]>exit
```

SNMP traffic is made of UDP packets addressed to ports 161 and 162. Incoming traffic on **vlan2** interface, matching the destination address, the egress interface (**vlan1**) and the **dport** is allowed to flow. All the remaining traffic (not matching the rules) is blocked.

Allow UDP to 1.2.3.4 ports 161 from any interface to vlan1

Allow UDP to 1.2.3.4 ports 162 from any interface to vlan1

```
A@VDS-TSW.05-20-CA>configure ip filter EXAMP_4
A@VDS-TSW.05-20-CA[config-filter EXAMP_4]>10 accept udp any any any
1.2.3.4 161 vlan1
A@VDS-TSW.05-20-CA[config-filter EXAMP_4]>15 accept udp any any any
1.2.3.4 162 vlan1
A@VDS-TSW.05-20-CA[config-filter EXAMP_4]>enable
A@VDS-TSW.05-20-CA[config-filter EXAMP_4]>exit
```

All the incoming traffic matching the destination address, the egress interface (**vlan1**) and the **dport** is allowed to flow. All the remaining traffic (not matching the rules) is blocked.

Allow UDP traffic from network 1.2.3.0/24 (on vlan2) and host 1.2.4.5 (on vlan3) to 239.10.1.1 (on vlan1) port 5000

```
A@VDS-TSW.05-20-CA>configure ip filter EXAMP_5
A@VDS-TSW.05-20-CA[config-filter EXAMP_5]>10 accept udp 1.2.3.0/24 any vlan2
239.10.1.1 5000 vlan1
A@VDS-TSW.05-20-CA[config-filter EXAMP_5]>20 accept udp 1.2.3.5 any vlan3
239.10.1.1 5000 vlan1
A@VDS-TSW.05-20-CA[config-filter EXAMP_5]>enable
A@VDS-TSW.05-20-CA[config-filter EXAMP_5]>exit
```

In this filter matching is performed, in addition to the destination address, the egress interface (**vlan1**) and the **dport** (5000), on **src** and **in-if** items also. All the remaining traffic, as in the previous example, is blocked.

Deny all IP traffic from vlan2, vlan3 ... to vlan1

```
A@VDS-TSW.05-20-CA>configure ip filter EXAMP_6
A@VDS-TSW.05-20-CA[config-filter EXAMP_6]>10 accept icmp-ping any any any any
A@VDS-TSW.05-20-CA[config-filter EXAMP_6]>11 accept icmp-ping any any local
A@VDS-TSW.05-20-CA[config-filter EXAMP_6]>20 deny ip any vlan2 any vlan1
A@VDS-TSW.05-20-CA[config-filter EXAMP_6]>21 deny ip any vlan3 any vlan1
A@VDS-TSW.05-20-CA[config-filter EXAMP_6]>90 accept ip any any any any
A@VDS-TSW.05-20-CA[config-filter EXAMP_6]>91 accept ip any any local
A@VDS-TSW.05-20-CA[config-filter EXAMP_6]>enable
A@VDS-TSW.05-20-CA[config-filter EXAMP_6]>exit
```

This is another example of “deny” filter; in other words the filter must deny some kind of traffic, allowing all the rest. This is why a couple of “accept all” rules are put at the end of the list in order to overcome the default “block all” policy.

Furthermore, rules 10 and 11 are put at the beginning of the list to allow for “ping” to flow. This kind of traffic would be otherwise blocked by rules 20 and 21. Remember that **ip** implies no protocol match and this is why “ping” traffic would be caught by these rules and therefore needs to be allowed well in advance.

Allow all IP traffic from vlan1 to vlan2, vlan3, ...

```
A@VDS-TSW.05-20-CA>configure ip filter EXAMP_7
A@VDS-TSW.05-20-CA[config-filter EXAMP_7]>10 deny icmp-ping any any any any
A@VDS-TSW.05-20-CA[config-filter EXAMP_7]>11 deny icmp-ping any any local
A@VDS-TSW.05-20-CA[config-filter EXAMP_7]>20 accept ip any vlan1 any vlan2
A@VDS-TSW.05-20-CA[config-filter EXAMP_7]>21 accept ip any vlan1 any vlan3
A@VDS-TSW.05-20-CA[config-filter EXAMP_7]>enable
A@VDS-TSW.05-20-CA[config-filter EXAMP_7]>exit
```

This filter has been interpreted in a “strict” way, differentiating between ICMP (“ping”) traffic (blocked) and IP traffic (allowed). This is the rationale behind rules 10 and 11, that block packets otherwise matching rules 20 and 21 (for the reason explained in the previous example).

Allow UDP from 1.2.3.4 vlan2 to 5.6.7.8 vlan1 port 5001

...

Allow UDP from 1.2.3.6 vlan2 to 5.6.7.8 vlan1 port 5001

```
A@VDS-TSW.05-20-CA>configure ip filter EXAMP_8
A@VDS-TSW.05-20-CA[config-filter EXAMP_8]>10 accept udp 1.2.3.4 any vlan2
                                                    5.6.7.8 5001 vlan1
...
A@VDS-TSW.05-20-CA[config-filter EXAMP_8]>20 accept udp 1.2.3.6 any vlan2
                                                    5.6.7.8 5001 vlan1
A@VDS-TSW.05-20-CA[config-filter EXAMP_8]>enable
A@VDS-TSW.05-20-CA[config-filter EXAMP_8]>exit
```

This examples highlights that in such scenario a rule is needed for each source host.

Allow incoming management traffic only from interface vlan1

```
A@VDS-TSW.05-20-CA>configure ip filter EXAMP_9
A@VDS-TSW.05-20-CA[config-filter EXAMP_9]>10 accept icmp-ping any vlan1 local
A@VDS-TSW.05-20-CA[config-filter EXAMP_9]>20 accept tcp any any vlan1 local 22
A@VDS-TSW.05-20-CA[config-filter EXAMP_9]>25 accept tcp any any vlan1 local 23
A@VDS-TSW.05-20-CA[config-filter EXAMP_9]>30 accept tcp any any vlan1 local 9901
A@VDS-TSW.05-20-CA[config-filter EXAMP_9]>35 accept tcp any any vlan1 local 9943
A@VDS-TSW.05-20-CA[config-filter EXAMP_9]>enable
A@VDS-TSW.05-20-CA[config-filter EXAMP_9]>exit
```

Management traffic is basically “ping”, ssh (port 22), telnet (port 23), REST (port 9901)

and REST on SSL (port 9943). Only local rules are defined since the target is the traffic destined to the device itself.

Chapter 13 - Port mirroring

Port mirroring monitors network traffic by sending copies of all incoming and outgoing traffic from one port to a monitoring port. Network administrators can use port mirroring as a debugging tool for analyzing network packets as they enter or exit a specific set of switch ports.

You can enable port mirroring by issuing the following command from *system administration* level:

```
port-mirror destination INTERFACE source LIST
```

where *INTERFACE* is the port where all packets will be sent to (*monitoring port*) and *LIST* is a comma delimited list of source ports (*mirrored ports*). Port mirroring can then be disabled by issuing the *no-form* of the command.

NOTE

Port mirroring configured with the `sysadmin` command is not part of the *running* configuration and it will be not saved in the *startup* configuration. When the device is rebooted, port mirroring setup is cleared.

Using `configure` as command root instead of `sysadmin`, will make the port mirroring setup being stored in the *running* configuration so it can be saved and restored at boot.

You can see the active port mirroring setup with the `sysadmin show port-mirror` command like in the following example:

```
A@SW-Std.03-00-CA>sysadmin show port-mirror
Active port mirroring:
Destination port: e1-1
Source ports: e1-0,e0-1,e0-2,e0-3
```

Chapter 14 - **Monitoring and diagnostics**

This chapter describes features and capabilities of the Consist Switch which can be used to obtain information about its counters and working parameters.

14.1 Logs

14.1.1 Runtime logs

The messages helpful in tracing the activity of the processes running on the Consist Switch can be view at any time on the CLI issuing the command:

```
[no] debug
```

Since the amount of data shown can be huge and cumbersome to read, it is possible to filter the messages basing on their source. We will call each source a *subsystem*.

To show only the messages related to one or more subsystems, issue the command `debug` followed by a `+` separated list of identifiers:

```
debug <subs1[+subs2[+...]]>
```

The order of the subsystems in the list does not matter.

The available subsystem identifiers can be listed by typing `debug` followed by the `TAB` or `?` key:

```
A@SW-Std.03-00-CA>debug
<subs1[+subs2[+...]]> Available subsystems:
- core      Device Core
- conf      Configuration
- coupled   Coupled Mode
- dev       Device management
- dhcp      Dynamic Host Configuration Protocol (DHCP)
- diag      Diagnostics
- dot1x     Port-based Network Access Control (IEEE 802.1x)
- groups    Link aggregation
- igmp      Internet Group Management Protocol (IGMP)
- iogw      Communications among subsystems
- lldp      Link Layer Discovery Protocol
- match     TTCMP Project Matching
- mon       Device Monitoring
- poe       Power over Ethernet (IEEE 802.3af/802.3at)
- route     Routing
- rstp      Spanning Tree Protocol (IEEE 802.1D/802.1w)
- snmp      Simple Network Management Protocol (SNMP)
- tcnopen   Train Real Time Data Protocol (TRDP network I/O)
- trdp      Train Real Time Data Protocol (TRDP logics)
- ttcmp     Train Topology and Configuration Management Protocol (TTCMP)
- switch    Switching
- vrrp      Virtual Router Redundancy Protocol (VRRP)
```

Some examples:

- view the messages related to the *link aggregation*, the *routing* and the *per-port DHCP server* (but no other):

```
A@SW-Std.03-00-CA>debug groups+route+dhcp
```

- view all the messages except the ones coming from the *Link Layer Discovery Protocol* and the *internal distributed properties* (need two commands):

```
A@SW-Std.03-00-CA>debug  
A@SW-Std.03-00-CA>no debug lldp+props
```

14.1.2 Log files

Many run-time messages are also written on log files and can be optionally sent to a remote logging system through the network. The log files maintained by the Consist Switch are:

- An operational log on a volatile medium; this file is cleared as soon as the device is switched off.

This log can be read with the command:

```
0@SW-Std.03-00-CA>show log
```

- A persistent log reserved for the most important events.

It can be viewed with:

```
0@SW-Std.03-00-CA>show log persistent
```

and cleared with:

```
A@SW-Std.03-00-CA>sysadmin delete log persistent
```

- A log used to save results from each Firmware Install/Upgrade Operation (see Konuendo Total Device Management). It can be read with:

```
0@SW-Std.03-00-CA>show log update
```

and cleared with:

```
A@SW-Std.03-00-CA>sysadmin delete log update
```

It is possible to send the log information to a remote *syslog* server with the command:

```
0@SW-Std.03-00-CA>configure system logging host ADDRESS
```

where *ADDRESS* is the IP address of the remote *syslog* server.

14.1.3 Subsystem full log on file

As already stated many messages are written on the operational log but not all, usually only the most important or related to significant events.

For most of the Consist Switch subsystems it is possible to have *all* the messages

written in the operational log, allowing a complete tracing of its activity since the system start-up. This feature can be enabled for one or more subsystems using the following command:

```
configure system logging debug <subs1[,subs2[,...]]>
```

Note that in this command the subsystem identifiers are separated by a comma (,) and it is mandatory to specify at least one subsystem.

The command can be cleared using its no-form:

```
no configure system logging debug
```

The available subsystem identifiers can be listed by typing `configure system logging debug` followed by the `TAB` or `?` key:

```
A@SW-Std.03-00-CA>configure system logging debug
<subs1[+subs2[+...]]> Available subsystems:
- coupled    Coupled Mode
- dev         Device management
- dhcp        Dynamic Host Configuration Protocol (DHCP)
- diag        Diagnostics
- dot1x       Port-based Network Access Control (IEEE 802.1x)
- groups      Link aggregation
- igmp        Internet Group Management Protocol (IGMP)
- lldp        Link Layer Discovery Protocol
- match       TTCMP Project Matching
- poe         Power over Ethernet (IEEE 802.3af/802.3at)
- route       Routing
- rstp        Spanning Tree Protocol (IEEE 802.1D/802.1w)
- tcnopen     Train Real Time Data Protocol (TRDP network I/O)
- trdp        Train Real Time Data Protocol (TRDP logics)
- ttcmp       Train Topology and Configuration Management Protocol (TTCMP)
- switch      Switching
- vrrp        Virtual Router Redundancy Protocol (VRRP)
```

14.2 Diagnostics

The Consist Switch provides *diagnostic information* that can be used by control units to detect and report potential anomalies. Such information can be provided in the form of Process Data (PD) transmitted with TRDP or IPTCom protocols and/or also sent over VDS Rail's TTCMP® protocol.

IMPORTANT NOTE

The description of the *data-sets* sent with TRDP/IPTCom or TTCMP® is released in specific documentation thus not covered by this manual. For more detailed information about the *data-sets*, please contact the VDS Rail Technical Support.

14.2.1 Configuration

The Consist Switch always collects diagnostic information but it does not send data unless suitably configured.

14.2.1.1 *Diagnostics over TRDP*

The following commands control *diagnostic data* publishing in the form of Process Data through TRDP protocol:

```
configure diagnostics trdp publish comid <comid>
```

It sets up the ComID (the identification number of the TRDP data set); only values greater than or equal to 1001 are allowed (1-1000 ComID values are reserved in accordance with IEC 61375). This command is optional (if missing, the default ComID value will be 1001).

```
configure diagnostics trdp publish destination address <a.b.c.d>
```

It sets up the Process Data destination address; *multicast* addresses are allowed if the device knows how to route them (see Layer 3 services on page 99). This command is mandatory.

```
configure diagnostics trdp publish interval <interval>
```

It sets up how frequently (in seconds) the PD is transmitted. This command is optional (if missing, the default interval value will be 5 seconds).

```
configure diagnostics trdp publish ip precedence <0-7>
```

It sets up the IP Precedence level of PDs. This command is optional (if missing, the default TRDP implementation value for PDs is used).

```
configure diagnostics trdp publish ip ttl <1-255>
```

It sets up the TTL of PDs. This command is optional (if missing, the default TRDP implementation value for PDs is used).

```
configure diagnostics trdp enable
```

This command finally starts the transmission and is mandatory.

Follows a configuration example to send data with ComID set to 1001 and 1 second rate to the address 10.0.0.75:

```
A@SW-Std.03-00-CA>configure diagnostics trdp publish comid 1001
A@SW-Std.03-00-CA>configure diagnostics trdp publish dest address 10.0.0.75
A@SW-Std.03-00-CA>configure diagnostics trdp publish interval 1
A@SW-Std.03-00-CA>configure diagnostics trdp enable
```

14.2.1.2 *Diagnostics over IPTCom*

The following commands control *diagnostic data* publishing in the form of Process Data through IPTCom protocol.

```
configure diagnostics iptcom publish comid <comid>
```

It sets up the ComID (the identification number of the IPTCom data set); only values greater than or equal to 1001 are allowed (1-1000 ComID values are reserved in accordance with IEC 61375). This command is optional (if missing, the default ComID value will be 1001).

```
configure diagnostics iptcom publish destination address <a.b.c.d>
```

It sets up the Process Data destination address; *multicast* addresses are allowed if the device knows how to route them (see Layer 3 services on page 99). This command is mandatory.

```
configure diagnostics iptcom publish interval <interval>
```

It sets up how frequently (in seconds) the PD is transmitted. This command is optional (if missing, the default interval value will be 5 seconds).

```
configure diagnostics iptcom publish ip precedence <0-7>
```

It sets up the IP Precedence level of PDs. This command is optional (if missing, the default IPTCom implementation value for PDs is used).

```
configure diagnostics iptcom publish ip ttl <1-255>
```

It sets up the TTL of PDs. This command is optional (if missing, the default IPTCom implementation value for PDs is used).

```
configure diagnostics iptcom enable
```

This command finally starts the transmission and is mandatory.

Follows a configuration example to send data with ComID set to 9999 and 4 second rate to the address 10.0.0.23:

```
A@SW-Std.03-00-CA>configure diagnostics iptcom publish comid 9999
A@SW-Std.03-00-CA>configure diagnostics iptcom publish dest address 10.0.0.23
A@SW-Std.03-00-CA>configure diagnostics iptcom publish interval 4
A@SW-Std.03-00-CA>configure diagnostics iptcom enable
```

14.2.1.3 **Diagnostics over TTCMP®**

The following commands control *diagnostic data* publishing in the form of TTCMP® diagnostic packet.

```
configure diagnostics ttcmp announce
```

It configures TTCMP® subsystem so that it sends a diagnostic packet immediately after transmitting each TTCMP® announcement packet. This command is optional.

```
configure diagnostics ttcmp interval <interval>
```

It configures TTCMP® subsystem so that it sends diagnostic packet based on time, depending on a defined period, expressed in seconds. This command is optional (if missing, the default interval value will be 5 seconds). A value of 0 is allowed and usually set in combination with `configure diagnostics ttcmp announce` when no timed sending is required.

```
configure diagnostics ttcmp enable
```

It enables TTCMP® diagnostic packet transmission. This command is mandatory.

14.2.2 Control Commands

It is possible to show the subsystem status with the command:

```
show diagnostics
```

The following is an example of a system configured for sending data with both IPTCom protocol, ComID 1001 and 1s rate, and with TTCMP® protocol, after each announcement. Fields of interest, according to the configured parameters, have been highlighted.

```
U@SW-Std.03-00-CA>show diagnostics

Diagnostic Data Collector
=====
Data collector name:      default
Build time:              2020-06-17 09:44:35 UTC
Version:                 sw-std - (no branch) - 97a5e3979
Module configuration:    completed
Collecting Diagnostic Data: started
=====
TTCMP protocol: enabled
Send timed              disabled
Send with announcements enabled
=====
TRDP protocol: disabled
=====
IPTCom protocol: enabled
ComId      1001      Notify      Request      Reply
Interval   1.00 s    not set    not set      not set
IP Prec.   default    default
TTL        default    default
Address    10.200.5.1 not set
Status     published not listening
```

14.3 SNMP

The device currently supports the Simple Network Management Protocol (SNMP) v1/v2c/v3. In the absence of any SNMP specific configuration, an SNMP v2c Agent is enabled by default with two usual communities defined: a *public* read-only community and a *private* read-write community.

```
0@SW-Std.03-00-CC>show running-config
...
...
snmp-server community public ro
snmp-server community private rw
snmp-server enable
```

The SNMP Agent configuration is available at the *global configuration* level - submenu `snmp-server`, as you can see asking for the submenu on-line help:

```

0@SW-Std.03-00-CA>configure snmp-server          (? pressed)
community          Set the SNMP server community value
contact            Set the SNMP contact value
description        Set the SNMP server description
enable             Enable SNMP services
engineid           Configure manually an SNMPv3 engine-ID for the local
                  agent
exit               Exit from SNMP subsystem level
group-access       Configure access capabilities for an SNMPv3 group
host              Set the SNMP server target host address for egressing
                  traps
location          Set the SNMP location value
name              Set the SNMP server name
traps-all-if      Configure emission of selected traps on all available
                  interfaces at one time
traps-global       Configure emission of global generic traps.
traps-ttcmp        Configure emission of TTCMP related traps
user              Configure an SNMPv3 user
view              Include the specified oid-tree value in the set of
                  allowed OIDs for the named view

```

Beside the configuration commands, the `enable` command is the one which effectively starts up the SNMP Agent if a minimum set of required parameters has been configured. Every SNMP configuration command issued on a running (enabled) SNMP Agent has instead immediate effect without the need of restarting the SNMP subsystem, apart from the `snmp-server engineID` command which is not effective until agent is restarted via `enable` command or at the next reboot.

14.3.1 Configuring SNMPv2 Community-Based Authentication

In the following we will describe each one of the above configuration parameters, but first keep in mind that, in order to start the SNMP subsystem, at least one community string or one SNMPv3 user must be defined in advance, as you can see from the following output obtained trying to start the SNMP subsystem without any settings already in place:

```

0@SW-Std.03-00-CA>configure snmp-server enable
SNMP Subsystem NOT STARTED !
At least ONE v1/v2c community OR v3 user MUST be configured for the SNMP
subsystem to be enabled.

```

The string defined by the `community` command is the means by which SNMP v1/v2c authenticates a user connection demanding an SNMP query: you can define how many communities you need in the switch but such community names must be unique; if you simply define a community name the SNMP subsystem will declare by default such community to be a *read-only* one and the access will be set *unlimited* to the whole MIB OID tree and without any further limitation regarding the IP source address of the incoming connection. As an example let us establish a couple of communities:

```

0@SW-Std.03-00-CA>configure snmp-server community vds-private rw
0@SW-Std.03-00-CA>configure snmp-server community public ro

```

and verify the added configuration lines:

```
0@SW-Std.03-00-CA>show running-config
....
interface vlan1
  ip address 172.16.16.111 255.255.0.0
  enable

snmp-server community public ro
snmp-server community vds-private rw
```

In this way you have just simply defined two communities: one with read-only `ro` access, and the other with full read-write `rw` access capabilities. If you need a finer grain control to the access of SNMP-exposed data, you can additionally specify a specific MIB OID which will be the root of the MIB tree accessible from that community, using the `oid` sub-command of `community` as in the following example:

```
0@SW-Std.03-00-CA>configure snmp-server community public rw oid .1.3.6.1.2.1.2.2
```

This last configuration line will *redefine* a new read-write community named *public* with restricted access to the *ifTable* of the standard *IF-MIB*; in fact the above OID could also have been specified using the symbolic specification format of the MIB OID tree for *ifTable*:

```
0@SW-Std.03-00-CA>configure snmp-server community public rw oid IF-MIB::ifTable
```

Additionally, note from the above example that, since the community name is the unique key of the configuration line, you can simply override a previously defined settings for the community *public* issuing a new and extended community command including also the additional OID limitation while changing *also* the read-write access mode for such community.

Another more flexible mean of restricting the user to access only a subset of the whole available SNMP data is to use the *view access control model*; later on you will see how you can define an SNMP *view* which includes a set of one or more sub-trees of the whole SNMP MIB tree and associates it with a user-chosen name. After having defined such a named view, you can easily bound such partition of the MIB tree to a defined community, thus effectively further restricting access to a controlled set of MIB objects for some class of users.

So, in the following example, if you suppose to have already created a view named *alfaCentauri*, issuing then the following command:

```
0@SW-Std.03-00-CA>configure snmp-server community public rw view alfaCentauri
```

you will establish a new access restrictions for the *public* community instead of the previous restriction based on OID identifier. The view and OID sub-commands for access control are mutually exclusive.

Finally you can decide to further restrict the access to a certain community not only depending on *what* they're asking, but also depending on *where* they're asking from: in other words you can additionally associate some source IP address restrictions with a defined community. So extending the above example:

```
0@SW-Std.03-00-CA>configure snmp-server community public rw view  
alfaCentauri access 172.16.16.1
```

this additionally restricts such community to the specified IP address. This kind of restriction works only for single IP address: you cannot specify multiple IP addresses of whole subnets while restricting access to SNMP data.

14.3.2 Configuring Views

As already said, a view represents a named subset of one or more sub-trees of the whole MIB tree: it is a useful tool to partition such tree and refer to a specific part of it to enforce some particular access restrictions.

In order to reference a view as a mean of access restriction inside a community configuration line, you must also define it, describing how it is made up and what it refers to. Using multiple `snmp-server view` commands you can create a named view and at the same time specify the exact sub-tree which you want to include or to exclude from the view itself. Let us make a few examples on this. The following creates a view which is composed *only* of the named *ifTable*:

```
0@SW-Std.03-00-CA>config snmp-server view alfaCentauri IF-MIB::ifTable included
```

Once such a view is associated with a community directive as stated below:

```
0@SW-Std.03-00-CA>configure snmp-server community public rw view  
alfaCentauri access 172.16.16.1
```

you will be sure that, from now on, the *public* community will have *read-write access only* to the data of the *IF-MIB::ifTable* and *only* when *public* community data queries are received from IP address 172.16.16.1.

On the other side, if you define the view as:

```
0@SW-Std.03-00-CA>configure snmp-server view alfaCentauri .1.3.6.1 included  
0@SW-Std.03-00-CA>configure snmp-server view alfaCentauri  
IF-MIB::ifTable excluded
```

you will end up with the opposite access restriction for the *public* community, in the sense that, from now on, every query received from 172.16.16.1 on behalf of the community *public* will be satisfied *only if not* regarding data included in the *IF-MIB::ifTable*.

14.3.3 Configuring SNMPv3 USM-Based Authentication

Apart from the above described community-based SNMPv2 cleartext authentication, the device supports also the more secure SNMPv3 authorization scheme based on SNMPv3 USM. Currently there is no support for alternate SNMPv3 authorization schemes like DTLS or SNMP-to-SSH tunneling.

In order to use the native SNMPv3 User Security Model scheme you must define:

- a **valid SNMPv3 user together with its authentication/privacy parameters**

while **specifying the group** it belongs to. Remember that:

- a user can *be part of only one group at a time*
- used authorization protocol can be MD5 or SHA1
- used privacy-protocol can be chosen among DES and AES
- authorization/privacy passwords must be 8 characters long minimum

```
0@SW-Std.03-00-CA>configure snmp-server user testUser group testGroup
md5 myAuthPass des myPrivPass
```

- the **access capabilities of the defined groups** of users in terms of security-level and read/write access views. Remember that the security level associated to a group can be:
 - **noauth**: neither authorization nor privacy is enforced for SNMPv3 requests coming from a user belonging to such a group.
 - **auth**: every request coming from users belonging to such a group is processed using the user's defined authorization protocol and password. No privacy encryption is required.
 - **priv**: when an **encrypted** request is received coming from a user belonging to a group having a security-level of *priv*, after the user is authenticated using the chosen authorization-protocol and password, the request is decrypted using the user's privacy-protocol and password.

```
0@SW-Std.03-00-CA>configure snmp-server group-access testGroup
auth readview iftable writeview konuendo
```

NOTE

Please note that the view you choose as *writeview* with the above command is a write-only view and not a read-write view.

- the **used views** as already explained:

```
0@SW-Std.03-00-CA>configure snmp-server view iftable
IF-MIB::ifTable included
0@SW-Std.03-00-CA>configure snmp-server view konuendo
KONUENDO-MONITOR-MIB::konEventTable included
```

In the above example we have defined the new SNMPv3-user *testUser* as belonging to SNMPv3-group *testGroup*, using MD5 as authorization-protocol with password *myAuthPass* and DES as privacy-protocol with *myPrivPass* as privacy-password; then we have defined the SNMPv3-group *testGroup* using a security level of *auth* and restricting such group access capabilities to view *iftable* for read request and to view *konuendo* for write request. At the end we have taken care to define the needed views as named in the *group-access* directive.

Additionally it is possible to manually set the *engineID* parameter using:

```
0@SW-Std.03-00-CA>configure snmp-server engineID myCustomEngineID
```

Just remember that, in case you have not manually set an *engineID* parameter, the

system, during the SNMP AutoDiscovery phase, will use an *engineID* which is **unique** to this device because automatically generated and tied to the MAC address of management interface.

IMPORTANT NOTE

In case you modify the *engineID* parameter with the SNMP Agent already running, the change is not enforced until next Agent restart.

To remove a previously defined user, group-access or *engineID* simply use these commands no-form:

```
0@SW-Std.03-00-CA>no configure snmp-server engineID
0@SW-Std.03-00-CA>no configure snmp-server user testUser
0@SW-Std.03-00-CA>no configure snmp-server group-access testGroup
```

14.3.4 Configuring General Server Data

Using the `snmp-server` configuration sub-commands you can also configure some common identification parameters regarding the SNMP server instance itself. Such additional configuration parameters are:

contact	Set the SNMP contact value
description	Set the SNMP server description
location	Set the SNMP location value
name	Set the SNMP server name

These sub-commands may be simply used to help customize your specific machine SNMP server for identification purposes. Consider that also these SNMP commands, as the ones above, take effect immediately if the SNMP Agent is running, BUT please note also that, instead, after issuing a no-form on these sub-commands, in order to restore their original values, you **MUST** restart the SNMP subsystem for such an action to have effect.

Moreover, these commands determines the values exposed by a set of well-known standard OIDs (RFC1213-MIB::sysName/sysContact/sysLocation/sysDescr), and when such OIDs are set through these commands they suddenly became read-only from the SNMP perspective: as long as one of these sub-commands about general server data is actively modifying the corresponding exposed OID value, such OID will be **NO MORE** modifiable through an external SNMP SET operation.

14.3.5 Configuring SNMP Notifications

Beside the classical query/response paradigm, an SNMP Agent is also able to send asynchronous notification toward a specified destination, which typically runs some sort of Network Management Entity used to collect data remotely.

These actions are what SNMP calls traps or informs: many kinds of notifications can be sent by an SNMP Agent on various types of events (both standard and proprietary), but regardless of the nature of such events a minimum set of system-wide notification-related parameters must be configured.

This is the purpose of the `snmp-server` sub-command `host` and related options, as briefly showed in the following example:

```

0@SW-Std.03-00-CA>configure snmp-server host                (? pressed)
<ipaddr>           Set the SNMP server target host address for egressing traps

0@SW-Std.03-00-CA>configure snmp-server host 172.16.16.1    (? pressed)
<community/user>   Set the SNMP server trap community or user to be used for
                   egressing notifications toward the specified host address
<CR>

0@SW-Std.03-00-CA>configure snmp-server host 172.16.16.1 trap-comm  (? pressed)
v1                 Set SNMP protocol Version 1 for egressing traps
v2c                Set SNMP protocol Version 2c for egressing notifications
v3                 Set SNMP protocol Version 3 for egressing notifications
<CR>

0@SW-Std.03-00-CA>configure snmp-server host 172.16.16.1 trap-comm v2c      (?)
informs            Set the SNMP server egressing notifications type to inform
traps              Set the SNMP server egressing-notifications type to trap
...

```

See CLI commands reference (page 217) for more information.

Briefly, the most important and mandatory required data are:

- **<ipaddr>**: which defines the IP address of the host that will be the target of the generated outgoing traps; if this is not set, no trap will be ever generated.
- **<community-user>**: a community or an SNMPv3 user on behalf of which traps are sent. This could be used by target host network management entities to classify or discard the received traps themselves.

If no other parameter is specified, the system will emit an SNMPv2 trap toward the specified *host/community* pair.

Additionally you can choose to specify also:

- **<port>**: the destination port to use in case the target host has been configured to use a non-standard port for trap reception.
- **traps/informs**: this token specifies the kind of trap to be sent: an old style non-Acknowledged SNMPv2c/v3 Trap or a new SNMPv2c/v3 Acknowledged Inform.
- **v1/v2c/v3**: the SNMP version to use for outgoing notifications. Depending on this choice, when you select SNMPv3 notification type, you will be additionally presented with the following v3-related notification commands:
 - **auth/noauth/priv**: the desired minimum security level for outgoing SNMPv3 notifications.
 - **md5 <authpass>/sha1 <authpass>**: the desired authentication algorithm and chosen password (minimum 8 characters).
 - **aes <privpass>/des <privpass>**: the desired encryption algorithm and chosen password (minimum 8 characters); this choice is available only if a minimum security-level of *priv* was selected.

- **remoteEngineID <engineID-String>**: the remote NMS *engineID* to be used for outgoing SNMPv3 traps (which cannot otherwise complete the SNMPv3 AutoDiscovery phase).

As soon as the above minimal configuration for outgoing traps is set, the system begins to emit the following traps without the need of further configuration:

- SNMP Agent *coldStart* (**SNMPv2-MIB::coldStart**): each time the agent is started or restarted (or after a successful boot).
- SNMP Agent *nsNotifyShutdown* (**NET-SNMP-AGENT-MIB::nsNotifyShutdown**): each time the agent is stopped or the system is rebooted.
- SNMP Agent *nsNotifyRestart* (**NET-SNMP-AGENT-MIB::nsNotifyRestart**): each time the configuration is changed while the agent is running it reloads the configuration.
- SNMP Agent *authenticationFailure* (**SNMPv2-MIB::authenticationFailure**): each time a v2c/v3 authorization fails and is rejected.

14.3.6 Konuendo Monitoring Facility

The system has an embedded and flexible monitoring engine which can be used to satisfy a wide-range of notification needs; one of this needs is surely the configurable emission of diagnostic traps.

Once a notification trap is enabled for some sort of event by the means of proper commands, a corresponding event is handled to the engine for monitoring and a full description of the registered entity is reported in the special SNMP table *KONUENDO-MONITOR-MIB::konEventTable* exported by the switch SNMP subsystem; as soon as such event fires, the SNMP subsystem takes care to intercept such inner engine-notification and packs an SNMP trap containing all the needed details while updating, when needed, the *konEventTable*.

This means that an updated list of all currently monitored events is always available for consultation via SNMP by a Network Management System (when SNMP agent is properly configured); the provided proprietary MIBs, *KONUENDO-MIB* and *KONUENDO-MONITOR-MIB*, contain a formal and comprehensive description of such table and the definition of the generic container-trap that is emitted upon a monitored fired event. In the following you will find a detailed explanation of the commands enabling the emission of specific traps (once the SNMP subsystem has been configured in a general way as stated in the previous paragraphs).

Moreover, the payload of such proprietary traps defined in *KONUENDO-MONITOR-MIB* is typically composed of:

- *konEntityName*: carrying the monitored entity internal reference as it appears also in *konEventTable.konEntity*.
- *konEntityValue*: carrying the value of the monitored *konEntityName* which triggered the condition and the subsequent outgoing notification.

14.3.7 Enabling SNMP Notifications

All the available SNMP notifications can be activated from the global configuration-level:

```
0@SW-Std.03-00-CA>configure snmp-server          (? pressed)

...
traps-all-if      Configure emission of selected traps on all available
                   interfaces at one time
traps-global       Configure emission of global generic traps.
traps-ttcmp        Configure emission of ttcmp related traps
```

and are grouped in families:

- **traps-global** - these traps are generic for the system and not tied to a specific interface:
 - *overtemp* (**KONUENDO-MONITOR-MIB::konOverTemperature / konNormalTemperature**): emitted when system operating temperature is above normal operating level.
 - *high-cpu-load* (**KONUENDO-MONITOR-MIB::konHighCpuLoad / konNormalCpuLoad**): emitted when system CPU computational level is above normal operating threshold for a sustained period of time (5 minutes).
 - *configuration-change* (**KONUENDO-MONITOR-MIB::konConfigChange**): a configuration command has been issued on the system.
 - *sysadmin-change* (**KONUENDO-MONITOR-MIB::konSysAdmChange**): a system-administration command has been issued on the system.
 - *new-root* (**BRIDGE-MIB::newRoot**): the bridge controlled by the sending agent has been elected as the new *rootBridge* of the calculated spanning-tree.
 - *topology-change* (**BRIDGE-MIB::topologyChange**): emitted when any NON EDGE port on the bridge controlled by the sending agent has transitioned to the Forwarding State.
- **traps-all-if** - these, instead, are per-interface traps (such as linkUp/linkDown) and from here you can activate such traps on all *applicable* interfaces at a time:
 - *link-status* (**IF-MIB::linkDown/linkUp**): emitted when one of the links goes Up or Down. In this case a standard *IF-MIB::LinkUp IF-MIB::LinkDown* trap is emitted.
 - *link-flap* (**KONUENDO-MONITOR-MIB::konLinkFlapping**): emitted when one link goes repeatedly Up/Down for more than 5 times within the last 10 seconds.
 - *link-poe* (**KONUENDO-MONITOR-MIB::konPoeUp/konPoeDown**): emitted when a PoE capable port starts/stops powering an attached device (**please note** that when a PoE-capable port's administrative status is set to 'disabled', the related PoE functionality is NOT concurrently turned off, so no trap is supposed to be emitted in this case).

- **traps-ttcmp** - these are traps related to the TTCMP subsystem:
 - *ttcmp-anomaly* (**KONUENDO-MONITOR-MIB::konTtcmpAnomalyDetected/konTtcmpAnomalyCleared**): emitted when TTCMP subsystem (if active) detects an anomaly or when an anomaly is cleared.

In case you wish to activate all the traps of a family at a time you simply use the keyword **ALL** when specifying enabled traps for that family:

```
0@SW-Std.03-00-CA>configure snmp-server traps-global ALL
```

and you will end up with all of the available traps for that family enabled. Please note that the command parsing is case sensitive and the **ALL** parameter must be all **uppercase** (**all** will not work).

14.4 LLDP

Link Layer Discovery Protocol (LLDP) is a standard protocol used by network devices for advertising their identity and capabilities on an Ethernet network.

Each LLDP enabled device transmits LLDP packets out from all ports on which the protocol is enabled. LLDP devices that receive these packets create and maintain a list of LLDP neighbours they are receiving information from.

If no LLDP packet is received from a known neighbour for a defined amount of time the device is automatically removed from the list.

LLDP packets are sent to a special multicast address that 802.1D compliant bridges do not forward. In this case neighbouring can be formed only between directly connected devices.

LLDP protocol is enabled by default on all Consist Switch models.

You can get a list of the current known neighbours using the command:

```
A@SW-Std.03-00-CA>show lldp neighbours
```

The result is a list of neighbour devices, not only switches, like this:

Local port	ChassisID	SysName	Capabilities
e1-7	00:1d:b3:52:46:80	HP ProCurve Sw	bridge/switch
e1-7	a0:59:3a:03:00:ca	SW-Std.04-01-CB	TTCMP bridge/switch

If you need more detailed information about neighbors you can use the *detail* option of the show command. In this case the result, for each neighbor, is like this:

```
A@SW-Std.03-00-CA>show lldp neighbours detail
LLDP neighbours discovered
=====
ChassisId: a0:59:3a:03:00:ca
System name: SW-Std.03-00-CA
Description: VDS Consist switch
Management addr: 10.200.100.1
Capabilities: TTCMP bridge/switch
Port id: e2-0 (if_name)
Port description:
TTL: 15 secs
Local interface: e1-7
TTL remaining: 11
```

14.4.1 Configuring LLDP options

You can globally disable the protocol from global configuration level using the command:

```
A@SW-Std.03-00-CA>no configure lldp run
```

or just disable it on a specific interface. For example:

```
A@SW-Std.03-00-CA>no configure interface e0-1 lldp enable
```

You can also set the frequency at which LLDP packets are transmitted:

```
A@SW-Std.03-00-CA>configure lldp timer 10
```

or define the number of seconds that neighbors should wait before removing this device from their neighbors list if no packet is received:

```
A@SW-Std.03-00-CA>configure lldp holdtime 120
```

You can get information about current LLDP settings using the `show lldp` command:

```
A@SW-Std.03-00-CA>show lldp
Global LLDP information:
  LLDP protocol is running
  LLDP holdtime advertised 15 secs
  LLDP packets sent every 5 secs
  TTCMP extensions are enabled
```

Chapter 15 - Multicast

This chapter describes Consist Switch multicast IP traffic capabilities and configuration.

Multicast packets are packets sent by one host to multiple hosts. On train networks there are many applications that use multicast packets to transmit information using this one-to-many relationship. Video surveillance, multimedia devices and passenger information systems are common applications that frequently use IP multicast streams. Furthermore several industrial protocols used by TCMS applications use a publisher/subscriber communication model based on multicast packets.

15.1 Multicast filtering

By default multicast packets are treated by switches as broadcast packets and so flooded out from all their ports.

With an increase of the number of multicast applications, this behavior could easily overload the whole network. In this case multicast filtering can be used to prune multicast traffic so that it travels only to those end destinations that require the traffic, reducing the amount of useless traffic on the network.

Depending on the complexity of the network, the number of multicast addresses to handle and the L3 protocol used, multicast forwarding rules can be statically configured or dynamically build by "snooping" some high level multicast protocol.

15.2 Static filtering

By default multicast filtering is disabled and multicast packets are flooded out from all switch ports as if they were broadcast packets.

You can change this behavior by disabling flooding of unknown multicast packets from all ports with the global command:

```
A@SW-Std.03-00-CA>no configure ip igmp snooping flood-unknown
```

In this way multicast filtering is enabled and multicast packets can be forwarded only to those ports that are connected with the multicast end-stations.

Alternatively you can operate on a per port basis enabling or disabling multicast filtering on a single ports, for example issuing the following interface command:

```
A@SW-std.03-00-CA>no configure interface e0-1 ip igmp-snooping flood-unknown
```

With multicast filtering active you must manually define the ports where multicast packets should be forwarded to.

You can do this adding one or more static entries to the layer 2 forwarding table or,

working at IP layer, associating an ip multicast address with a specific port.

In the first case you can use the global command:

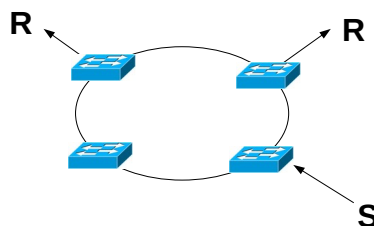
```
configure mac-address <l2-mcast-addr> interface <if-name> vlan <vid>
```

In the second case you can use the interface command:

```
config interface <if> ip igmp-snooping static-group <mcast-group> [vlan <vid>]
```

In this case the associated vlan can be defined only on trunk ports and not on access ports.

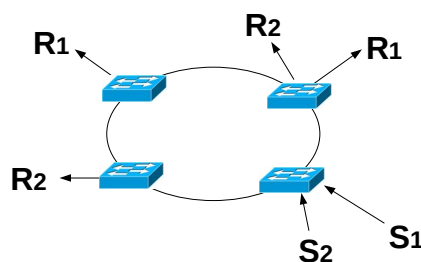
As an example we could make some examples based on the following topology:



Here we have a single multicast source (S) with two multicast receivers (R). You could configure switches as follow:

- globally disable unknown multicast flooding on all switches
- enable unknown multicast flooding on ports where the receivers are connected as well on all “backbone” ports.

In this case the static configuration is quite simple but, as soon as you add a second multicast source (S2) and a couple of receivers things become immediately more complex.



In fact, in this case, beside disabling unknown multicast flooding on all switches you should explicitly configure the forwarding rules for both streams on all backbone and edge ports connected with receivers.

15.3 IGMP snooping

Today most of the network traffic is based on IPv4 protocol. The Internet Engineering Task Force (IETF) has defined IGMP (Internet Group Management Protocol) a protocol used by hosts and multicast routers to establish multicast group memberships.

When enabled on a switch, IGMP snooping listens for **join/leave** multicast group

messages, transmitted by end devices, and dynamically updates the Layer 2 forwarding table.

In this way multicast packets can be forwarded only to ports connected with devices that have registered their interest on a specific multicast stream avoiding unnecessary flooding to the other ports.

The Consist Switch supports IGMP version 1, 2 and 3.

This registration must be periodically renewed by the end devices otherwise, after a configurable amount of time, it will be automatically aged out.

For this reason, periodical membership queries should be transmitted by a multicast router or by a switch acting as a *querier* and forwarded to all ports by the other switches; the igmp responses coming from the end devices are used to keep membership information up to date.

15.3.1 Configuring IGMP snooping

By default IGMP snooping is disabled and multicast traffic is flooded out from all forwarding ports.

IGMP snooping can be enabled by issuing the global command:

```
A@SW-Std.03-00-CA>configure ip igmp snooping enable
```

When IGMP snooping is enabled, join/leave requests are used by the switch to forward multicast traffic only to interfaces that are connected to active interested receivers.

Each host that wants to receive a multicast stream must send a JOIN request for that specific group address; when it wants to stop receiving the stream it must send a LEAVE request. The switch make use of these requests for updating its forwarding rules.

Incoming multicast packets that are not associated with any switch port (unknown), depending on the *flood-unknown* global or per port setup, could be forwarded to some ports or are silently dropped.

As already mentioned, the active multicast forwarding table must be periodically refreshed; the aging time for registered groups is set by default to 260 seconds but can be changed to a value between 100 and 1000 seconds using the following command:

```
configure ip igmp snooping aging-time <100-1000>
```

You can view the current status of the IGMP snooping subsystem issuing the command:

```
@SW-Std.03-00-CA>show ip igmp

IGMP snooping is enabled
-Robustness: 2 sec
-Aging time: 100 sec
-Immediate leave: disabled
-TCN ignore: disabled
-TCN flood query count: 2
-TCN query solicitation: disabled

Unknown multicast flooding is enabled
Flood to cpu is disabled

IGMP querier is disabled
-Querier IGMP version: 2
-Query interval: 125 sec
-Querier timeout: 70 sec
-Max response time 10 sec

Active IGMP queriers:

  Address          Vlan  Type      Port
  -----
  192.168.4.31      1    igmp v2   e0-1

Active mrports:

  Name  Type
  -----
  e0-1  dynamic

Registered groups:

  Group address  Port  Vlan  Type      Expires  Last reporter
  -----
  239.10.10.1    e0-0  1     igmp v2   01:24    192.168.4.111
  239.255.255.250 e0-0  1     igmp v2   01:28    192.168.4.211
  224.0.0.251    e0-0  1     igmp v2   01:25    169.254.136.108
```

The active configuration parameters, the active queriers and the list of registered multicast addresses together with their associated ports and VLAN ID are reported on the console.

15.3.2 IGMP immediate leave processing

When an IGMP version 2 leave message is received on a port, a timer (*leave timer*) is started. If no membership report for the specific multicast address is received on the port for the duration of the leave timer, the multicast address is removed from the port, otherwise the timer is stopped and the multicast registration remains active.

This timer is necessary to allow other receivers that might be connected to the same ports to respond to the specific IGMP query transmitted by the multicast router upon reception of the leave message.

By enabling IGMP immediate leave processing, the switch immediately stop forwarding a multicast stream to a port as soon as an IGMP version 2 leave message is received from that port.

IGMP immediate leave can be enabled by issuing the command:

```
A@SW-Std.03-00-CA>configure ip igmp snooping immediate-leave
```

You should not use the immediate leave processing feature where multiple hosts are

connected to a single port because the leave process could affect other hosts registered for receiving for the same group.

15.4 Configuring IGMP querier

Multicast membership must be periodically renewed by the end devices otherwise, as soon as the associated *aging timer* expires, the multicast address is removed from the port and the flow is no more forwarded to the end device.

Membership renewal is normally triggered by IGMP queries transmitted by a multicast router. If there is no multicast router available on the network the switch itself can be configured to operate as multicast querier issuing the command:

```
A@SW-Std.03-00-CA>configure ip igmp querier enable
```

VLAN1 management interface must be properly configured with an IP address otherwise no QUERY message is transmitted.

Only one querier should be active at a time. If the querier service is enabled on more than one switch, an election is automatically performed and the switch with the *lower ip address* assigned to the management interface becomes the *active querier* for the LAN while the querier activity on the other switches is suspended.

Note that the interval between query messages must be less than the aging timer of *join* messages. The default value of *query interval* is 125 seconds while the default value of *aging time* is 260 seconds. You can change these values using the commands:

```
A@SW-Std.03-00-CA>configure ip igmp snooping aging-time 120  
A@SW-Std.03-00-CA>configure ip igmp querier query-interval 60
```

which, in this example, set the aging time and the query interval at 120 and 60 seconds respectively.

15.5 Using multiple devices

Things get much more complicated when multiple switches or other devices are used.

The IGMP forwarding rules and the possible issues that may occur with redundant topologies must be understood for operate correctly.

15.5.1 IGMP forwarding

To determine how to forward multicast traffic the IGMP snooping subsystem need to known how the active multicast router/querier can be reached. In fact IGMP standard defines that unknown multicast flows as well as IGMP join and leave messages should be delivered to the multicast router for further processing.

This is why switch ports where IGMP query messages are received are automatically defined as mrouter (multicast router) interfaces.

You can also statically configure an interface to be a multicast router interface. A

statically configured interface is not subject to aging and does not require an IGMP querier for IGMP snooping to learn about the interface. You can have a mix of statically configured and dynamically learned interfaces on a switch by issuing the following interface command:

```
A@SW-Std.03-00-CA>configure interface e0-1 ip igmp-snooping mrouter
```

15.5.2 Handling STP topology changes

If you are using a redundant topology, special care should be given on path changes introduced by the spanning tree process, otherwise multicast streams path update will rely only on general queries transmitted by the active querier and so the process could require up to *query interval* seconds.

By default, whenever a topology change notification (TCN) is noticed, the IGMP process starts flooding multicast streams out from all ports until a predefined number of General Query is received.

By default the number of general queries for which multicast traffic is flooded is two but it can be changed using the command:

```
configure ip igmp snooping tcn-flood-query-count <1-10>
```

At the same time of multicast flooding the current spanning-tree root send a special IGMP leave message, called Query Solicit, with a group address of 0.0.0.0. When the active querier or the multicast router receives the Query Solicit message, it send out a General Query which all IGMP listeners respond to. This immediately update the snooping information throughout the network allowing the multicast data traffic to be recovered almost instantly

Query solicitation is active by default on the root bridge but can enabled on any switch, whether or not it is the spanning-tree root, by issuing the following global command:

```
configure ip igmp snooping tcn-query-solicit
```

To get the network to converge faster, use the command

```
configure ip igmp querier max-response-time <1-30>
```

and set a low response time value, such as one or two seconds, so that the clients will respond immediately with a report as a response to the IGMP Queries

Reaction to TCN events is enabled by default but can be disabled by issuing the following global command:

```
configure ip igmp snooping tcn-ignore
```

15.5.3 IGMP filters

Multicast IGMP packets (join/leave) can be filtered globally or on per port basis. Filtering IGMP join requests can be useful to limit the flowing of one or more streams

through some parts of the L2 network.

IGMP filters can be configured globally using the global command:

```
configure ip igmp snooping filter <maddress> <mask>
```

or on per interface basis using the interface command:

```
configure interface <if-name> ip igmp-snooping filter <maddress> <mask>
```

On both commands <maddress> is the multicast group whose IGMP related messages should be ignored and dropped.

The optional <mask> parameter allow the definition of a range of addresses to be filtered like in the following example:

```
configure ip igmp snooping filter 239.10.10.0 255.255.255.0
```

will filter all the IGMP packets related to the addresses in the range 239.10.10.0 to 239.10.10.255.

Chapter 16 - **Power over Ethernet (PoE)**

16.1 Power Over Ethernet

The 8 ports labeled *e1-0* to *e1-7* may be optionally equipped with Power over Ethernet (PoE).

Each port can detect and power all PoE defined classes (0 to 4), with a maximum power delivered per port of 15.40 W (class 0, 3 and 4).

For the 14 ports device models this maximum is raised to 30.0 W.

If the electric power for PoE is supplied by an external 48 Vdc power supply, the total amount of power provided is limited by the power supply maximum capacity.

In the models where the PoE is supplied by an internal power supply, the total amount of power provided cannot exceed 60 W ca.

When a PoE network device is connected to a port and its power request is compatible with the system capabilities, it is powered on and the PoE green LED is switched on.

If a PoE device requests more power than it is allowed for its class, the power delivering is cut off and the PoE LED will turn red. Every 30 seconds ca. the system will try to power on the device.

If the total amount of the power requested is greater than the power available from the internal supply the system tries to maintain the power delivery within the limit by means of a controlled switching off of PoE devices.

The policy used is a fixed priority one, connected with the port number: the lesser the port number the greater the priority, so, for example, port *e1-2* has a greater priority than port *e1-4*.

So the system will try to switch off the PoE powered devices starting with port *e1-7*, going back to port *e1-0*. As soon as the total power requested will go back within the limit, the process will stop.

Every 30 seconds ca. devices will be tentatively switched back on repeating the process from the beginning.

All critical events are logged.

It is possible to verify the PoE status with the command:

```
A@SW-Std.03-00-CA>show poe
Port    Mode    Status  Class    Power (W)
-----
e1-0    auto    off     -         0.00
e1-1    auto    on      class 3   15.67
e1-2    auto    off     -         0.00
e1-3    auto    on      class 1    3.96
e1-4    auto    off     -         0.00
e1-5    auto    off     -         0.00
e1-6    auto    off     -         0.00
e1-7    auto    on      class 2    2.32
```

16.2 PoE mode

The PoE subsystem behavior, for each PoE port, may be modified in the following way:

- **mode on:** the power supply is always active, not depending on the port link status (default).
- **mode off:** the power supply is always disabled
- **mode as-link:** the power supply follows the port link status, so it is active when the link is up, it is disabled when the port link is down.

The PoE port mode may be modified with the command:

```
A@SW-Std.03-00-CA>configure interface e1-0 poe mode on
```

where the *mode* qualifier may be **on**, **off** or **as-link**.

Chapter 17 - **System administration**

17.1 Gathering system information

You can get information about current system version with the command:

```
O@SW-Std.03-00-CA>show version
Consist Switch by VDS Rail.
Firmware Version: 6.4.1
Build Release: 62003
```

A summary of the device vital info is produced by the command:

```
O@SW-Std.80-08-F0>show system

Device type:      Consist Switch
Model Code:      V143E
Board Revision:   D
System UUID:      a7ff52be-5532-4e6d-aa66-684a56b5fba0
Primary MacAddr:  a0:59:3a:80:c1:f0
Serial Number:    163103
Bypass:          2
PoE:             available
RTC addon:       NOT present

Free memory:      452640 kB
System Temp:      50.5C

Running hours:    18472.0
Uptime: 4 days, 7:27, load average: 0.12, 0.12, 0.09

Bootloader Rel:   2010.03.vdssw11

The system is running in Normal Mode
```

Internal device temperature can be obtained with the command:

```
O@SW-Std.03-00-CA>show system temperature
System Temperature: 49.0C (Max: 80.0C)
```

Current date and time can be obtained with the command:

```
O@SW-Std.03-00-CA>show system time
Tue 1 Feb 2011 10:00:24 [UTC]
```

Memory usage information can be obtained with the command:

```
O@SW-Std.03-00-CA>show system memory
```

	total	used	free	shared	buffers
Mem:	515156	54244	460912	0	380
Swap:	0	0	0		
Total:	515156	54244	460912		

Persistent storage usage information can be obtained with the command:


```
0@SW-Std.03-00-CA>show system flash
Disk          State      Size      Used      Avail      Used%
Firmware      ro        73.5M     31.8M     37.9M      46%
Configuration ro/rw     13.4M     84.0K     12.6M      1%
User          rw        332.1M    32.0K     327.4M     0%
```

17.2 Importing, exporting and deleting system configuration

System configuration can be exported to an external network resource using the command sequence:

```
save startup-config
sysadmin save configfile URL
```

where *URL* must be a valid URL with *ftp* or *scp* network scheme, pointing to a directory (**not a file**) of a reachable network resource.

This action will upload a file in Konuendo Archive Data (KAD) format to the given network resource. This file can be used to restore the configuration or to deploy it to other devices by means of a couple of mechanisms. One can copy the KAD file to a network service, as explained in *Managing device firmware*, and use the command:

```
sysadmin load URL
```

where *URL* contains the reference to the file just copied. The KAD file has notion of its own contents, so the *load* operation is able to tell, just looking inside the KAD file, if the file is a configuration dump file or an OS image one and behave accordingly.

These load operations **do not** modify the switch behavior. To make the action effective, a specific activation command must be issued:

- for O.S. Image loading:

```
sysadmin set default image x.y.z
```

- for configuration loading:

```
save startup-config
```

NOTE

The device must be rebooted for these operations to take effect.

Otherwise, the KAD file may be copied on a USB key prepared for system maintenance (see *Konuendo Total Device Management* section for details)

You can also save the configuration file in plain text format by appending the `plain` keyword to the above command:

```
sysadmin save configfile URL plain
```

The startup configuration file can be deleted from the non volatile memory using the

command:

```
A@SW-Std.03-00-CA>sysadmin delete startup-config
```

Chapter 18 - **REST Interface**

18.1 VDS Rail Software Platform REST Interface

Every VDS Rail device belonging to the Konuendo product line is equipped with a REST programmatic interface.

Representational State Transfer (REST) is a style of software architecture for distributed hypermedia systems such as the World Wide Web.

The subset included in the VDS Rail products permits the user interaction and the device configuration.

Each request, in form of a well defined URI, gets an XML answer through the HTTP protocol.

The XML nature of the results is a guarantee of easy parsing and understanding of the returned data.

The queries may be HTTP or HTTPS based.

The queries may be embedded in the user applications by means of well known web interaction libraries.

The request structure is:

```
http://device_ip_address:9901/exec_command?query=configuration_command
```

or, in case of HTTPS:

```
https://device_ip_address:9443/exec_command?query=configuration_command
```

where:

- **device_ip_address** is the IP address of the device to query and
- **configuration_command** is the command to send with the blanks substituted with the char '+'.
For example: "show version" will be written as `show+version`.

18.2 Examples

Here are some query examples performed on network devices from a web browser in order to have a quick look at the XML formatted results:

```
Q: http://10.200.2.100:9901/exec_command?query=show+spanning-tree
A:
<answer status="OK">
  <query>show spanning-tree</query>
  <bridge>
    <bridge_instance force_ver="RSTP" running_status="running.">
      <root_bridge>
        <bridge_id>32768/00:1d:b3:52:46:80</bridge_id>
        <root_port root_cost="200000">e2-0</root_port>
        <timers HT="2" MA="20" FD="15">
          Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
        </timers>
      </root_bridge>
    </bridge_instance>
    <bridge>
      <bridge_id prio="32768" mac-addr="a0:59:3a:80:08:f0">
        32768/a0:59:3a:80:08:f0
      </bridge_id>
      <timers HT="1" MA="10" FD="15">
        Hello Time 1 sec Max Age 10 sec Forward Delay 15 sec
      </timers>
    </bridge>
    <bridge_ports>
      <port>
        <name>e2-0</name>
        <port_id>128.13</port_id>
        <role>ROOT</role>
        <status>FWD</status>
        <path_cost>200000</path_cost>
        <port_uptime>105817</port_uptime>
        <port_mode>rstp,p2p,autoEdge</port_mode>
      </port>
    </bridge_ports>
  </bridge_instance>
</bridge>
</answer>
```

```

Q: https://10.200.2.100:9443/exec_command?query=show+interface e2-0 details
A:
<answer status="OK">
  <query>show interface e2-0 det</query>
  <device name="e2-0">
    <property type="integer" key="l2_txqlen">0</property>
    <property type="string" key="in_good_octets">32368097</property>
    <property type="string" key="l2_speed_force">Auto</property>
    <property type="string" key="in_jabber">0</property>
    <property type="string" key="l2_flags">
      UP BROADCAST RUNNING MULTICAST
    </property>
    <property type="string" key="out_fcs_errors">0</property>
    <property type="string" key="in_bad_octets">0</property>
    <property type="string" key="hist_128_255bytes">11314</property>
    <property type="string" key="in_undersize">0</property>
    <property type="string" key="single">0</property>
    <property type="string" key="l3_ipv4_netmask"/>
    <property type="string" key="in_broadcast">54133</property>
    <property type="string" key="in_fragments">0</property>
    <property type="string" key="l3_ipv4_brd"/>
    <property type="string" key="late">0</property>
    <property type="string" key="in_fcs_error">0</property>
    <property type="integer" key="nat_enabled">0</property>
    <property type="string" key="deferred">0</property>
    <property type="integer" key="l2_mtu">1500</property>
    <property type="string" key="hist_256_511bytes">50031</property>
    <property type="string" key="if_phys_status">up</property>
    <property type="string" key="in_pause">0</property>
    <property type="string" key="family">netdev</property>
    <property type="string" key="hist_1024_max_bytes">17647</property>
    <property type="string" key="in_multicast">96683</property>
    <property type="string" key="if_log_status">enabled</property>
    <property type="string" key="out_broadcast">17586</property>
    <property type="integer" key="default_cos">0</property>
    <property type="string" key="in_unicast">794</property>
    <property type="string" key="hist_64bytes">104430</property>
    <property type="string" key="l2_brd">ff:ff:ff:ff:ff:ff</property>
    <property type="string" key="out_octets">14497510</property>
    <property type="integer" key="l2_mdicross">0</property>
    <property type="string" key="excessive">0</property>
    <property type="string" key="out_multicast">21088</property>
    <property type="string" key="rx_packets">151610</property>
    <property type="string" key="l2_phycapab">
      10 100 1000 baseT Half/Full Duplex
    </property>
    <property type="integer" key="if_index">15</property>
    <property type="integer" key="l3_ipv4_prefixlen">0</property>
    <property type="string" key="hist_65_127bytes">7243</property>
    <property type="string" key="qos_trust">both</property>
    <property type="string" key="l2_dupmode">Full</property>
    <property type="integer" key="queues_sched">0</property>
    <property type="string" key="l3_ipv6_addr"/>
    <property type="string" key="if_type">switched Gigabit Ethernet</property>
    <property type="string" key="tx_packets">39576</property>
    <property type="string" key="in_rx_error">0</property>
    <property type="string" key="l2_speed">1000 Mb/s</property>
    <property type="integer" key="if_type_id">10</property>
    <property type="string" key="link_gw_addr"/>
    <property type="string" key="collisions">0</property>
    <property type="string" key="l2_dupmode_force">Auto</property>
    <property type="string" key="cos_override">Disabled</property>
    <property type="integer" key="uflags">769</property>
    <property type="integer" key="vlan_id">0</property>
    <property type="string" key="in_oversize">0</property>
    <property type="string" key="l2_qdisc">mq</property>
    <property type="string" key="out_unicast">902</property>
    <property type="string" key="l2_addr">a0:59:3a:80:08:f0</property>
    <property type="string" key="out_pause">0</property>
    <property type="string" key="if_name">e2-0</property>
  </device>
</answer>

```

```
<property type="string" key="l3_ipv4_addr"/>
<property type="integer" key="kflags">70211</property>
<property type="string" key="hist_512_1023bytes">521</property>
<property type="string" key="multiple">0</property>
<property type="string" key="l2_autoneg">Enabled</property>
</device>
</answer>
```

Chapter 19 - **TTCMP®**

The **Train Topology and Configuration Management Protocol** (TTCMP®) is a **Layer 2 protocol** that has been developed to give an answer to some important requirements of the modern Ethernet networks designed (or to be designed) for most of the trains.

The main aim of this protocol is to simplify the installation and the maintenance of active and passive network devices (switches, gateways and end devices in general) made by VDS Rail.

Basically, the user should be able to design the network layout using an off-line tool that runs on a standard workstation. This activity will produce a description (**TTCMP project**) of the whole network with details of network devices, their interconnections, their names and network addresses, the kind of end devices connected, their network address and the ports used, etc...

After the network has been physically built, the devices running the TTCMP® know the network topology in terms of nodes and interconnections.

A project can be loaded on one of the active devices by means of the TTCMP® and, from there, be automatically spread to all the others active devices. At this point each device is able to identify itself on the project (*matching*) and extract and apply its configuration details.

After this step, the TTCMP® goes on with its monitoring activities to assure that the current network topology is actually reflecting the TTCMP project design.

Project deployment, discovery and monitoring of the physical network topology are the **main TTCMP® activities**.

The advantages provided by this protocol can be summarized as:

- no need to manually configure each TTCMP® capable device, the configuration is extracted from the project file;
- real time check of the network topology, the active topology is compared with the topology described in the project and all the discrepancies are signaled;
- easy replacement of faulty devices, any faulty device can be replaced with brand new (not configured) ones and the TTCMP project file is then automatically received and applied by the new one;
- easy upgrade of the network project, a new release of the project can be loaded on one of the devices and immediately diffused to all the other active devices.

19.1 Operating principles

The TTCMP® is able to be executed at **layer 2** of the ISO/OSI stack, without the need of any layer 3 addressing (it can be executed on not configured devices). Moreover it runs

only on Ethernet physical layer.

The first level of discovery is based on the standard Link Layer Discovery Protocol (LLDP). More information on section 14.4 at page 131. By means of this protocol, each device discovers other LLDP devices that are directly connected to its ports and collects some information about them (name, description, management IP address and other).

The second level of discovery is performed spreading out all these information (and a lot of others), in a concise form, to all the other TTCMP® enabled devices of the network. The protocol is based on a series of messages that every TTCMP® capable device may broadcast and receive.

19.1.1 Terms and definitions

For the purposes of this chapter, the following terms and definitions apply:

- **Anomaly**
Inconsistency between the discovered topology and the one defined by the TTCMP project.
- **Confirmation**
The action of permanently store ("freeze") the configuration of a TTCMP enabled device on its permanent storage.
- **Consist**
Train set, rake of coaches, single vehicle or a group of vehicles which are not separated during ordinary train operation. For multi-coach operative mode, each coach is also a consist.
- **Consist network**
Communication network interconnecting communication devices in one consist.
- **End Device (ED)**
Unit connected to a Consist Network by means of a switch.
- **Fault**
Error caused by some device failure.
- **Matching**
Action performed by every TTCMP enabled device in order to identify itself inside the TTCMP project.
- **TTCMP project**
Description of all the consist network details.
- **Topology**
Cable interconnection and number of devices in the Consist network.

19.1.2 Project loading

The TTCMP® constantly monitors the network topology, even without any active project, registering device, link and end devices activity, appearance or disappearance. A TTCMP project reflecting the actual consist network design must be supplied in order to start the active phases of the protocol.

A project may be loaded by means of an USB-pen drive, as detailed on the chapter 29 (AutoUSB subsystem) on page 211, with the `project_diffusion` or `parametric_project_diffusion` actions, or using a TTCMP library based tool like **KonfStation** or **Kobserver** (ask VDS Rail representative for details about these tools and the TTCMP library). Once the project has been loaded, it becomes the *active one*⁴ and it is spread out to every TTCMP® enabled device.

The next phase is the *matching*: during this phase every device tries to identify itself on the project and, in case of success, it extracts and applies the proper configuration.

19.1.3 Confirmation

When all the switches are properly identified and configured, the network can be put in a different state called **confirmed**; switching to this state every switch stops trying to identify itself from the current topology and saves on flash its configuration details ready to be used at every next boot. After the network has entered the **confirmed** state, all the devices are fully configured and operative.

The **confirmed** state, that “freezes” the current configuration, is reached only if the device receives a specific TTCMP message or if the designated `sysadmin` command is executed on its shell (the details are described in the next sections).

19.2 Configuration

The use of TTCMP® is not mandatory; it does not affect the normal behavior of the network even if left enabled but not used. It runs on top of LLDP and both are enabled by default.

In order to turn off the TTCMP® issue:

```
A@SW-Std.81-85-20>no configure lldp ttcmp extensions enable
```

and verify the status of the subsystem with:

```
A@SW-Std.81-85-20>show lldp
Global LLDP information:
  LLDP protocol is running
  LLDP holdtime advertised 15 secs
  LLDP packets sent every 5 secs
  TTCMP extensions are disabled
A@SW-Std.81-85-20>show ttcmp
```

4 Each device can store many projects at the same time, but only one of them can be the *active project*.

```
Global TTCMP information:
TTCMP protocol version 2 extensions are disabled
TTCMP subsystem status is: idle
TTCMP packets signature is: disabled
TTCMP matching mode is: default
Consist identifier: undefined
Coach identifier: not set
Composition Head flag: not set
```

No active project

```
Device replacement:
Automatic project confirmation: enabled
Automatic cst-id election:      enabled
```

To switch it back on issue:

```
A@SW-Std.81-85-20>configure lldp ttcmp extensions enable
```

and verify the status of the subsystem with:

```
A@SW-Std.81-85-20>show lldp
Global LLDP information:
LLDP protocol is running
LLDP holdtime advertised 15 secs
LLDP packets sent every 5 secs
TTCMP extensions are enabled
```

```
A@SW-Std.81-85-20>show ttcmp
Global TTCMP information:
TTCMP protocol version 2 extensions are enabled
TTCMP subsystem status is: idle
TTCMP packets signature is: disabled
TTCMP matching mode is: default
Consist identifier: undefined
Coach identifier: not set
Composition Head flag: not set
```

No active project

```
Device replacement:
Automatic project confirmation: enabled
Automatic cst-id election:      enabled
```

NOTE

If LLDP is switched off with the command `no configure lldp run`, TTCMP® can no longer run.

19.2.1 Subsystem status

The status of the TTCMP® subsystem can be inspected by means of the following command:

```
A@SWITCH-3-B>show ttcmp
```

```
Global TTCMP information:
TTCMP protocol version 2 extensions are enabled
TTCMP subsystem status is: idle
TTCMP packets signature is: disabled
TTCMP matching mode is: default-fast
Consist identifier: undefined
Coach identifier: 3 (Coach Type 3 1st-2nd class slave coach without WC)
Composition Head flag: not set

Active project: 2002 Rev. 24 Serial F84CA7F1
Active context: 0
Multi-coach project: Yes

Nodes discovered: 10/2
Nodes matched on project: 10/2
Nodes with anomalies: 10

Topology status is: confirmed

Local TTCMP information:
Matched on project: Yes
Anomalies detected: Yes

Device replacement:
Automatic project confirmation: enabled
Automatic cst-id election: enabled
```

and this is the detail of each field:

- **TTCMP protocol version** reports the TTCMP® version the device firmware is using and whether or not the TTCMP® subsystem is enabled and running.
- **TTCMP subsystem status** reports what the TTCMP® subsystem is currently doing; possible value are:
 - transmitting project data when sending project data on the network;
 - receiving project data once the first project data packet is received;
 - new project data available once the last project data packet is received;
 - resetting active project when the reset action of the active project is in progress;
 - ignore ttcmp announces (this status is actually never used);
 - confirming topology when the confirmation action has been started from the local device;
 - resetting configuration when the reset action of the active project is in progress due to an error;
 - idle (none of the above).
- **TTCMP packets signature** reports whether or not sent TTCMP packets are signed and received ones are verified.
- **TTCMP matching mode** reports the matching algorithm in use:
 - default is the algorithm that tries to understand which device of the project is the local one selecting the one that better fits the actual connectivity status;

- `default-fast` is the one used for manually forcing a match (it works if the hostname of the device is set in advance before switching to this algorithm);
- `diffusion` is the one used in the deployment of the multi-coach projects;
- `no-end-devices` is like the default one but it does not take into account the end devices;
- `none` when no algorithm is selected.
- **Consist identifier** is the Consist UUID announced by the device; the value undefined means that the factory UUID of the device is used.
- **Coach identifier** reports the Coach identification number (if set) and the optional description of the coach.
- **Composition Head flag** reports whether or not the Composition Head flag is set in the local device (see Coupling solutions section for further information).
- **Active project** reports the name, revision and serial number of the active TTCMP project (if any).
- **Active context** is the identification number of the project context in use (this feature is a legacy of the TTCMP® version 1 and it is actually unused on current firmwares).
- **Multi-coach project** Yes if the project defines multiple coaches; No otherwise.
- **Nodes discovered** reports the number of nodes discovered in the topology against the total number of TTCMP devices provided by the project for the consist (or coach, if in multi-coach mode).
- **Nodes matched on project** reports the number of nodes announcing the status of *matched* against the total number of TTCMP devices provided by the project for the consist (or coach, if in multi-coach mode).
- **Nodes with anomalies** reports the number of nodes announcing an *error* or *anomaly*.
- **Topology status** confirmed if the configuration has been “frozen”; unconfirmed otherwise.
- **Matched on project** Yes if the device found a full match on the active project, P1 if the device found only a partial match, No otherwise; the optional string `delayed configuration` is printed when the device found a match very early but the configuration is not applied until the *matched* status is not announced by a significant number of other devices.
- **Anomalies detected** Yes in case of faults and/or anomalies; No otherwise.
- **Automatic project confirmation** enabled (by default) if the device can turn to the *confirmed* status without receiving a specific command; this feature is used when the replacement of a faulty device took place on a confirmed network (the brand new device automatically turn to confirmed if it finds a match on the project and all the other devices are already *confirmed*).

- **Automatic cst-id election** enabled (by default) if the device can change its Consist UUID making its own the one with lower value received from other devices belonging to the same consist; see Coupling solutions section for further information about this feature.

19.2.2 Topology

The whole network topology, as seen by means of the TTCMP® actions, can be examined with the following command:

```
A@SWITCH-3-B>show topology
```

SysName	Chassis-id	Coach-id	Port	Match	Err	Project	C
SWITCH-1-A	a0:59:3a:80:c2:20	1	e0-0	Yes	No	2002/24	Y
SWITCH-1-A	a0:59:3a:80:c2:50	1	e0-1	Yes	No	2002/24	Y
SWITCH-1-B	a0:59:3a:81:7a:70	1	ist0	Yes	No	2002/24	Y #
SWITCH-1-B	a0:59:3a:80:31:60	1	ist0	Yes	No	2002/24	Y
SWITCH-2-A	a0:59:3a:81:7a:90	2	ist0	Yes	Yes	2002/24	Y
SWITCH-2-B	a0:59:3a:80:27:40	2	e0-0	Yes	No	2002/24	Y
SWITCH-3-A	a0:59:3a:80:c1:c0	3	ist0	Yes	No	2002/24	Y
SWITCH-3-B	a0:59:3a:80:31:30	3	none	Yes	No	2002/24	Y #
SWITCH-4-A	a0:59:3a:80:31:50	4	ist0	Yes	No	2002/24	Y
SWITCH-4-B	a0:59:3a:00:01:02	4	e0-1	Yes	Yes	2002/24	Y

This is the meaning of each column:

- **SysName** is the hostname of the device as seen by the Operating System.
- **Chassis-id** is the device ID as announced by the TTCMP® subsystem; VDS Rail devices use the MAC Address as device ID for TTCMP®.
- **Coach-id** is the Coach identification number; this field is meaningful only for networks where a multi-coach project is in use, otherwise it can be set to 0.
- **Port** is the name of the interface of the local switch from which the announcements sent by the detected device are received.

The last four fields are meaningful only if a TTCMP project is active on the network:

- **Match** shows if the current switch is matched with the active project.
- **Err** is set to Yes in case of faults and/or anomalies on the device.
- **Project** is the name of the active project followed by its revision number.
- **C** is Y when the topology is confirmed, N otherwise.

If present at the end of a topology line, the hash character indicates that there is a mismatch between the Consist ID and/or confirmation status announced by the device and the values of the local one. The actual kind of mismatch can be inspected looking at the topology details:

```
A@SWITCH-3-B>show topology detail
```

```

....
long output
....
=====
System name: SWITCH-4-B
ChassisID: a0:59:3a:00:01:02
Description: VDS Train switch
Device capabilities: TTCMP bridge/switch
Management address: 10.0.5.159
Consist-id: 48d11d08-332e-41ff-b147-33c89a641bd8
Coach-id: 4
Active project: 2002 rev. 24
Project matching status: Yes
Anomalies detected on this device: Yes
Topology confirmed on this device: Yes
Conflicts detected on this device: None
Device seen on local interface e0-1

Port  Status  STP  Type  Dev  Mac-addr          Err  Code
-----
e0-0  Up        fwd  ttcmp  sw   a0:59:3a:80:31:30  N
e0-1  Up        fwd  ttcmp  sw   a0:59:3a:80:31:60  N
e0-2  Up        fwd  ttcmp  sw   a0:59:3a:80:31:50  N
e0-3  Up        fwd  ttcmp  sw   a0:59:3a:80:31:50  N
e1-0  Down      blk                Y      2
e1-1  Down      blk                Y      2
e1-2  Down      blk                Y      2
e1-3  Down      blk                Y      2
e1-4  Down      blk                Y      2
e1-5  Down      blk                Y      2
e1-6  Down      blk                Y      2
e1-7  Up        fwd  mac    ed   a0:36:9f:d8:82:4e  Y      1
e2-0  Down      blk                Y      2
e2-1  Down      blk                Y      2

```

The meaning of each field:

- **System name** is the hostname of the device as seen by the Operating System.
- **ChassisID** is the device ID as announced by the TTCMP® subsystem; VDS Rail devices use the MAC Address as device ID for TTCMP®.
- **Description** is a textual description of the device; at present time the possible values are:
 - VDS Konuendo Vehicular Gateway;
 - VDS Train switch;
 - VDS Train Backbone router;
 - VDS Integrated TBN;
 - VDS Coupling Node;
 - VDS generic device.
- **Device capabilities** are the capabilities announced by the device; at present time the possible values are:
 - TTCMP router/gateway;
 - router/gateway;
 - TTCMP bridge/switch;

- bridge/switch;
- generic network device.
- **Management address** is the IP address of the device's management interface.
- **Consist-id** is the Consist UUID announced by the device (can be unknown).
- **Coach-id** is the Coach ID value set on the device or None⁵.
- **Active project** : is the name of the active project followed by its revision number.
- **Project matching status** Yes if the device found a full match on the active project, P1 if the device found only a partial match, No otherwise.
- **Anomalies detected on this device** Yes if the matched device found any anomaly in regard to the topology described in the project, No otherwise.
- **Topology confirmed on this device** Yes if the configuration of the matched device has been confirmed, No otherwise.
- **Conflicts detected on this device** are the issues detected by TTCMP® and announced by the device; at present time the possible values are None or a combination of one or more of the following:
 - Coach ID: the device detected a conflict between its own Coach ID value and the one announced by the other devices belonging to the same coach;
 - Composition Head: the device detected that the Composition Head flag is also set and announced by other coaches;
 - Lock CRC: the device detected that its own Composition Lock CRC differs from the one announced by the other devices belonging to the same coach;
 - Lock Position: the device detected that its own Composition Lock Position value differs from the one announced by the other devices belonging to the same coach;
 - Lock Orientation: the device detected that its own Composition Lock Orientation differs from the one announced by the other devices belonging to the same coach;
 - Circular Reference: the device is not able to detect the current Train Composition.

Except for the last one, all these conflicts are meaningful only for networks where a multi-coach project is in use.

- **Device seen on local interface** is the interface of the local device where the remote device announcements are received.

This is the meaning of each column of the table reporting information about the

⁵ The value not available can be displayed on devices whose firmware does not support multi-coach projects.

interfaces of the remote device:

- **Port** is the name of the interface.
- **Status** is the interface link status (Up or Down).
- **STP** is the port status as seen from the Rapid Spanning Tree Protocol:
 - fwd when *forwarding*;
 - lrn when *learning*;
 - blk when *blocking*;
 - dis when *disabled*.
- **Type** is the kind of device connected to this port, from a network point of view; it can be:
 - ttcmp if the connected device is TTCMP® capable;
 - ll dp if the connected device is LLDP capable;
 - mac for any other end device.
- **Dev** gives some hint about the connected device; possible values are gw (Gateway), sw (Switch), hb (Hub) and ed (end device).
- **Mac-addr** is the MAC address of the connected device (if any); it can happen that many devices seem to be directly connected to the interface (i.e. using a hub or an unmanaged switch); when they are all end devices the value ff:ff:ff:ff:ff:ff is reported.
- **Err** is set to Y in case of anomalies; in that case **Code** specifies the type of error with a value that can be:
 - 1 when a device is detected but it is not present in the project;
 - 2 when the project provides for a device on the interface but no device has been detected;
 - 4 when the detected device does not match the type provided by the project (i.e. an end device is expected but a switch has been detected);
 - 8 when the connected device is TTCMP capable but it is not the one provided by the project; it can happen in three cases:
 - the connected device is unconfigured or configured manually (no project);
 - the project on the connected device is different from the local one;
 - the by-pass of the actual connected device is closed; this can be temporary if the connected device is booting up or permanent when it is switched off due to a fault.

19.2.3 Project handling

The list of TTCMP projects loaded on the device can be shown with the command:


```
A@SW-Std.03-00-DC>sysadmin show projects
```

```
test_release:1
test_release:2 [current]
```

```
Free projects space: 327.3M
```

The [current] tag, when present, indicates that the project is the one currently *active*.

As already stated a project may be loaded and spread out to every TTCMP® enabled device by means of an USB-pen drive or using a TTCMP library based tool.

This is an example of how the output of `show topology` changes after the diffusion of the project:

```
A@SW-Std.03-00-DC>show topology
```

SysName	Chassis-id	Coach-id	Port	Match	Err	Project	C
SW-Std.03-00-F8	a0:59:3a:03:00:f8	None	e2-0	No	No	/0	N
SW-Std.07-00-CA	a0:59:3a:07:00:ca	None	e2-0	No	No	/0	N
SW-Std.03-00-DC	a0:59:3a:03:00:dc	None	none	No	No	/0	N
SW-Std.03-00-DA	a0:59:3a:03:00:da	None	e2-0	No	No	/0	N
SW-Std.03-00-FC	a0:59:3a:03:00:fc	None	e0-2	No	No	/0	N
SW-Std.03-00-FA	a0:59:3a:03:00:fa	None	e2-0	No	No	/0	N
SW-Std.03-00-D8	a0:59:3a:03:00:d8	None	e2-0	No	No	/0	N

....
project deployment by an external TTCMP library based tool
....

```
A@SW-Std.03-00-DC>show topology
```

SysName	Chassis-id	Coach-id	Port	Match	Err	Project	C
sw9	a0:59:3a:03:00:da	None	e2-0	Yes	No	test_release/2	N
sw8	a0:59:3a:03:00:d8	None	e2-0	Yes	No	test_release/2	N
sw10	a0:59:3a:03:00:dc	None	none	Yes	No	test_release/2	N
sw24	a0:59:3a:03:00:f8	None	e2-0	Yes	No	test_release/2	N
sw20	a0:59:3a:07:00:ca	None	e2-0	Yes	No	test_release/2	N
sw25	a0:59:3a:03:00:fa	None	e2-0	Yes	Yes	test_release/2	N
sw26	a0:59:3a:03:00:fc	None	e0-2	Yes	No	test_release/2	N

A further investigation will give us hints about the error reported on the last but one row:

```
A@sw10>show topology detail
```

```

....
long output
....
=====
System name: sw25
ChassisID: a0:59:3a:03:00:fa
Description: VDS Train switch
Device capabilities: TTCMP bridge/switch
Management address: 192.168.2.125
Consist-id: 48d11d08-332e-41ff-b147-33c89a641bd8
Coach-id: None
Active project: test_release rev. 2
Project matching status: Yes
Anomalies detected on this device: Yes
Topology confirmed on this device: No
Conflicts detected on this device: None
Device seen on local interface e2-0

Port  Status STP  Type  Dev Mac-addr          Err Code
-----
e0-0  Up      blk             Y      1
e0-1  Down    blk             N
e0-2  Down    blk             N
e0-3  Up      fwd ttcmp sw  a0:59:3a:03:00:f8 N
e1-0  Down    blk             N
e1-1  Down    blk             N
e1-2  Down    blk             N
e1-3  Down    blk             N
e1-4  Down    blk             N
e1-5  Down    blk             N
e1-6  Up      fwd mac      ed 00:40:8c:ac:60:39 N
e1-7  Down    blk             N
e2-0  Down    blk             N
=====
....
long output
....

```

An error is reported on port e0-0 with value 1; this means that on that interface a device is connected, but its presence is not provided by the project. As in this case, erroneous or malicious connections to the network can be easily discovered.

Other notable TTCMP project related commands are:

```
A@SW-Std.03-00-DC>sysadmin reset global project
```

that sets all the TTCMP enabled reachable devices in the network to a “no active project” status (all the devices discard the configuration obtained from the project) and makes the TTCMP® subsystem going on with the monitoring activities only.

The same action can be limited to the current switch only:

```
A@SW-Std.03-00-DC>sysadmin reset local project
```

NOTE

Until all the devices composing the Network are not started and operational, the automatic discovery process is somewhat more cautious because it has to take into account the presence of some potential closed hardware bypass.

As soon as the totality of the devices is operational, the discovery process is expedited.

19.2.4 Project confirmation

The TTCMP project confirmation action (configuration “freezing”) can be triggered from one of the TTCMP enabled switches of the network by means of the command:

```
A@sw10>sysadmin set topology status confirmed
```

The permanent change to the *confirmed* status is acknowledged by the LED n. 3 on the device that turns steady green.

19.2.5 Operations feedback

Some information about the status of the TTCMP® subsystem are visually presented to the user by means of some of the Consist Switch’s LEDs:

- LED n. 2
 - off
TTCMP disabled
 - slow blinking green
TTCMP running but no match found (either no active project or switch trying to match)
 - fast blinking green
The switch is matched and is waiting for the other switches to match; configuration will be loaded in seconds
 - steady green
The switch is matched and the configuration applied
 - steady red
Fault or anomaly detected
- LED n. 3
 - steady green
The switch status is *confirmed*
- LED n. 4

- steady red

The status of the switch is *confirmed* but it cannot be configured since the project does not define any device matching the current *hostname*

19.3 Commands summary

The following is the reference list of TTCMP® available commands from the CLI. Parameter names are in all-caps italic. Optional arguments are enclosed in square brackets.

All commands, except for the show group, support the no-form.

```
configure lldp ttcmp extensions enable
```

Turn on/off the TTCMP® subsystem.

```
configure ttcmp auto confirm enable
```

Turn on/off the automatic TTCMP® confirmation of new devices inserted into an already confirmed network.

```
configure ttcmp auto cst-id enable
```

Turn on/off the automatic selection of the announced Consist UUID among the ones announced by the devices belonging to the same consist.

```
show ttcmp
```

Display information about TTCMP protocol.

```
show topology [ detail [ device <NAME> ] ]
```

List a table where all the TTCMP® capable devices in the network segment are shown together with their TTCMP® information.

The optional `detail` parameter produces a verbose output.

The produced information is limited to a specific switch if a `device` clause is specified.

```
show counters protocol ttcmp
```

Display various counters related to each different TTCMP® message type.

```
configure snmp-server traps-ttcmp <TRAP-TYPE>
```

Set one or more SNMP trap to be emitted by the device.

Currently the only TTCMP® trap defined is `ttcmp-anomaly`.

```
clear counters protocol ttcmp
```

Clear all the TTCMP® counters.

```
clear topology
```

Clear all the topology information gathered by the device.

```
sysadmin show projects
```

List all the TTCMP projects loaded in the device.

```
sysadmin set global project <PROJECT-ID>
```

The TTCMP project identified by **PROJECT-ID** is selected, spread to all other TTCMP capable devices and set as the current *active* project.

```
sysadmin set topology match-mode default|default-fast|diffusion|none
```

Select and activate a TTCMP® matching algorithm.

```
sysadmin set topology status confirmed
```

Device configuration is permanently saved on flash so that no subsequent matching activity is needed.

```
sysadmin reset global project
```

Disable the current *active* project on every switch in the network.

```
sysadmin reset local project
```

Disable the current *active* project on the local switch only.

```
sysadmin set cst-id [uuid/]<VALUE>
```

Set the consist identifier (Consist ID).

If the optional **uuid/** prefix is specified then **VALUE** is parsed as a UUID. Otherwise **VALUE** is supposed to be plain text and the ASCII values of its composing characters are stored.

```
sysadmin reset cst-id
```

Reset the consist identifier (Consist ID).

The factory UUID becomes the value announced by the device.

```
sysadmin delete project <PROJECT-ID>
```

The specified TTCMP project is deleted from the device.

Chapter 20 - **Coupled-Switch[©]** mode

20.1 Introduction

The **Coupled-Switch[©]** mode is an advanced configuration option that allows the aggregation of two switches into a single switching system named **Coupled-Switch[©]** that can then be connected to other **Coupled-Switches[©]** resulting in a ladder physical topology.

A **Coupled-Switch[©]** is perfectly suitable to provide a solution to the constraints commonly found on trains allowing the design of loop-free topologies overcoming the limitation of classical spanning tree designs and providing a deterministic sub-second recovery time in case of single or multiple link or switch failure.

Coupled-Switch[©] mode offers the following key advantages:

- no need of spanning-tree protocol
- deterministic sub-second recovery time independent from network size
- no need for any special adapter or special cable for connecting the pair
- handle multiple concurrent switch/cable failures
- doubled bandwidth available over the ladder backbone
- integrated with the TTCMP management protocol allows the replacement of a faulty member with the automatic configuration of the new device

Since the switch pair share their data plane but not the control plane both members remain autonomous, use their own CPU, have their own management IP address and are viewed as separate devices on the TTCMP topology tables.

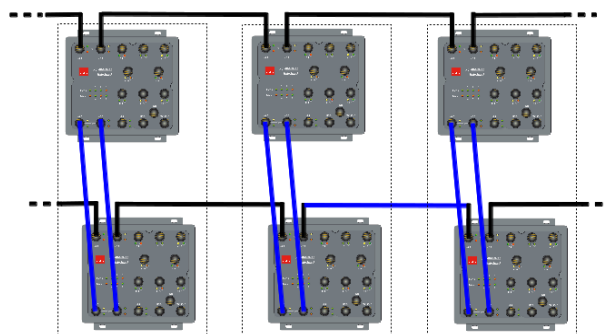
20.2 How to connect the switches

The ports used to form a **Coupled-Switch[©]** pair are fixed.

Ports e0-2 and e0-3 are used to link together the switch pair: port e0-2 on master device should be connected with port e0-2 on slave device, and port e0-3 on master with port e0-3 on slave. These Ethernet ports are aggregated together and they are used as a backplane to connect the two switching engines.

Ports e0-0 and ports e0-1, on both members, are used to connect to another **Coupled-Switch[©]**. Each of these ports, commonly referred as “backbone” ports, is aggregated with the correspondent port on the partner device forming two distinct distributed groups, one for each direction of the ladder.

The resulting structure is depicted in the following drawing.



In order to guarantee the maximum level of resiliency, also in case of a power failure, the backbone lines can be protected using a hardware bypass between the two backbone ports.

20.3 Coupled-Switch[®] Coupling mode configuration

The only command required to enable the **Coupled-Switch[®]** mode is the following:

```
configure coupled-mode role master|slave [left-only|right-only]
```

When this command is run, some changes on the active configuration are automatically executed:

- structural ports e0-0, e0-1, e0-2 and e0-3 are removed from spanning tree control
- ports e0-2 and e0-3 are linked together and the ist0 (Inter-Switch Trunk) interface is created
- backbone ports e0-0 and e0-1 are disabled, waiting for members synchronization
- a synchronization protocol is started on the Inter-Switch Trunk.
- as soon as the coupled partners are synchronized, or after a predefined timeout period, the backbone ports are re-enabled.

All the information related to the **Coupled-Switch[®]** mode status can be obtained issuing the `show coupled mode` command, like in the following example:

```
A@SW-Std.03-00-DC>show coupled-mode

Coupled mode is enabled
Node type: middle/master
Backbone aggregation protocol: static
```

Inter-switch connections

```
=====
Partner address: a0:59:3a:03:00:fa
Connection 0 (e0-2) is up
Connection 1 (e0-3) is up
Synchronization status: SYNC_OK
```

Distributed ports

```
=====
(L - Local port   P - Partner port)
  name      type      line      status
-----
L  e0-0     backbone  up        active
P                up        active
L  e0-1     backbone  up        active
P                up        active
```

The most important field is the synchronization status, SYNC_OK means that the **Coupled-Switch**® is fully operational and that it can start operating as a single switching engine.

The possible values for this field are reported on the following table:

Value	Meaning
SYNC_PRE0	Initial state, ready to start the synchronization protocol.
SYNC_SO	First step of the initial handshake, waiting for partner acknowledge.
SYNC_CHECK	Some trouble on IST links or on partner member.
SYNC_CHECK_SENT	This member is attempting to reconnect with its partner.
SYNC_OK	Fully operational.

If we consider a **Coupled-Switch**® pair as a reference point, then it may be connected with other **Coupled-Switch**® pairs on the right and/or on the left side.

A right connected Coupled-Switch® have its backbone ports (either *e0-1* or *e0-0*) connected with the *e0-0* ports of the reference Coupled-Switch® pair.

A left connected, on the other hand, have its backbone ports (either *e0-1* or *e0-0*) connected to the *e0-1* ports of the reference.

If no **Coupled-Switch**® pairs will be connected on the left of the reference, it shall be configured as “coupled right” (adding the `right-only` option in the command). Vice-versa if no other pairs will be connected on its right side, the reference shall be configured as “coupled left” (adding the `left-only` option in the command).

20.4 Backbone aggregation protocol

The backbone ports *e0-0* and *e0-1* of each coupled-switch members are aggregated with their respective ports of the partner device in a distributed group. Both members

are active and the backbone traffic is distributed across them using a hashing function of the source and destination MAC addresses.

The status of these ports is also reported by the `show coupled` command.

These distributed groups can be statically or dynamically aggregated. By default the aggregation protocol for the backbone lines is “static”, meaning that the bonding/unbonding of group members relies only on their physical link status. This could lead to some reachability issues when a **Coupled-Switch[®]** member, protected with a hardware bypass, is switched off since on its neighbors the backbone link remain up.

In these cases the **CMAP** protocol (*Coupled Mode Aggregation Protocol*) can be enabled on backbone lines. The main role of this protocol is to detect and resolve the issue described on the previous section. Each **Coupled-Switch[®]** have a unique identifier, automatically defined by combining the MAC addresses of both members. This identifier is transmitted inside the protocol payload allowing neighbors to detect the status of asymmetrical connection, due to the bypass action, and set to stand-by the backbone port connected through the bypass.

Beside of this the CMAP protocol is also capable to detect connectivity issues on the rx/tx path that does not cause a physical link failure.

The CMAP protocol is enabled by issuing, on both members, the command:

```
A@SW-Std.03-00-DC>configure coupled-mode cmap enable
```

When enabled, protocol status information is reported on the output of the `show coupled` command:

```
[...]
CMAP status
=====
backbone port e0-0: port is down
backbone port e0-1: running
```

Note that if the `right-only` or `left-only` option is not set, the protocol is started on both sides of the **Coupled-Switch[®]**.

20.5 Coupled-Switch[®] Dual home

A high performance end-device, e.g. a media streaming server, may need to be connected to the network with redundant links for high performance and high-availability reasons.

In the **Coupled-Switch[®]** mode two ports can be aggregated not only locally, taking advantage of the Link Aggregation features (cfr. Link Aggregation), but also in a distributed way, so that the end-device could benefit of both the redundancy and the aggregation of the two links.

The server is said to be “dual homed”.

Statistically, the traffic along the ladder in a **Coupled-Switch[®]** topology is sent on both rails so the total throughput of such a topology is more or less 2 times the nominal speed of the ports.

Dual homed end-devices may benefit from this fact being able to actually deliver data at the full speed through the two links, maintaining the ability to send all the traffic on one of the two links in case of a fault of the other line.

The member ports of a dual home connection must be the same on the two switches of a **Coupled-Switch[®]** pair. Not all the ports may be part of a dual home pair but only *e1-0*, *e1-1*, *e1-2*, *e1-3*, *e1-4*, *e2-0* and *e2-1* (if available).

The type of aggregation realized by default on the dual home pair is “static, balanced”, so it can be used with any kind of balanced packet sending set in the end-device, as for example the “balance-xor” mode in the Linux bonding driver.

The command to set a port in dual home mode is:

```
A@SW-Std.03-00-CA>configure interface e2-0 switchport dual-home
```

This command must be input on both the switches of a **Coupled-Switch[®]** pair.

During the setup phase a LED will blink to signal the synchronization phase. As long as the two ports are joint, non more LED flashing will happen.

NOTE

Due to the nature of the aggregation mode used, the *e0-0* and the *e0-1* ports, if enabled, may be used as a dual home pairs. So, for example, in the above picture, instead of a **Coupled-Switch[®]** pair on the *e0-0* ports couple of the rightmost pair, a dual homed media streaming server could be connected. The same could be done with the *e0-1* ports couple of the leftmost **Coupled-Switch[®]** pair.

20.5.1 Support for LACP protocol on dual home ports

Beside being configured for static aggregation, dual home ports can also be configured for dynamic negotiation with the partner device by means of the standard LACP protocol.

However, since Coupled-Switch[®] architecture does not support the spanning-tree protocol, unbounded members are automatically moved to STP blocking state in order to avoid Ethernet loops.

We refer to LACP on dual-home ports as DLACP (Distributed LACP), due to the *distributed* nature of this kind of setup.

LACP can be enabled while configuring the dual home ports using the following command on both the coupled members:

```
configure interface <if-name> switch port dual-home lacp [fast | slow]
```

The DLACP subsystem behavior when the coupled members are out-of-sync can be set with the command:

```
configure coupled-mode dlacp mode-on | mode-sync
```

It specifies whether or not LACP packets should be transmitted/received if the



Coupled-Switch® is out-of-sync. The default mode is `mode-sync` (packets *not* transmitted/received of out-of-sync).

This setup aim to force member unbonding on partner device in case of missing coupled synchronization.

The `show coupled-mode` command reports all the information related to the status of the DLACP subsystem.

As depicted in the following example, if the DLACP protocol succeeds in negotiating the aggregation with the partner, the status of the link is reported as `bndl`, otherwise it is reported as `standby`.

```
show coupled-mode
Coupled mode is enabled
Node type: middle/master
Backbone aggregation protocol: CMAP
DLACP global mode: on

Inter-switch connections
=====
Partner address: a0:59:3a:80:c2:10
Connection 0 (e0-2) is up
Connection 1 (e0-3) is up
Synchronization status: SYNC_OK

Distributed ports
=====
(L - Local port   P - Partner port)
name      type      line      status
-----
L e0-0      backbone  up        active
P          backbone  up        active
L e0-1      backbone  up        active
P          backbone  up        active
L e1-3      dual-home down       -
P          dual-home down       -
L e1-4      dual-home down       -
P          dual-home down       -
L e2-0      dual-home up         standby
P          dual-home up         bndl

CMAP status
=====
backbone port e0-0: running
backbone port e0-1: running
```

Chapter 21 - **Coupling solutions**

This chapter describes the solutions available on Consist Switch for supporting the joining of two or more trains together.

These solutions are referred as “TTCMP for Backbone” or **T4B**.

21.1 **TTCMP for Backbone (T4B)**

TTCMP for Backbone (T4B) takes advantage of the capabilities offered by the TTCMP protocol to provide a solution to one of the most common requirements of the railway operators: *handling the coupling/decoupling of two or more trains*.

21.1.1 **TCDS**

The **T4B** main component is the *TTCMP Train Composition Discovery Service (TCDS)*; this is the service in charge of the automatic discovery of the active **consists** composition.

Consists are sequences of indivisible items (cars, coaches, trains).

Since each consist has its own *Ethernet Consist Network (ECN)*, the result of a new composition will always be a new bigger or smaller ethernet network.

T4B provides two different modes for interconnecting ECN's:

- plain;
- routed.

In **plain** mode all the ECN's are directly connected and share a common broadcast domain and IP network for each one of the defined VLANs.

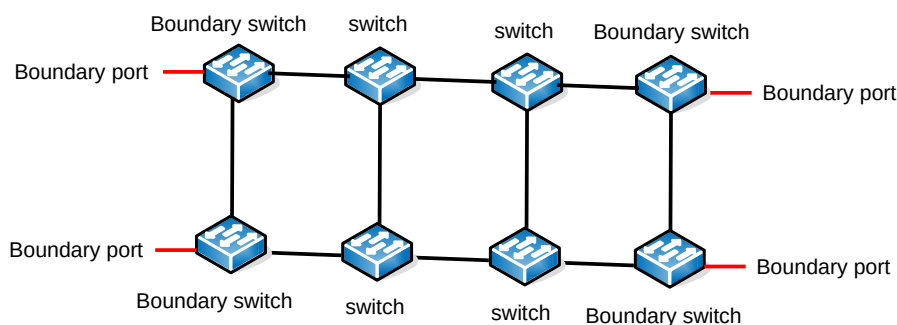
In **routed** mode all the ECN's are logically separated into different IP networks so inter-consist traffic must be somehow routed at IP level.

The service works properly if the following conditions are enforced on switches configuration and train topology:

- the train composition is made of a linear sequence of consists;
- all the switches in the same consist shares a common consist identifier (**cst-id**);
- ethernet ports connecting different consists are configured as “**boundary ports**”;
- each boundary port must be statically associated with a specific direction **DIR1** or **DIR2** that will be used to understand each consist orientation respect to the current composition reference direction;
- the boundary port attribute can be configured on standard access ports associated to any VLAN as well as on ports configured as trunks;

- a switch with one or more ports configured as boundary port is named **"boundary switch"**.

The following picture summarizes the key elements and their roles:



The TCDS service can be started on any node using the command:

```
configure tcds enable
```

TCDS service startup is mandatory in all the boundary switches. Optionally it can also be enabled in all the other switches in order to have the composition information available in every node. In non boundary switches TCDS does not play any active role but just listens and collects the messages exchanged by the boundary switches.

21.1.2 Consist boundaries

As stated before, the ports that can be used for coupling purposes constitute the boundary of a consist and must be configured accordingly using the following interface command:

```
configure interface NAME tcds boundary-port direction dir1 | dir2
```

If IEC61375-2-5 compatibility is required, the boundary ports can also be labeled with one of the two supported lines (A or B) using the following interface command:

```
configure interface NAME tcds line-id A | B
```

Depending on the coupling logic used (plain or routed), boundary ports could also be configured as access ports for a specific VLAN or as trunk ports capable of carry traffic for multiple VLANs.

Note that boundary ports only inhibit TTCMP packets from entering the consist while they do not interfere in any way with all the other network traffic entering or exiting the port.

This behavior can be altered by the command

```
configure tcds ttcmp forwarding
```

that allows all the TTCMP traffic to enter a boundary port and flow through the consist while preserving the correct handling of the packets. This is useful, for example, when the complete TTCMP network topology is required to be known by any device,

regardless of the consist (i.e. in a train configured using a multi-coach project).

All the consist switches must be configured with the same consist-id since they are all members of the same "entity". Both the available formats (*string* and *uuid*) for configuring the consist-id can be used for this purpose.

21.2 Local consist discovery

The first activity performed by the TCDS service at startup is the local consist properties discovery (boundary switches, boundary ports and associated directions).

This activity is based on the periodical transmission of Ethernet messages from each of the configured boundary switches. These messages are called "*Internal CST announcements*". They are transmitted within the consist and do not cross its boundaries.

Each node with an active TCDS service collects the data carried by these announcements and builds a dynamic representation of the local consist.

The information describing the local consist can be inspected using the command:

```
show tcds local
```

that reports the local consist status as follows:

```
Using automatic consist-id election
  Consist-id election process is completed
  Selected consist-id is: 00d10061-2498-4762-9783-5a2f41855d55

Boundary switches discovered inside this consist: 4
Active TCDS mode is: plain
Forwarding of TTCMP traffic on boundaries is: enabled
Discovery position: current 3 - locked 0 (unlocked)
Device name: B-SW-8 - Type: boundary switch
Sending internal CST announcements at slow frequency

Gathered neighbour information for direction DIR2:
  CstUUID: 32fe4b1b-2599-4283-addd-1e07de7232ae
  Bsw data: A(e0-1) ---> C-SW-10 A(e0-0)

Local consist boundary status:

Direction DIR1
=====
sw-name      port  ID  status  neighbour cst-id
-----
B-SW-6       e0-0  A   up-b-fwd  5bb9e0f2-7e08-49e9-9abb-eea3ab5c4495
B-SW-7       e0-0  B   up-b-fwd  5bb9e0f2-7e08-49e9-9abb-eea3ab5c4495

Direction DIR2
=====
sw-name      port  ID  status  neighbour cst-id
-----
B-SW-8       e0-1  A   up-b-fwd  32fe4b1b-2599-4283-addd-1e07de7232ae
B-SW-9       e0-1  B   up-b-fwd  32fe4b1b-2599-4283-addd-1e07de7232ae

Train composition table:
  CRC 0x048b4f79 - stable since 3 hours 43 minutes 5 seconds
```

```
Advertised CST-IDs:
DIR1 5bb9e0f2-7e08-49e9-9abb-eea3ab5c4495
OWN  00d10061-2498-4762-9783-5a2f41855d55
DIR2 32fe4b1b-2599-4283-addd-1e07de7232ae
```

```
Dropped External CST announcements: 0
Dropped Internal CST announcements: 0
```

21.2.1 Automatic consist-id election

In all the situations where the user does not want to deal with consist-id assignment, an automatic consist-id election is applied. If there is no consist-id configured, the boundary switches automatically use the UUID value burned into their EEPROM for their internal CST announcements. The lowest UUID value learned from the internal CST announcements will then be used as the common consist-id.

NOTE

In order to operate properly, this election mechanism needs that none of the boundary switches is configured with a user defined consist-id.

The internal CST announcements are initially transmitted at a “fast” frequency, to speed-up the discovery process, that is turned to “slow” when the discovery/election process is completed.

To prevent a complete reelection process at each boot, the information about the elected consist-id is permanently saved on each device. In case of a boundary switch replacement, the election process is automatically restarted if an internal CST announcement with a lower consist-id is received.

21.3 Coupling local detection

Coupling begins when one of the boundary ports transitions from down to up. Neighbor detection is based on standard LLDP packets exchanged between boundary ports.

These packets report to the connected neighbor some local consist information; if this information is missing the coupling is not considered valid and an error condition is associated to the boundary port. In the following example the code 'E' on the “status” column means “**Error**” (the others are the link status, up or down, and the spanning tree port status, blk or fwd).

```

...
...
Direction DIR2
=====
sw-name          port    ID   status  neighbour  cst-id
-----
SW-C             e2-0      up-E-fwd
SW-D             e2-0      up-E-blk

```

Otherwise, when the received information is correct, the `show tcds local` command reports the consist-id of the connected consist as depicted in the following example:

```

...
...
Direction DIR2
=====
sw-name          port    ID   status  neighbour  cst-id
-----
SW-C             e2-0      up-b-fwd  2bd16b36-bbed-4521-857f-b883078db682
SW-D             e2-0      up-b-blk  2bd16b36-bbed-4521-857f-b883078db682

```

In this case the 'b' code means "**B**oundary ok".

21.4 Composition discovery

While coupled, each boundary switch periodically transmits from its connected boundary ports an "*External CST announcement*". These announcements are forwarded to all the switches of the network and allow the receivers to build a complete representation of the active train composition.

The core of the discovery logic is the **Train Composition Table**. This table is filled with the information contained in the "*External CST announcements*" (received from all the consists) and represents the current composition as an ordered and oriented consists list, starting from the extremity with the lowest consist-id and moving towards the opposite side.

Consists are numbered starting from the end side with the lowest consist-id, to which is assigned the number 1. A reference direction is defined so that DIR1 points in the direction of the consist number 1.

The current active composition can be retrieved using the command:

```
show tcds composition
```

which reports not only the list of discovered consists but also the overall status of the discovery process as depicted in the following picture:

```

Discovery information
-----
TCDS discovery is complete
Active TCDS mode is plain
TTCMP forwarding is enabled
Number of discovered consists: 5
CRC computed on local table: 0x048b4f79

```


Active Train Composition							
CstNum	Cst-id	Coach ID (Description)		Orient.	CRC		
					H	Discovery	Lock
	1	12345678-9abc-def0-1234-56789abcdef0	2 (---)	Inverse	0x048b4f79	N	0 N 0x00000000
	2	32fe4b1b-2599-4283-addd-1e07de7232ae	5 (---)	Inverse	0x048b4f79	N	0 N 0x00000000
*	3	00d10061-2498-4762-9783-5a2f41855d55	4 (---)	Inverse	0x048b4f79	N	0 N 0x00000000
	4	5bb9e0f2-7e08-49e9-9abb-eea3ab5c4495	3 (---)	Inverse	0x048b4f79	N	0 N 0x00000000
	5	4127a97a-f4d7-4ad4-8f2b-ce379e430d89	1 (---)	Inverse	0x048b4f79	N	0 N 0x00000000

In every active consist a CRC for the active train composition table is computed and transmitted to all the other consists by means of the External CST announcements. When all the consists share the same CRC value, the discovery process can be considered complete.

The consist we are connected to is reported with a * character at the beginning of the line.

21.5 TCDS routed mode

While TCDS plain mode is intended to support only the discovery of the current coupled consists without playing any active role in controlling inter-consist network traffic (the result is always a single Layer-2 Ethernet domain), TCDS routed mode introduces a Layer-3 separation among the interconnected consists. This means that the IP traffic between hosts on different consists must be *routed* in order to reach its destination. This behavior, together with an automatic IP address translation, allows to connect together several trains that share the same IP address space and makes local consist end-devices accessible from the other consists using “train wide” translated addresses.

This mode mimics the standard IEC-61375-2-5 logic without requiring a dedicated cabling for the train-wide Ethernet backbone.

When routed mode is enabled, the TCDS sub-system handles traffic that flows among the different consists using a user-defined VLAN (the *backbone-vlan*). At least one of the switches must be configured with the role of “virtual TBN” (vTBN) in order to provide the automatic routing and R-NAT (Railway NAT) services.

The steps required to properly configure and enable the TCDS service in routed mode are summarized below:

- configure TCDS for routed mode on all the switches

```
configure tcds mode routed
```

- create and configure the “backbone” VLAN on all the switches

```
configure vlan 100 description "backbone vlan"
configure tcds backbone-vlan 100
```

- on boundary switches, configure all the boundary ports as *access ports* for the selected backbone VLAN

```
configure interface e1-1 switch port mode access vlan 100
configure interface e1-1 tcds boundary-port direction DIR1
```

- on the nodes that should act as virtual TBN, create the SVI interface associated to the backbone VLAN; this interface will be automatically configured for IP routing and NAT after each coupling/uncoupling event

```
configure interface vlan100
configure interface vlan100 enable
configure tcds vtbn enable
```

- configure an IP address on the SVI interface facing the local consist network; the default value (10.0.0.1/18) comes from the standard IEC addressing schema, however it can be adapted to an own custom schema by redefining the base templates (as explained next)
- enable IP routing

```
configure ip routing enable
```

- finally enable the TCDS protocol on all the nodes

```
configure tcds enable
```

When this setup is in place, the vTBNs automatically reconfigure the IP address of the backbone virtual interface, the R-NAT rules and the inter-consist routes whenever a change in the active composition is detected.

The IP addresses used by TCDS for configuring the backbone network are built starting from some templates (by default following the rules defined in IEC61375-2-5). These templates can be customized using the commands listed below:

```
configure tcds vtbn consist-network <a.b.c.d/n>
```

This command defines the IP network that is used inside the consist and it is used in building the R-NAT rules; by default it is equal to 10.0.0.0/18.

```
configure tcds vtbn ip-address-template <template>
```

This template is used in building the IP address assigned to the backbone SVI interface. Its default value is 10.128.0.[etbn-id]/18 where *etbn-id* is a placeholder for the vTBN position in the current composition.

```
configure tcds vtbn train-network-template <template>
```

This last template is used in building the backbone address assigned to each consist network. The default is 10.128.[subnet-id<6>].0/18 where *subnet-id* is the placeholder for the position of the consist network in the active composition.

For example, assuming that two consists configured with the default templates are connected, a device with local address 10.0.0.20 will be able to reach the corresponding device located in the other consist using the backbone address 10.128.64.20 or 10.128.128.20 depending on the composition order and direction (which is driven by the consist-UUID value as explained before).

When adopting not standard addressing schemes, the customized templates must be carefully selected paying attention to avoid overlaps in used network address spaces.

Moreover, since the automatic R-NAT applies a 1-to-1 address translation, the network mask applied to the local consist network must be the same used for the backbone addresses.

For example, when configuring the network templates as follows:

```
configure tcds consist-network 172.16.1.0/24
configure tcds vtbn train-network-template 192.168.[subnet-id].0/24
```

a device with local consist address `172.16.1.10` will be able to reach the corresponding device located in the other consist using the backbone address `192.168.1.10` or `192.168.2.10`.

Additional custom multicast routes can be automatically installed in the vTBN configuration whenever a change in the active composition is detected. This is achieved adding, for each custom multicast route, one command like

```
configure tcds vtbn mroute <IN-PORT> <ADDRESS> <NETMASK>
```

where `IN-PORT` is the ingress port and `ADDRESS` is the multicast address to match applying the specified `NETMASK`.

21.6 Composition Lock

The *Composition Lock* operation triggers the computation of the position of each consist in the active composition using the data of the Train Composition Table and permanently stores it on the device memory and file-system until a *Composition Unlock* is performed.

The stored information is an integer number representing the position (starting from 1) of the consist respect to one end of the train. Also the CRC of the Composition Table at the time of the Lock operation is saved for later reference.

This information survives reboots and it is used at startup, in the configuration phase, for dynamically building the configured IP addresses that are marked as *position dependent*.

An IP address is marked as *position dependent* if preceded by the @ character. Any address in the configuration can be marked in this way. For example:

```
configure interface vlan1 ip address @172.16.0.1 255.255.255.0
```

21.6.1 Position dependent addresses mask

The stored *position number*, if available, is applied at startup to every *position dependent* marked address found in the configuration (it does not matter if static or by TTCMP project). How the position of the consist must be applied for building the IP addresses is specified by the *address composition mask* using the following command

```
configure tcds address composition mask <a.b.c.d>
```

where the `a.b.c.d` parameter is a sort of netmask with the bits set only in the portion of the address where the *position number* is applied.

Supposed the address space of the consist to be `172.16.0.0/16`, the requirement is to apply the *position number* to the 3rd octet (i.e. an address built in a device belonging to the 1st consist of the composition will be `172.16.1.X`, in a device belonging to the 2nd consist will be `172.16.2.X` and so on). In order to obtain such behavior the *address composition mask* to be configured is `0.0.255.0` with the command

```
configure tcds address composition mask 0.0.255.0
```

NOTE

If no *position number* is available (the composition is Unlocked) or when the *address composition mask* is missing, every *position dependent* marked address is applied as it is.

21.6.2 Composition Head flag

The reference direction of the active composition can be reversed respect to the actual train orientation (since the first consist of the Train Composition Table is the far end one with the lowest UUID). As a consequence the order of the *position number* assignment (ascending or descending) respect to the actual train orientation is unpredictable. To overcome this issue, a fixed starting point for the enumeration can be set by means of the *Head flag*.

The *Head flag* is an information shared by a consist and used in the *Composition Lock* stage to point the consist enumeration in ascending or descending order respect to the current Train Composition Table. The value of this flag is included in both the “Internal” and “External CST announcements”.

If the first or the last consist of the composition (not more than one) advertises the *Head flag*, at the *Composition Lock* the *position number* 1 is assigned to that consist and the enumeration will be in ascending or descending order accordingly.

The command that sets the flag in a device is

```
sysadmin set composition head
```

while the one that removes the flag is

```
sysadmin reset composition head
```

The status of the flag is visible in the output of the `show ttcmp` command:

```
A@C-SW-1>show ttcmp

Global TTCMP information:
  TTCMP protocol version 2 extensions are enabled
  TTCMP subsystem status is: idle
  TTCMP packets signature is: disabled
  TTCMP matching mode is: default
  Consist identifier: undefined
  Coach identifier: 5 (no description)
  Composition Head flag: not set
  ...

A@H-SW-2>show ttcmp

Global TTCMP information:
  TTCMP protocol version 2 extensions are enabled
  TTCMP subsystem status is: idle
  TTCMP packets signature is: disabled
  TTCMP matching mode is: default
  Consist identifier: undefined
  Coach identifier: 1 (no description)
  Composition Head flag: set
  ...
```

Even though it is a good practice to set the value on all the Consist Switchs installed in the consist that is required to be the “head consist” (*position number 1*), it is sufficient to set it on one of its *boundary switches*.

21.6.3 Lock/Unlock operations

The *Composition Lock* is the operation responsible of permanently storing the *position number* referred to the current composition on each Consist Switch belonging to the network and having the TCDS enabled. The position will be used later in building the IP addresses found in the device configuration and marked with the char @ replacing a part of the address with the position (as defined by the *address composition mask*).

The position number is stored and used every time the device boots up, until a the *Composition Unlock* operation is performed. The *Unlock* deletes the position information that, from the next boot, is not available any more.

The *Composition Lock/Unlock* status and related operations are the following:

- **reset**: is the default status of a brand new device and can be set by the command

```
sysadmin reset composition lock
```

It is the only value that is not set train-wide when the command is executed; it automatically turns to **unlocked** or **locked** when the device is installed as replacement to a broken one in an already configured train.

- **unlocked**: a position number is not stored nor used in building the IP addresses marked with the @ char; if the management address of the device is marked, it is dynamically changed according to the current position in the composition. This status can be set train-wide issuing the following command on a boundary switch of any consist

```
sysadmin set composition unlock
```

- **Locked**: a position number is stored and used in building the IP addresses marked with the @ char, including the management address of the device (that does not change dynamically if the actual composition changes); the position is computed according to the presence of the *Head flag* or, alternatively, according to the actual order of the table. This status can be set train-wide issuing the following command on a boundary switch of any consist

```
sysadmin set composition lock [ direct | reversed | forced ]
```

- When the **direct** optional token is specified, the presence of any *Head flag* is ignored and the position is computed according to the order of the table.
- When the **reversed** optional token is specified, the presence of any *Head flag* is ignored and the position is computed reversely respect to the order of the table.
- The **forced** optional token can be used only on devices belonging to the first or the last consist of the composition; when specified, the presence of any *Head flag* is ignored, the position of the current consist is set to 1 and the position of the other consists is computed accordingly.

The *local* status of the *Composition Lock* can be queried by the command

```
show tcds local
```

that reports the local *Unlocked* status as follows:

```
Using automatic consist-id election
Consist-id election process is completed
Selected consist-id is: 00d10061-2498-4762-9783-5a2f41855d55

Boundary switches discovered inside this consist: 4
Active TCDS mode is: plain
Forwarding of TTCMP traffic on boundaries is: enabled
Coach identifier: 3 (First Class)
Composition Head flag: not set
Discovery position: current 3 - locked 0 (unlocked)
Device name: B-SW-8 - Type: boundary switch
Sending internal CST announcements at slow frequency
...
```

It reports a local *Locked direct* status as follows:

```
...
Discovery position: current 2 - locked 2 (direct)
...
```

and a local *Locked reversed* status as follows:

```
...
Discovery position: current 2 - locked 4 (reversed)
...
```

The *train wide* status of the *Composition Lock* can be queried by the command

```
show tcds composition
```

that reports the status of all the consists, as shown in the following example:

```
Discovery information
-----
TCDS discovery is complete
Active TCDS mode is plain
TTCMP forwarding is enabled
Number of discovered consists: 5
CRC computed on local table: 0x048b4f79

Active Train Composition
-----
CstNum  Cst-id
-----
Orient.  CRC
Coach ID (Description) H Discovery Lock
-----
1 12345678-9abc-def0-1234-56789abcdef0 Inverse 0x048b4f79
2 (--- ) N 5 R 0x048b4f79
2 32fe4b1b-2599-4283-addd-1e07de7232ae Inverse 0x048b4f79
5 (--- ) N 4 R 0x048b4f79
* 3 00d10061-2498-4762-9783-5a2f41855d55 Inverse 0x048b4f79
4 (--- ) N 3 R 0x048b4f79
4 5bb9e0f2-7e08-49e9-9abb-eaa3ab5c4495 Inverse 0x048b4f79
3 (--- ) N 2 R 0x048b4f79
5 4127a97a-f4d7-4ad4-8f2b-ce379e430d89 Inverse 0x048b4f79
1 (--- ) Y 1 R 0x048b4f79
...
```

The value in the **H** column is always **N** (none) and it turns to **Y** when the consist advertises the *Head flag*. The **Discovery Lock** column has 3 values: the first is the *position number*, the second is the *orientation* respect to the composition at the time of the Lock operation (**S** for direct, **R** for reversed, **N** if the Lock is not set), the last is the CRC of the Train Composition Table at the time of the Lock operation.

The device can be configured to reboot automatically after the *Composition Lock* operation ended successfully. When the following command is present in the configuration, the device reboots after the specified number of minutes has passed since the last successful *Composition Lock* operation.

```
configure tcds lock reboot timeout <MINUTES>
```

If a *Composition Unlock* operation is performed in the mean time, the reboot timer is cleared and no reboot happens.

21.7 Multiple coaches in TTCMP Projects

A TTCMP Train Project statically describes all the devices present in the network and their interconnection and has the limitation that it can describe only a static network topology in which the composition in terms of devices and interconnections is fixed.

This limit has been overcome by the introduction of the *Multi-coach TTCMP projects*. These kind of projects describe the train topology in terms of sub-sets of the train network, called *Coaches*. Each Coach, that is an indivisible unit, has no fixed position in the in the actual train composition and can be present multiple times and with different orientation.

Each *Coach* is treated by the TCDS in all respects as a *Consist*; thus all its devices must share the same Consist id, that must be unique at train level.

21.7.1 Coach Identifier and Description

The way the Consist Switch can recognize at deploy time to which kind of *Coach* described in a Multi-coach TTCMP Project it belongs to is the *Coach Identifier*.

The *Coach Identifier* is an integer value ranging from 1 to 63 that marks a device as belonging to a certain type of coach (i.e. locomotive, 1st class car, 2nd class car, dining car, etc...); the value 0 means that no identifier has been set. An optional *Coach Description* human readable string can be set together with the Identifier; the length of this string is limited to 60 characters.

The Coach Identifier and its optional Description can be set by the commands

```
sysadmin set coach-id <0-63> [ description "Coach Description" ]
```

and removed by

```
sysadmin reset coach-id
```

If the Description contains white spaces, it must be enclosed in double quotes.

The values set in a device can be inspected with the `show ttcmp` command:

```
A@C-SW-1>show ttcmp

Global TTCMP information:
  TTCMP protocol version 2 extensions are enabled
  TTCMP subsystem status is: idle
  TTCMP packets signature is: disabled
  TTCMP matching mode is: default
  Consist identifier: undefined
  Coach identifier: 5 (Traction Unit)
  Composition Head flag: not set
  ...

A@H-SW-2>show ttcmp

Global TTCMP information:
  TTCMP protocol version 2 extensions are enabled
  TTCMP subsystem status is: idle
  TTCMP packets signature is: disabled
  TTCMP matching mode is: default
  Consist identifier: undefined
  Coach identifier: not set
  Composition Head flag: not set
  ...
```

The *train wide* status of the *Coach Identifier* can be queried by the command

```
show tcds composition
```

that reports the status of all the consists, as shown in the following example:

```
Discovery information
-----
  TCDS discovery is complete
  Active TCDS mode is plain
  TTCMP forwarding is enabled
  Number of discovered consists: 5
  CRC computed on local table: 0x048b4f79
```




Active Train Composition						

CstNum	Cst-id	Coach ID (Description)		Orient.	CRC	

	1	12345678-9abc-def0-1234-56789abcdef0	2 (2nd class + bikes)	Inverse	0x048b4f79	
	2	32fe4b1b-2599-4283-addd-1e07de7232ae	4 (2nd class	Inverse	0x048b4f79	
			)		N 5 R	0x048b4f79
*	3	00d10061-2498-4762-9783-5a2f41855d55	4 (2nd class	Inverse	0x048b4f79	
			)		N 3 R	0x048b4f79
	4	5bb9e0f2-7e08-49e9-9abb-eaa3ab5c4495	3 (1st class	Inverse	0x048b4f79	
			)		N 2 R	0x048b4f79
	5	4127a97a-f4d7-4ad4-8f2b-ce379e430d89	1 (Locomotive	Inverse	0x048b4f79	
			)		Y 1 R	0x048b4f79
...						

Chapter 22 - Managing device firmware

The system can hold a number of different firmware images. Images are stored in the device flash memory and the number of images that a system can hold depends on the flash size.

At any time on the system there are at least two images:

- the *default* image, which holds the firmware the system will load and boot at startup
- a hidden *fallback* image, used by the system in the boot up sequence, should the other image load fail

You can list the images installed on the system with the command `sysadmin show images`, like in the following example:

```
A@SW-Std.03-00-CA>sysadmin show images
sw-std_1.5.0-4938.ksi [11.0M]
sw-std_1.5.0-4943.ksi [11.0M] [Default image]

Free image space: 37.5M
```

The command shows also the amount of free space left on the flash.

In order to load a firmware image on the system, you can use the command

```
sysadmin load URL
```

where *URL* points to a Konuendo Archive Data (KAD) file on a reachable network resource. Valid network schemes in the URL are *ftp*, *http*, *tftp* and *scp*. In case a network resource should need authentication to be accessed, username and password credentials must be specified in the URL. For example:

```
A@SW-Std.03-00-CA>sysadmin load scp://user:password@192.168.1.1/path/to/image.kad
```

To change the *default* image, use the command

```
sysadmin set default image NAME
```

where *NAME* is the name of one of the images listed by the `sysadmin show images` command.

To remove an image from the system, use the command

```
sysadmin delete image NAME
```

where *NAME* is the name of one of the images listed by the `sysadmin show images` command.

The *default* image can be removed from the system provided that you set another image as *default* before deleting it.

Chapter 23 - **Technical specification**

23.1 Specifications

23.1.1 Product Specific

23.1.1.1 *13 ports model*

- Available configurations:
 - 13 Fast Ethernet ports
 - 5 Gigabit Ethernet + 8 Fast Ethernet ports
- Optional IEEE 802.3af (PoE) on 8 FE ports
- Up to 2 Hardware Bypass

23.1.1.2 *14 ports model*

- Available configurations:
 - 14 Fast Ethernet ports
 - 6 Gigabit Ethernet + 8 Fast Ethernet ports
- Optional IEEE 802.3at (PoE+) on 8 FE ports
- Up to 3 Hardware Bypass

23.1.1.3 *19 ports model*

- Available configurations:
 - 4 10Gigabit Ethernet + 11 Gigabit Ethernet + 4 Fast Ethernet ports
 - 15 Gigabit Ethernet + 4 Fast Ethernet ports
 - 19 Fast Ethernet ports
- Optional IEEE 802.3at (PoE+) on 8 FE ports or 4 GbE + 4 FE ports
- Up to 2 Hardware Bypass

23.1.1.4 *28 ports model*

- Available configurations:
 - 5 10Gigabit Ethernet + 11 Gigabit Ethernet + 12 Fast Ethernet ports
 - 16 Gigabit Ethernet + 12 Fast Ethernet ports

- Optional IEEE 802.3at (PoE+) on 4 GbE + 12 FE ports
- Up to 2 Hardware Bypass

23.1.2 Common (all products)

- IEEE-802.3 layer 2 managed switch
- M12 circular connectors
- IPv4 protocol supported
- Spanning tree (STP 802.1D) and Rapid Spanning Tree (RSTP 802.1w)
- Link aggregation protocol (LACP 802.3ad)
- Up to 4096 802.1Q VLANs
- DHCP option 82 handling
- Integrated per-port DHCP server
- IGMP versions 1, 2 and 3 snooping
- SNMP v1,v2c,v3 device management
- Extended RMON counters
- Link Layer Discovery Protocol (LLDP 802.1ab) with LLDP-MED extensions
- 802.1X port authentication
- RADIUS authentication
- DSCP/802.1p Class of Service
- 4 output hardware queues for each port
- Strict priority or weighted (WRR) scheduler
- Ingress rate limiting
- In-band (SSH, Telnet) and out-of-band (console) CLI interface for device management
- In-band and out-of-band firmware upgrade
- Fallback firmware image for maximum reliability
- Proprietary Train Topology and Configuration Management Protocol (TTCMP©) technology for support of project-based device configuration management, including network discovery and auto-configuration

23.2 Approvals/compliance

Standard	Description	Details
EN 50155	Railway applications: electronic equipment used on rolling stock	-
EN 50121-3-2	Railway applications: electromagnetic compatibility. Part 3.2: rolling stock equipment	-
EN 61000-4-2	Electromagnetic compatibility (EMC) - Part 4-2: Testing and measurement techniques - Electrostatic discharge immunity test	Contact discharge: 6 kV Air discharge: 8 kV
EN 61000-4-3	Electromagnetic compatibility (EMC) - Part 4-3: Testing and measurement techniques - Radiated, radio-frequency, electromagnetic field immunity test	80 – 1000 MHz: 20 V/m 1400 – 2100 MHz: 10 V/m 2100 – 2500 MHz: 5 V/m
EN 61000-4-4	Electromagnetic compatibility (EMC) - Part 4: Testing and measurement techniques Section 4: Electrical fast transient/burst immunity test	Power line: 2 kV Data line: 2 kV
EN 61000-4-5	Electromagnetic compatibility (EMC) - Part 4-5: Testing and measurement techniques - Surge immunity test	Power line: 1 kV line/line Power line: 2 kV line/earth
EN 61000-4-6	Electromagnetic compatibility (EMC) - Part 4-6: Testing and measurement techniques - Immunity to conducted disturbances, induced by radio-frequency fields	Line-conducted interference voltages: 10 Vrms
EN 60068-2-1	Environmental testing - Part 2-1: Tests - Test Ad: cooling test	-
EN 60068-2-2	Environmental testing - Part 2-2: Tests - Test Bd: dry heat test	-
EN 61373	Shock & Vibration	Category 1 class B

Standard	Description	Details
EN 50124-1+A1	Clearances and Creepage distances	
EN 60950-1	Information technology equipment - Safety - Part 1: General requirements	-
IEEE P802.3at	PoE, Data Terminal Equipment Power via the Media Dependent Interface Enhancements	-

Table 6: Compliancy to standards

23.3 Inter-networking standards

Standard	Description
IEEE 802.3u	Fast Ethernet 802.3ab – Gigabit Ethernet
IEEE 802.1Q	Tagged VLANs
IEEE 802.1D	Spanning Tree Protocol
IEEE 802.1w	Rapid Spanning Tree
IEEE 802.1X	Port-based network access control
IEEE 802.1AB	Link Layer Discovery Protocol (LLDP)
IEEE 802.3ad	Link Aggregation Protocol (LACP)
IEEE 802.3at	Power over Ethernet (PoE)

Table 7: Implemented standards

Chapter 24 - **Support and warranty**

Support

Contact information:

VDS Rail S.r.l.

Via Arno, 23-25

50019 – Sesto Fiorentino (FI) – Italy

E-mail: support@vdsrail.com

Website: <http://www.vdsrail.com/support>

Warranty

Please refer to the *Warranty* section in the following web page:

<http://www.vdsrail.com/ordering/info/#warranties>

Chapter 25 - Device LED status and boot sequence

25.1 Service LEDs

On the front panel of the switch there are 8 service LEDs (POWER, FAULT and six numbered LEDs) used to notify specific system states or events.

Each network port has two LEDs, one for the link state and one for PoE state (if supported by the product).

POWER LED is steady green if the system has been correctly connected to a proper power source and is operating.

FAULT LED notifies critical conditions in which device operation is compromised and possible replacement is needed.

LEDs numbered from 1 to 6 are intended to provide a visual feedback of the current logic state of the device. This will enable you to acknowledge at a glimpse if the switch logic is working properly or not.

LED	Function
1	System state
2	TTCMP [®] state
3	Reserved
4	Coupled-switch [®] state
5	Reserved
6	Miscellaneous info (boot fw status, transient ops, etc)

Table 8: System LEDs

25.2 Ports LEDs

Each network port has one or two bi-color LEDs:

- a *green/yellow* one (all ports) which indicates that the network link is up (steady) or that traffic is flowing through that link (blinking).

Its actual color gives information about the link speed, with the green color used for the port nominal speed (100 Mbps for FE ports and 1 Gbps for GbE ports).

For FE ports the LED will be yellow for a link speed of 10 Mbps. For GbE ports the

LED will be yellow for a link speed of 100 Mbps, green-yellow for a speed of 10Mbps.

- a *green/red* one used for PoE signaling (PoE equipped ports only).

A steady green indicates that the port is supplying power to a PoE-capable device while a red PoE LED signals a problem in the power supply phase.

This LED is marked with a special symbol on the panel.

25.3 LEDs and boot sequence

When you connect the device to the power source, the POWER LED becomes steady green and the system starts the boot sequence:

1. internal logic is being warmed up and FAULT LED and LED #1 become steady red for one second
2. the system then performs a general hardware check switching on and off every second all the LEDs (except for POWER) in the 2 possible colors
3. when the system is ready to boot, all LEDs go off, except for FAULT LED and LED #1 which become steady red again
4. while the system is loading the firmware image, the FAULT LED blinks red, while LED #1 changes from steady red to blinking red, then to blinking green
5. if bootstrap succeeded, FAULT LED is off and LED #1 is steady green

POWER LED state will remain unchanged during the above sequence.

25.4 LED n.6

A blinking red LED n.6 is a signal of some corrective action made during the boot phase.

In detail, if the red LED is blinking slow, the first recovering action has been taken to boot the device properly. The system is working correctly, but it is advisable to operate a firmware update or re-install to fix the situation.

If the red LED is blinking fast, the second level of actions has been performed, meaning that the device has gone through some serious degradation. The system is still working properly, but some action (firmware update or re-install) to fix the situation has to be taken immediately, because the system cannot stand a further corruption.

Chapter 26 - **Configuring the console**

This chapter describes how to connect a PC to the Consist Switch in order to perform offline maintenance operations or out-of-band access to the command-line interface.

The required setup for connecting a terminal emulator to the device console varies depending on the operating system in use on the PC. You can follow the instructions given in the section below which applies to your platform.

The following settings apply to *all platforms*:

speed:	115200 bps
data bits:	8
parity:	none
stop-bits:	1
terminal type:	VT100

Before setting up and running the terminal emulator, you need to attach the USB-to-console cable to the *CONSOLE M12* connector on the Consist Switch.

IMPORTANT NOTE

Starting with **S/N 151350**, the interaction with the console driver for both Linux and Windows OS has changed.

In the following chapters we will refer to the different approaches as the *old* and the *new* method.

26.1 Terminal emulator setup on Linux

Old method

The software drivers needed to connect to the *CONSOLE* port of the Consist Switch are normally found in recent Linux distributions. After plugging the cable you should be able to find a new virtual communication port on the system.

You can easily find out the name of this new port using the `dmesg` command:

```
# dmesg
.....
[ 165041.483643] usb 2-4: Number of endpoints 2
[165041.483645] usb 2-4: Endpoint 1 MaxPacketSize 64
[165041.483647] usb 2-4: Endpoint 2 MaxPacketSize 64
[165041.483649] usb 2-4: Setting MaxPacketSize 64
[165041.485738] usb 2-4: FTDI USB Serial Device converter now attached to ttyUSB0
```

In the above example the virtual communication port has been named `/dev/ttyUSB0`.

With this information it is possible to use one of the standard Linux terminal emulation application (like Minicom or Kermit, for example) to access the user interface of the Consist Switch.

New method

The software drivers needed to connect to the *CONSOLE* port of the Consist Switch are normally found in recent Linux distributions. After plugging the cable you should be able to find a new virtual communication port on the system.

You can easily find out the name of this new port using the `dmesg` command:

```
# dmesg
.....
[44191.465444] usb 1-1.2: Number of endpoints 2
[44191.465446] usb 1-1.2: Endpoint 1 MaxPacketSize 64
[44191.465447] usb 1-1.2: Endpoint 2 MaxPacketSize 64
[44191.465448] usb 1-1.2: Setting MaxPacketSize 64
[44191.465947] usb 1-1.2: FTDI USB Serial Device converter now attached to ttyUSB0
[44191.465966] ftdi_sio 1-1.2:1.1: FTDI USB Serial Device converter detected
[44191.465992] usb 1-1.2: Detected FT2232C
[44191.465994] usb 1-1.2: Number of endpoints 2
[44191.465997] usb 1-1.2: Endpoint 1 MaxPacketSize 64
[44191.465999] usb 1-1.2: Endpoint 2 MaxPacketSize 64
[44191.466001] usb 1-1.2: Setting MaxPacketSize 64
[44191.467073] usb 1-1.2: FTDI USB Serial Device converter now attached to ttyUSB1
```

In the above example two virtual communication ports has been created, named `/dev/ttyUSB0` and `/dev/ttyUSB1`.

The port to be used is the one with the greater number, so in this case is the **`/dev/ttyUSB1`**.

With this information it is possible to use one of the standard Linux terminal emulation application (like Minicom or Kermit, for example) to access the user interface of the Consist Switch.

26.2 Terminal emulator setup on Windows

Old method

Use the *RES_1.0* package in *vdssw-ftdi-driver.zip* archive to install the Console-to-USB driver. The archive can be downloaded from the VDS Rail web site.

On Windows Host, check 'Ports' from the 'Device Manager'. It should display some new USB serial ports with a COM port number. The highest COM port number is used for

console. Steps to verify this:

- ⇒ Right click on *My Computer*.
- ⇒ Select *Properties* option.
- ⇒ Select *Hardware* tab.

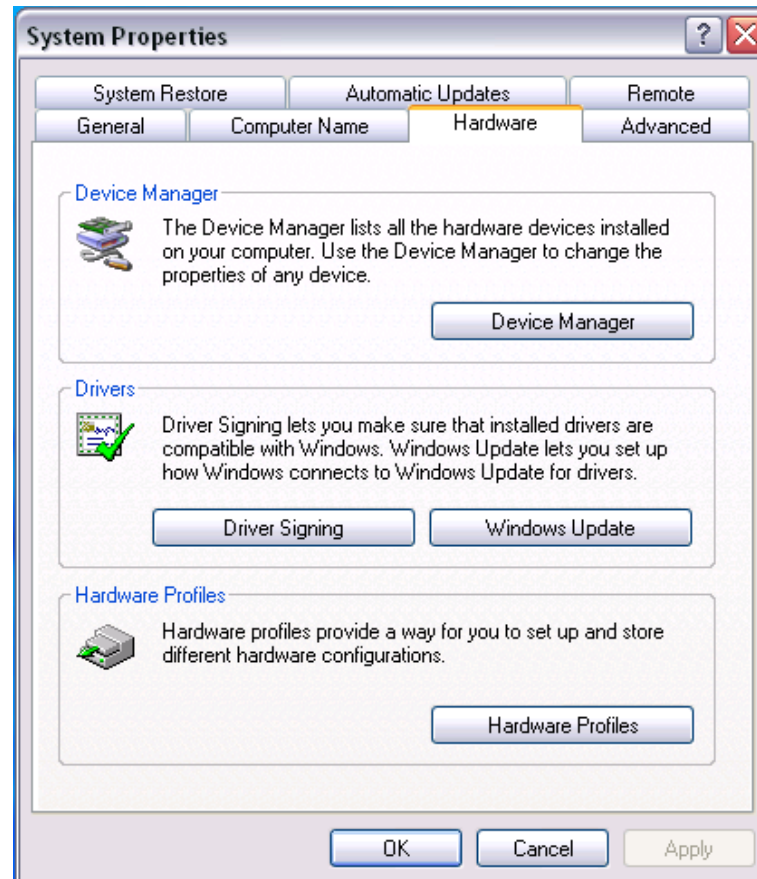


Figure 26.1: The System Properties dialog window

- ⇒ Click on *Device Manager* button.
- ⇒ Expand *Ports (COM & LPT)* label as shown in the figure below. As you can see in the figure, *COM4* is the USB serial port and thus should be the one to be used as our console port.

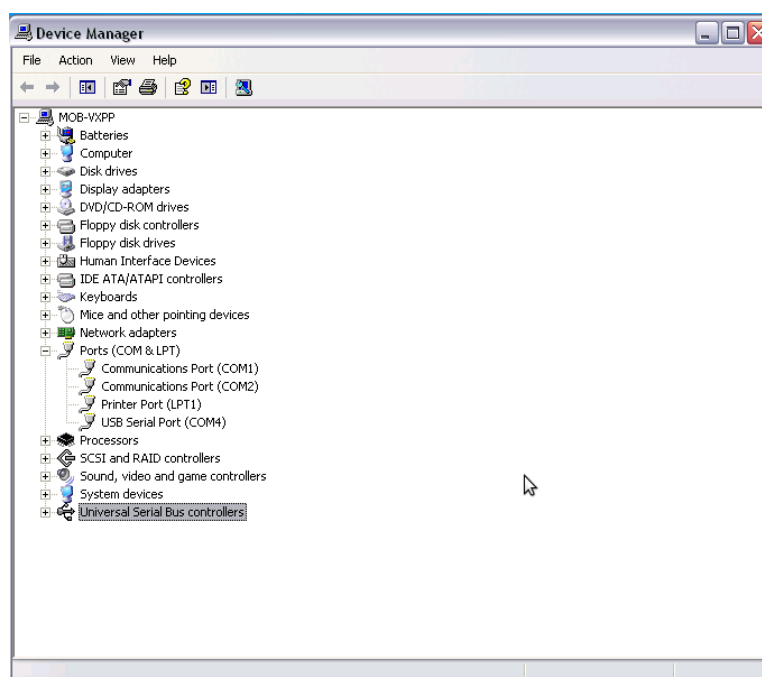


Figure 26.2: The Device Manager dialog window

- ⇒ Launch *Hyper Terminal* application. Select COM port for console as indicated by previous point. Set the communication parameters and the terminal type as outlined above.

New method

On a Windows based PC a driver must be installed. This can be done in 2 different ways:

- If the PC is connected to Internet, the first time a console is connected, a guided procedure automatically downloads and installs the needed software.
- Otherwise the driver installation can be done starting from the driver package. This package can be downloaded from the manufacturer's site (<http://www.ftdichip.com/Drivers/VCP.htm>) selecting the VCP driver as "setup executable".

This setup must be launched and the driver installation process takes place.

Once finished you can connect to the Consist Switch Console.

In both cases, two serial device will be created, usually called COM3 and COM4. Only one of the two is working, generally the last one, and through this COM port it will be possible to interact with the switch console.

On the PC a terminal emulator must be used. A recommended application is Putty (<http://www.putty.org/>).

The session may be configured in the following way:

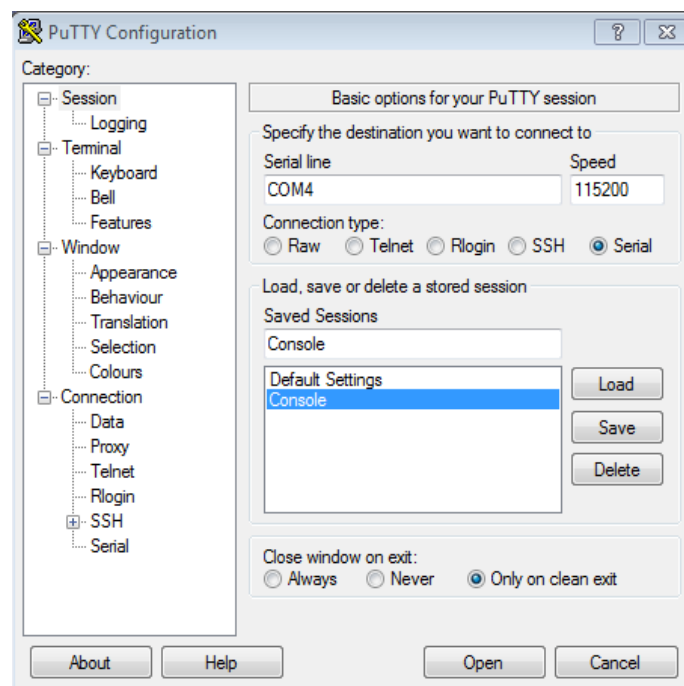


Figure 26.3: PuTTY configuration dialog window

Chapter 27 - Password Recovery

This chapter describes how to restore the Consist Switch original factory passwords in case you forget yours.

NOTE

All the other saved configurations are left untouched.

To execute this procedure follow these steps:

- ⇒ While connected to the console, power cycle the system.
- ⇒ When the system reboots, wait until the following message appears on the console:

```
Enter password - autoboot in 2 seconds...
```
- ⇒ Insert the password `vds`
- ⇒ At the boot-loader prompt (`vds:`), execute the command:

```
vds: run rescue
```
- ⇒ The system will boot and then automatically reboot.
- ⇒ At the end of the operation the system will have all the factory set passwords restored.

NOTE

During the operation some harmless messages may be output on the screen (eg: "umount: can't umount ...").

Chapter 28 - **Konuendo Total Device Management**

The *Konuendo Total Device Management* (KTDM) is a technology that gives you the possibility to handle the maintenance operations starting from a switched-off device.

The operations available are, among the others, software installations, software upgrade/downgrade, configuration (or other data) injection, custom operations, etc.

KTDM needs to be run from a USB pen-drive prepared as described below and may be used in a menu-based fashion or in a total unattended way.

KTDM is distributed in the form of a ZIP archive whose content must be extracted into an empty pen-drive base directory.

The USB pen-drive (preferably USB 2.0 compliant) must have at least 128 MB of free space and, in case of multiple partitions, only the first one will be used.

The files contained in the archive are, at least:

- `bigbang-a.ksi`, the KTDM main driver.
- `ipart.kad`, the system image to be loaded on the device.
- `u-boot.kad`, the bootloader image.

Other files may optionally be present according to the operation needs.

28.1 Firmware upgrade procedure

- ⇒ Prepare the USB pen-drive as detailed before.
- ⇒ Turn off the Consist Switch
- ⇒ Plug the previously prepared pen-drive into the system maintenance port (labeled *MNT-H* or *MGMT*)
- ⇒ Turn on the device
- ⇒ After the power-on phase, the green LEDs labeled 1 to 6 should be switching on and off in a circular fashion.
- ⇒ When the operation is complete, all the LEDs will be steady green.
- ⇒ At this point unplug the USB pen-drive and the Consist Switch will automatically reboot.
- ⇒ In case of trouble, all the LEDs will be turned to steady red. In this case, rerun the operation and, if the problem persists, contact VDS Rail representatives for help.



WARNING

A POWER LOSS DURING THE FIRMWARE WRITING STEP CAN BE VERY DANGEROUS AND MAY COMPROMISE THE FIRMWARE IMAGE.

28.1.1 Firmware upgrade without losing saved device configuration

The operation described in the previous paragraph clears all the data on the Consist Switch.

The following instructions let you perform the firmware upgrade operation without losing the Consist Switch saved configuration:

- ⇒ Prepare the USB pen-drive as detailed before.
- ⇒ Plug the pen-drive into the system maintenance port (labeled *MNT-H* or *MGMT*).
- ⇒ From the *admin* account, execute the command

```
A@SW-Std.03-00-CA>sysadmin save configfile usb:///
```

and then

```
A@SW-Std.03-00-CA>sysadmin reboot
```

- ⇒ Say “Y” to the “Press Y to confirm, N otherwise:” question.
- ⇒ The Consist Switch will reboot.
- ⇒ After the power-on phase, the green LEDs labeled 1 to 6 should be switching on and off in a circular fashion.
- ⇒ When the operation is complete, all the LEDs will be steady green.
- ⇒ At this point unplug the USB pen-drive and the Consist Switch will automatically reboot.

In case of trouble, all the LEDs will be turned to steady red. In this case, rerun the operation and, if the problem persists, contact VDS Rail representatives for help.

WARNING

This operation saves the Consist Switch configuration on the pen-drive, as a by-product, in a file called `CFG-20xyyyzztt.kad`.

If the pen-drive has to be reused for another upgrade operation, be sure to remove this `CFG-20xyyyzztt.kad` file from the pen-drive.

Chapter 29 - **AutoUSB subsystem**

29.1 Description

AutoUSB is a mechanism to interact with the Consist Switch at run time to conduct various activities like firmware upgrade, Consist-Project injection and network configuration, simple reset or reboot actions, etc.

Those actions are pre-defined and are activated by the simple connection of a special prepared USB pen-drive to the **MNT-H** or **MGMT** connector.

This mechanism is extremely powerful, so, if needed, it can be disabled with a simple configuration command.

Later a list of all the defined actions is presented.

29.2 USB pen-drive setup

1. The USB pen-drive must have at least one VFAT partition bigger than 128 MB.

The name of this first partition must be: **VDS_KACTION**

On a Windows machine the partition Label may be easily changed from a File Explorer with a Rename operation.

On a Linux computer one way to rename the partition label is using the tool called **gparted**.

2. The possible actions are distributed, on request, via a **Zip** file.
3. Once this zip file is extracted in the main folder of the first USB pen-drive partition, a folder called **apps** is created.

This folder contains a few files, some called **K_ACTION...** are the actual procedures while others, **INFO_....txt** files, are help files.

4. Depending on the function required, the relevant **K_ACTION...** file has to be copied to the main folder. So, for example, if we want to just reboot a device, the main USB pen-drive folder will contain the subfolder **apps** and the function **K_ACTION.x.y.simple_reboot**.
5. Some other file might be needed depending on the required function. See the list below for specific instructions.

29.3 Operation

Once the USB pen-drive is set-up, the activation is very easy.

1. Plug the USB pen-drive in the connector labeled as **MNT-H** or **MGMT** by means of the M12-USB adaptation cable.
2. When the function starts, the LED n. 6 starts to blink green.
3. When the LED n.6 turns steady green, the function completed its actions and the USB pen-drive may be disconnected.

WARNING:

While this mechanism is very easy and powerful, it show little or no feed-back on the on-going operations, except for the LED n.6 code.

If a better understanding of the actions progression is requested, a richer approach is achieved using a diagnostic/configuration station loaded with the **KonfStation** tool by VDS Rail (ask VDS Rail representatives for details).

29.4 Functions

Name	Action	Requirements
K_ACTION.x.y.simple_reboot	It simply reboots the device.	
K_ACTION.x.y.device_total_reset_reboot	Saved configuration and Network Projects are deleted. The device is then rebooted.	
K_ACTION.x.y.factory_boot	The system reboots with the factory hidden firmware image.	
K_ACTION.x.y.ttcmp_reset	On a TTCMP-based network all the devices are reset to an idle state and the current Project is reset to NULL.	
K_ACTION.x.y.ttcmp_local_reset	Only the device receiving the pen-drive is reset.	
K_ACTION.x.y.ttcmp_local_reset_reboot	Only the device receiving the pen-drive is reset and then rebooted.	
K_ACTION.x.y.project_diffusion	Configure a TTCMP-based network.	A text file called K_SWITCH_ID containing the name of the device we are inserting the pen-drive into as designed in the KAD file. A KP_*.kad file containing the Consist-Project description as produced by the TTCMP® network editor (NED).
K_ACTION.x.y.project_diffusion_check	Configure a TTCMP-based network, optionally set the Consist-ID and confirm (set) the configured setup. Red LED n.6 report if	See the INFO_project_diffusion_check.txt help file for details.

	configuration is failed.	
K_ACTION.x.y.parametric_project_diffusion	Configure a TTCMP-based network from a parametric kad file.	See the INFO_parametric_project_diffusion.txt help file for details.
K_ACTION.x.y.ttcmp_confirm	Once the configuration is applied on a TTCMP-based network, a confirmation action is performed to freeze the configured setup.	Attention: The topology confirmation may fail if the Consist-ID is not equally set (or left undefined) on every network device.
K_ACTION.x.y.single_sw_config_confirm	The configuration is diffused to a single switch, the configuration is confirmed and then the switch is rebooted.	A text file called K_SWITCH_ID containing the name of the device we are inserting the pen-drive into as designed in the KAD file. A KP_*.kad file containing the Consist-Project description as produced by the TTCMP® network editor (NED).
K_ACTION.x.y.project_extraction_2usb	The current Project kad file, if any, is copied on the USB pen-drive itself.	
K_ACTION.x.y.fw_update	Upgrade the device firmware.	An image file ipart.kad must be present in the pen-drive.
K_ACTION.x.y.generic_cmd	Execute a list of command.	In a file called LIST_K_ACTION the commands are listed one per line. Any command valid for the CLI may be used except for the "sysadmin load save" family.
K_ACTION.x.y.generic_kad_load	The first kad file found in the pen-drive is loaded.	
K_ACTION.x.y.show_fw_rel	It shows which FW release is installed on the device. The lower LED (4, 5 and 6) will blink with a pattern that specifies the release installed. Each led specifies one of the 3 number defining the release. No blink is for zero. Other numbers are shown with a regular number of blinks with a 4 seconds pause before restarting, so a value of 3 is 3 blinks, 4 secs of pause, 3 blinks etc.	

WARNING:

If the pen-drive is set-up on Windows machines, beware that a **.ini** or **.txt** suffix is often silently added by the operating system to the file names. This suffix must NOT be present at all.

Chapter 30 - External configuration

The *External* configuration mode allows to rely on configuration setup stored in an external USB pen-drive, so that no detail is maintained in the switch internal storage.

If the *External* configuration mode has been chosen, an USB pen-drive is needed in order to save the configuration commands externally.

The USB pen-drive (preferably USB 2.0 compliant) must have at least 128 MB of free space and, in case of multiple partitions, only the first one will be used.

Follow these steps for the *External* configuration mode:

1. Configure the Consist Switch the way you need.
2. Save the configuration with the command

```
A@SW-Std.03-00-CA>save running-config
```

3. Do not switch off the device.
4. Plug the pen-drive into the system maintenance port (labeled *MNT-H* or *MGMT*).
5. From the CLI issue the command

```
A@SW-Std.03-00-CA>sysadmin save configfile usb:/// external
```

The configuration now has been saved on the external pen-drive so, as long as the pen-drive is plugged in, the switch will boot with the externally saved configuration details, no matter how the switch will be reconfigured afterward.

If a new configuration setup is needed, the procedure must be re-done starting from point 1.

If the USB pen-drive is unplugged, the Consist Switch will boot with the internally saved configuration, if any.

If the USB pen-drive is plugged into another Consist Switch, the same considerations apply for the new device.

WARNING

During a boot with a pen-drive plugged in, a message is printed on the console. This message:

```
** Unable to read "bigbang-a.ksi" from usb 0:1 **  
is totally harmless.
```

Chapter 31 - CLI commands reference

The following is the reference list of available commands from the CLI. Parameter names are in all-caps *italic*. Optional arguments are enclosed in square brackets.

All commands support the *no-form*.

```
auto-logout [SECONDS]
```

Enable auto-logout. Default is 10 seconds.

```
clear counters protocol lldp
```

Clear the packet counters for the LLDP protocol.

```
clear counters protocol ttcmp
```

Clear the packet counters for the TTCMP protocol.

```
clear ip dhcp bindings
```

Clear all active DHCP server/snooping bindings.

```
clear ip dhcp bindings interface NAME
```

Clear the active DHCP server/snooping bindings for the given interface.

```
clear mac-address
```

Clear all MAC addresses detected from network and stored in memory.

```
clear mac-address interface NAME
```

Clear all MAC addresses detected from the given network interface.

```
clear topology
```

Clear the list of nodes collected by means of the TTCMP protocol.

```
configure aaa-domain radius DOMAIN-NAME acct-serv primary  
                           address ADDRESS secret SECRET
```

Define an aaa-domain named *DOMAIN-NAME* of type radius *accounting primary* using the provided *ADDRESS* and *SECRET*.

```
configure aaa-domain radius DOMAIN-NAME acct-serv primary  
                           address ADDRESS secret SECRET port PORT
```

Define an aaa-domain named *DOMAIN-NAME* of type radius *accounting primary* using the provided *ADDRESS* and *SECRET* and non standard port *PORT*.

```
configure aaa-domain radius DOMAIN-NAME acct-serv backup  
                           address ADDRESS secret SECRET
```

Define an aaa-domain named *DOMAIN-NAME* of type radius *accounting backup* using the provided *ADDRESS* and *SECRET*.

```
configure aaa-domain radius DOMAIN-NAME acct-serv backup  
                           address ADDRESS secret SECRET port PORT
```


Define an aaa-domain named *DOMAIN-NAME* of type radius *accounting backup* using the provided *ADDRESS* and *SECRET* and non standard port *PORT*.

```
configure aaa-domain radius DOMAIN-NAME auth-serv primary
                                     address ADDRESS secret
SECRET
```

Define an aaa-domain named *DOMAIN-NAME* of type radius *authentication primary* using the provided *ADDRESS* and *SECRET*.

```
configure aaa-domain radius DOMAIN-NAME auth-serv primary
                                     address ADDRESS secret SECRET port
PORT
```

Define an aaa-domain named *DOMAIN-NAME* of type radius *authentication primary* using the provided *ADDRESS* and *SECRET* and non standard port *PORT*.

```
configure aaa-domain radius DOMAIN-NAME auth-serv backup
                                     address ADDRESS secret
SECRET
```

Define an aaa-domain named *DOMAIN-NAME* of type radius *authentication backup* using the provided *ADDRESS* and *SECRET*.

```
configure aaa-domain radius DOMAIN-NAME auth-serv backup
                                     address ADDRESS secret SECRET port PORT
```

Define an aaa-domain named *DOMAIN-NAME* of type radius authentication *backup* using the provided *ADDRESS* and *SECRET* and non standard port *PORT*.

```
configure coupled-mode cmap enable
```

Enable Coupled Mode Aggregation Protocol (CMAP) on backbone ports.

```
configure coupled-mode dlacp mode-on
```

Configure Distributed LACP mode of operations: LACPDU tx/rx handled even if coupled status is out-of-sync.

```
configure coupled-mode dlacp mode-sync
```

Configure Distributed LACP mode of operations: LACPDU tx/rx stopped if coupled status is out-of-sync.

```
configure coupled-mode role master|slave [left-only|right-only]
```

Configure device as a master or slave for coupled-mode configuration; optionally the device can be set to be a left-end or a right-end of the ladder topology.

```
configure diagnostics iptcom|trdp|ttcmp enable
```

Enable the transmission of diagnostics packets by means of the given protocol (IPTCom, TRDP or TTCMP).

```
configure diagnostics iptcom|trdp notify|publish comid COMID
```

Set the *COMID* value for publishing or notifying the diagnostics packets by

means of the given protocol (IPTCom or TRDP).

```
configure diagnostics iptcom|trdp reply comid COMID request comid COMID
```

Set the *COMID* values for obtaining the diagnostics packets in the request/reply fashion by means of the given protocol (IPTCom or TRDP).

```
configure diagnostics iptcom|trdp notify|publish  
destination address ADDRESS
```

Set the destination IP *ADDRESS* for publishing or notifying the diagnostics packets by means of the given protocol (IPTCom or TRDP).

```
configure diagnostics iptcom|trdp notify|publish|reply ip precedence DCSP
```

Set the IP Precedence (*DCSP*) level (in the range 0-7) for publishing, notifying or requesting the diagnostics packets by means of the given protocol (IPTCom or TRDP).

```
configure diagnostics iptcom|trdp notify|publish|reply ip ttl TTL
```

Set the IP Time To Live (*TTL*) value (in the range 1-255) for publishing, notifying or requesting the diagnostics packets by means of the given protocol (IPTCom or TRDP).

```
configure diagnostics iptcom|trdp publish interval SECONDS
```

Set the send period for publishing the diagnostics packets by means of the given protocol (IPTCom or TRDP).

```
configure diagnostics ttcmp announce
```

Enable sending the diagnostics packets immediately after the TTCMP announcements.

```
configure diagnostics ttcmp interval INTERVAL
```

Enable sending the diagnostics packets with the given time period (*INTERVAL* seconds).

```
configure interface INTERFACE
```

Enter specific configuration level for given network interface.

When specified *INTERFACE* does not exist yet, and assumes some special value, a new virtual interface is created when possible. Current admissible *INTERFACE* special values are: *grpX* for the creation of aggregated group virtual interfaces, where *X* is a value between 0 and 6.

```
configure interface grpNN group members INTERFACE [INTERFACE ...]
```

Assign the list of switched-interfaces to group *grpNN* as members. Up to 6 links can be specified as *INTERFACE*.

```
configure interface grpNN group mode lacp [fast|slow]
```

Set the link aggregation mode for *grpNN* to be managed by LACP, optionally selecting the slow or fast period for sending LACPDUs.

```
configure interface grpNN group mode static
```

Set the static link aggregation mode for *grpNN*.

```
configure interface INTERFACE description DESCRIPTION
```

Set the *DESCRIPTION* text describing the given *INTERFACE*.

```
configure interface INTERFACE dhcp-server  
                                client-address ADDRESS  
[NETMASK]
```

Set the client-address property of per-port DHCP server for a given network interface. If the *NETMASK* argument is omitted, the natural one will be used.

```
configure interface INTERFACE dhcp-server  
                client-address ADDRESS [NETMASK] client-id ascii|hex CLIENT-  
ID
```

Set *CLIENT-ID* as Client Identifier (DHCP option 61) selector for the given *ADDRESS*.

```
configure interface INTERFACE dhcp-server  
                client-address ADDRESS [NETMASK] hostname  
HOSTNAME
```

Set *HOSTNAME* as Hostname (DHCP option 12) selector for the given *ADDRESS*.

```
configure interface INTERFACE dhcp-server  
                client-address ADDRESS [NETMASK] vci CLASS-  
ID
```

Set *CLASS-ID* as Vendor Class Identifier (DHCP option 60) selector for the given *ADDRESS*.

```
configure interface INTERFACE dhcp-server  
                client-address ADDRESS [NETMASK] vlan  
VID
```

Set *VID* as VLAN selector for the given *ADDRESS*.

```
configure interface INTERFACE dhcp-server client-address-pool FIRST  
LAST
```

Define a pool of IP addresses available to clients. *FIRST* is be the first address of the pool. *LAST* is be the last address of the pool.

```
configure interface INTERFACE dhcp-server client-boot-file FILENAME
```

Set the boot file name field (file) value to *FILENAME*.

```
configure interface INTERFACE dhcp-server client-boot-file FILENAME  
                                client-id CLIENT-  
ID
```

Set *CLIENT-ID* as Client Identifier (DHCP option 61) selector for the given boot file *FILENAME*.

```
configure interface INTERFACE dhcp-server client-boot-file FILENAME  
                                hostname  
HOSTNAME
```

Set *HOSTNAME* as Hostname (DHCP option 12) selector for the given boot

file *FILENAME*.

```
configure interface INTERFACE dhcp-server client-group NAME [vlan VID]
```

Set the client-group property of per-port DHCP server for a given network interface. Optionally a VLAN identifier *VID* can be specified.

```
configure interface INTERFACE dhcp-server client-name NAME [vlan VID]
```

Set the client-group property of per-port DHCP server for a given network interface. Optionally a VLAN identifier *VID* can be specified.

```
configure interface INTERFACE dhcp-server  
fallback address-pool FIRST  
LAST
```

Define a pool of IP addresses to be used as fallback for clients. *FIRST* is be the first address of the pool. *LAST* is be the last address of the pool.

```
configure interface INTERFACE dhcp-server fallback forward-mode
```

Enable forwarding DHCP requests on fallback.

```
configure interface INTERFACE dot1x aaa-domain DOMAIN-NAME
```

Associate the aaa-domain named *DOMAIN-NAME* to the given interface.

```
configure interface INTERFACE dot1x aaa-domain local-db
```

Associate the internal local users database to the given interface.

```
configure interface INTERFACE dot1x eapol-version 1|2
```

Specify the EAPOL version in use on the given interface.

```
configure interface INTERFACE dot1x host-mode multi|single
```

Select which host authentication mode in use on the given interface. When *multi* is selected only first client must authenticate, otherwise (*single*) each client must authenticate.

```
configure interface INTERFACE dot1x port-control  
auto|force-authorized|force-  
unauthorized
```

Select the port-control 802.1X management status. When *auto* is selected 802.1X exchange will take care of port authentication, otherwise no 802.1X exchange will be performed and the given interface status is always forced to authorized or un-authorized.

```
configure interface INTERFACE dot1x reauth-timer VALUE
```

Select 802.1X re-authentication timeout for the given interface. *VALUE* is expressed in seconds and must be in the range 0-86400.

```
configure interface INTERFACE dot1x reauthenticate station MAC-ADDRESS
```

Force 802.1X re-authentication of the specific authenticated supplicants (identified by *MAC-ADDRESS*) of the given interface.

```
configure interface INTERFACE dot1x reauthenticate station all
```

Force 802.1X re-authentication of all known authenticated supplicants of the given interface.

```
configure interface INTERFACE duplex MODE
```

Set the duplex mode for the given network interface. *MODE* can be one of the following: *auto*, *full*, *half*.

```
configure interface INTERFACE enable
```

Enable the given network interface.

```
configure interface INTERFACE fast-link-detection enable
```

Enable fast link-down detection mechanism available on Gigabit ports.

```
configure interface INTERFACE flow-control enable
```

Enable IEEE 802.3 Flow Control for the given *INTERFACE*.

```
configure interface INTERFACE ip address ADDRESS [NETMASK]
```

Set IPv4 address on given interface. If the *NETMASK* argument is omitted, the natural one, derived from the IP address class, will be used.

```
configure interface INTERFACE ip address dhcp
```

Obtain IPv4 address for given interface from a DHCP server on the network.

```
configure interface INTERFACE ip dhcp-option82 enable
```

Enable DHCP option82 insertion on the given *INTERFACE*.

```
configure interface INTERFACE ip dhcp-snooping drop-packets
```

Drop DHCP packets entering from the given *INTERFACE*.

```
configure interface INTERFACE ip dhcp-snooping enable
```

Enable DHCP snooping on the selected *INTERFACE*.

```
configure interface INTERFACE ip dhcp-snooping trusted-ports
```

Trust incoming DHCP replies on the given *INTERFACE*.

```
configure interface INTERFACE ip igmp-snooping enable
```

Enable IGMP snooping on the selected *INTERFACE*.

```
configure interface INTERFACE ip igmp-snooping filter ADDRESS [NETMASK]
```

Drop IGMP packets for selected multicast *ADDRESS*. An optional *NETMASK* can be specified for dropping packets from an address range.

```
configure interface INTERFACE ip igmp-snooping flood-unknown
```

Flood unknown multicast packets out from the given *INTERFACE*.

```
configure interface INTERFACE ip igmp-snooping mrouter
```

Configure the given *INTERFACE* as a multicast router port.

```
configure interface INTERFACE ip igmp-snooping static-group ADDRESS  
[vlan  
VLAN]
```

Configure an IGMP static multicast group for the given *INTERFACE*, optionally specifying a non default vlan.

```
configure interface INTERFACE ip ospf dead-interval VALUE
```

Interval after which a neighbor is declared dead. *VALUE* is expressed in seconds and must be in the range 1-65535.

```
configure interface INTERFACE ip ospf dead-interval  
minimal hello-multiplier  
VALUE
```

Set the dead-interval to 1 second with fast sub-second HELLO. *VALUE* is the number of HELLOs to send per second and must be in the range 1-10.

```
configure interface INTERFACE ip ospf hello-interval VALUE
```

Time between HELLO packets. *VALUE* is expressed in seconds and must be in the range 1-65535.

```
configure interface INTERFACE ip ospf priority VALUE
```

Router priority. *VALUE* must be in the range 0-255.

```
configure interface INTERFACE ip ospf retransmit-interval VALUE
```

Time between retransmitting lost link state advertisements. *VALUE* is expressed in seconds and must be in the range 3-65535.

```
configure interface INTERFACE ip proxy-arp enable
```

Enable the proxy-arp feature on the selected *INTERFACE*.

```
configure interface INTERFACE lldp enable
```

Enable LLDP on given network interface.

```
configure interface INTERFACE mdi-mode MODE
```

Set the MDI mode for the given network *INTERFACE*. *MODE* can be one of the following: *auto*, *mdi*, *mdi-x*.

```
configure interface INTERFACE mtu VALUE
```

Set the Ethernet MTU value for given network interface.

```
configure interface INTERFACE poe mode on|off|as-link
```

Configure the PoE mode for the given *INTERFACE*, respectively to be always active, always inactive or activated basing on the link status.

```
configure interface INTERFACE qos default cos VALUE [override]
```

Set the QoS CoS value for traffic coming from the given network interface. If the optional override argument is present, all incoming packets will be tagged with the given CoS value, otherwise only packets with missing CoS value will be tagged by the switch.

```
configure interface INTERFACE qos prio-queues QUEUE
```

Set the number of queues that should be served using a strict priority algorithm. *QUEUE* can be 1, 2, 3, 4, 5, 6 or *all*.

```
configure interface INTERFACE qos trust TYPE
```

Set the QoS trust policy for a traffic coming from the given network interface. *TYPE* can be one of the following: *cos*, *dscp*, *both*.

```
configure interface INTERFACE spanning-tree edge MODE
```

MODE can be one of the following: *auto*, *on*, *off*.

```
configure interface INTERFACE spanning-tree enable
```

Enable or disable the spanning-tree protocol on this port.

```
configure interface INTERFACE spanning-tree port-cost VALUE
```

Set the per-port specific spanning-tree port-cost. *VALUE* must be a value in range 1-200000000.

```
configure interface INTERFACE spanning-tree port-priority VALUE
```

Set the per-port specific spanning-tree port-priority. *VALUE* must be a value in range 0-240, in steps of 16.

```
configure interface INTERFACE speed VALUE
```

VALUE can be one of the following: *auto*, 10, 100, 1000.

```
configure interface INTERFACE switchport dual-home
```

Set the given interface as a static distributed aggregated link.

```
configure interface INTERFACE switchport dual-home lacp [fast|slow]
```

Set the given interface as a distributed aggregated link managed by LACP. Optionally, the frequency of LACPDU can be tuned using the *fast* or *slow* additional arguments.

```
configure interface INTERFACE switchport mode access
```

Set the specified interface as access-port for Default Native VLAN 1.

```
configure interface INTERFACE switchport mode access vlan ID
```

Set the specified interface as access-port for the specified VLAN ID. *ID* must be a value in range 1-4096.

```
configure interface INTERFACE switchport mode trunk
```

Set the specified interface as trunk-port using the default-native VLAN 1 as native-untagged VLAN for this trunk.

```
configure interface INTERFACE switchport mode trunk native vlan ID
```

Set the specified interface as trunk-port using the specified VLAN ID as native-untagged VLAN for this trunk. *ID* must be a value in range 1-4096.

```
configure interface INTERFACE switchport mode trunk native vlan ID
                                prune allow <xx[,yy][,ww-
zz]>
```

Set the specified interface as trunk-port using the specified VLAN ID as native-untagged VLAN for this trunk, while using the specified VLANs to define the set of allowed VLANs on this trunk. *ID* must be a value in range 1-4096.

```
configure interface INTERFACE switchport mode trunk native vlan ID
                                prune deny <xx[,yy][,ww-
zz]>
```

Set the specified interface as trunk-port using the specified VLAN ID as native-untagged VLAN for this trunk, while using the specified VLANs to define the set of denied VLANs on this trunk. *ID* must be a value in range 1-4096.

```
configure interface INTERFACE switchport mode trunk prune
                                allow <xx[,yy][,ww-
zz]>
```

Set the specified interface as trunk-port using the default-native VLAN 1 as native-untagged VLAN for this trunk, while using the specified VLANs to define the set of allowed VLANs on this trunk.

```
configure interface INTERFACE switchport mode trunk prune
                                deny <xx[,yy][,ww-
zz]>
```

Set the specified interface as trunk-port using the default-native VLAN 1 as native-untagged VLAN for this trunk, while using the specified VLANs to define the set of denied VLANs on this trunk.

```
configure interface INTERFACE switchport rate-limit
                                broadcast VALUE [K|M|
G]
```

Define a rate limit for broadcast packets. By default the rate *VALUE* is expressed in Mbps; optionally the value can be expressed in kbps.

```
configure interface INTERFACE switchport rate-limit
                                broadcast VALUE K|M|G burst auto|
BURST
```

Define a rate limit for broadcast packets with explicit committed burst size *BURST* expressed in Bytes. The keyword *auto* will trigger the automatic computation of an optimal burst size (but carrying out this operation requires up to some seconds).

```
configure interface INTERFACE switchport rate-limit inbound VALUE [K|M|
G]
```

Define a rate limit for inbound packets. By default the rate *VALUE* is expressed in Mbps; optionally the value can be expressed in kbps.

```
configure interface INTERFACE switchport rate-limit
                                inbound VALUE [K|M|G] burst auto|
BURST
```


Define a rate limit for inbound packets with explicit committed burst size *BURST* expressed in Bytes. The keyword *auto* will trigger the automatic computation of an optimal burst size (but carrying out this operation requires up to some seconds).

```
configure interface INTERFACE switchport rate-limit inbound VALUE K|M|G
                                                    burst auto|BURST prio
PRIO
```

Define a priority based rate limit for inbound packets with priority *PRIO* (in range 1-8) and automatic or explicit committed burst size *BURST* expressed in Bytes.

```
configure interface INTERFACE switchport rate-limit
                                                    inbound VALUE K|M|G prio
PRIO
```

Define a priority based rate limit for inbound packets with priority *PRIO* (in range 1-8).

```
configure interface INTERFACE switchport rate-limit
                                                    outbound VALUE [K|M|G|
PPS]
```

Define a rate limit for outbound packets. By default the rate *VALUE* is expressed in Mbps; optionally the value can be expressed in kbps or in PPS.

```
configure interface INTERFACE switchport rate-limit profile PROFILE
```

Assign and apply the rate-limit profile *PROFILE* to the given *INTERFACE*.

```
configure interface INTERFACE tcds boundary-port direction dir1|dir2
```

Set the given *INTERFACE* as a TCDS boundary port for the given direction.

```
configure interface INTERFACE tcds line-id A|B|C|D
```

Associate the specified IEC 61375 line identifier to the given *INTERFACE*.

```
configure interface INTERFACE ttcmp enable
```

Permit TTCMP on given network *INTERFACE*.

```
configure interface INTERFACE vrrp advertise INTERVAL
```

Set VRRP advertisement interval in seconds (allowed range 1-255).

```
configure interface INTERFACE vrrp advertise msec INTERVAL
```

Set VRRP advertisement interval in milliseconds (allowed range 20-999).

```
configure interface INTERFACE vrrp instance-id ID
```

Associate the VRRP instance identifier *ID* (in range 1-4) to the given *INTERFACE*.

```
configure interface INTERFACE vrrp ip ADDRESS
```

Set the VRRP IP *ADDRESS* for the given *INTERFACE*.

```
configure interface INTERFACE vrrp priority PRIORITY
```

Set the VRRP priority (allowed range 1-254) for the given *INTERFACE*.

```
configure interface INTERFACE vrrp version 2|3
```

Set the VRRP version in use on the given *INTERFACE*.

```
configure interface INTERFACE vrrp vrid VRID
```

Set the VRRP Virtual Router Identifier (*VRID*) in use on the given *INTERFACE*.
The allowed range is 1-255.

```
configure ip dhcp-option82 circuit-id INFO
```

INFO can be one of the following: *if-name*, *if-num*.

```
configure ip dhcp-option82 enable
```

Enable the insertion of DHCP option82 data.

```
configure ip dhcp-option82 remote-id INFO
```

INFO can be one of the following: *hostname*, *ip-addr*, *mac-addr*.

```
configure ip dhcp-server enable
```

Enable per-port DHCP server.

```
configure ip dhcp-server group NAME
```

Enter specific configuration level for the given DHCP group.

```
configure ip dhcp-server group NAME default-router ADDRESS
```

Set *default-router* property for the given DHCP group. *ADDRESS* must be a valid IPv4 address.

```
configure ip dhcp-server group NAME dns-server ADDRESS [ADDRESS]
```

Set *dns-server* property for the given DHCP group. *ADDRESS* must be a valid IPv4 address. An additional fallback *ADDRESS* can be specified optionally.

```
configure ip dhcp-server group NAME lease-time TIME
```

Lease period (minutes) for DHCP clients associated with the given group (in range 1-1440).

```
configure ip dhcp-server group NAME netmask NETMASK
```

Set *netmask* property for the given DHCP group. *NETMASK* must be a valid IPv4 netmask address.

```
configure ip dhcp-server group NAME next-server ADDRESS
```

Set the IP *ADDRESS* of the next server field (siaddr) for the given DHCP group.

```
configure ip dhcp-server group NAME server-id ADDRESS
```

Set the IP *ADDRESS* associated to the DHCP server for the given DHCP group.

```
configure ip dhcp-server group NAME sname HOSTNAME
```

Set the server name field (sname) value to *HOSTNAME* for the given DHCP

group.

```
configure ip dhcp-server group NAME
                                static-route ADDRESS NETMASK
GATEWAY
```

Set a static route to a host or network (specified by *ADDRESS* and *NETMASK*) with for the given DHCP group via the given *GATWEAY* address.

```
configure ip dhcp-server group NAME time-server ADDRESS
```

Set *time-server* property for the given DHCP group. *ADDRESS* must be a valid IPv4 address.

```
configure ip dhcp-snooping enable
```

Enable the DHCP snooping service.

```
configure ip filter NAME
```

Manage the definition of the filter called *NAME*.

```
configure ip filter NAME RULE# accept|deny icmp-ping|ip SRC-ADDRESS|any
                                INGRESS-INTERFACE|any DEST-ADDRESS|any EGRESS-INTERFACE |
any
```

Define a rule, identified with the number *RULE#*, for the given filter. It can be a rule accepting or denying the matching packets, matching IP or ICMP routed traffic. The matching traffic is defined by the source address (*SRC-ADDRESS*), ingress interface (*INGRES-INTERFACE*), destination address (*DEST-ADDRESS*) and egress interface (*EGRESS-INTERFACE*).

```
configure ip filter NAME RULE# accept|deny icmp-ping|ip SRC-ADDRESS|any
                                INGRESS-INTERFACE|any
local
```

Define a rule, identified with the number *RULE#*, for the given filter. It can be a rule accepting or denying the matching packets, matching IP or ICMP locally destined traffic. The matching traffic is defined by the source address (*SRC-ADDRESS*) and ingress interface (*INGRES-INTERFACE*).

```
configure ip filter NAME RULE# accept|deny tcp|udp
                                SRC-ADDRESS|any SRC-PORT|any INGRESS-INTERFACE|any
                                DEST-ADDRESS|any DEST-PORT|any EGRESS-INTERFACE|any
```

Define a rule, identified with the number *RULE#*, for the given filter. It can be a rule accepting or denying the matching packets, matching TCP or UDP routed traffic. The matching traffic is defined by the source address (*SRC-ADDRESS*), source port (*SRC-PORT*), ingress interface (*INGRES-INTERFACE*), destination address (*DEST-ADDRESS*), destination port (*DEST-PORT*), and egress interface (*EGRESS-INTERFACE*).

```
configure ip filter NAME RULE# accept|deny tcp|udp
                                SRC-ADDRESS|any SRC-PORT|any INGRESS-INTERFACE|any local DEST-PORT|any
```

Define a rule, identified with the number *RULE#*, for the given filter. It can be a rule accepting or denying the matching packets, matching TCP or UDP locally destined traffic. The matching traffic is defined by the source address (*SRC-ADDRESS*), source port (*SRC-PORT*), ingress interface (*INGRES-*

INTERFACE) and destination port (*DEST-PORT*).

```
configure ip filter NAME enable
```

Enable the filter called *NAME*.

```
configure ip igmp querier enable
```

Enable the IGMP querier service.

```
configure ip igmp querier max-response-time TIME
```

Set the maximum response time advertised in IGMP queries in seconds.
TIME must be an integer value in the range 1-30.

```
configure ip igmp querier query-interval VALUE
```

Set the interval between IGMP query messages. *VALUE* must be an integer value in the range 20-600.

```
configure ip igmp querier query-timeout VALUE
```

Set the timeout for trigger an IGMP querier re-election. *VALUE* must be an integer value in the range 60-300.

```
configure ip igmp querier version 1|2|3
```

Configure the IGMP version of the queries.

```
configure ip igmp snooping aging-time VALUE
```

Set the aging time of IGMP membership registrations. *VALUE* must be an integer value in the range 100-1000.

```
configure ip igmp snooping enable
```

Enable the IGMP snooping service.

```
configure ip igmp snooping filter ADDRESS [NETMASK]
```

Drop IGMP packets for the given multicast *ADDRESS*. An address range can be used specifying the optional *NETMASK*.

```
configure ip igmp snooping flood-to-cpu
```

Flood multicast packets to the CPU port.

```
configure ip igmp snooping flood-unknown
```

Enable flooding of unknown multicast packets.

```
configure ip igmp snooping immediate-leave
```

Enable the immediate leave of multicast group registrations.

```
configure ip igmp snooping tcn-flood-query-count VALUE
```

Configure multicast flooding duration. *VALUE* must be an integer value in the range 1-10.

```
configure ip igmp snooping tcn-ignore
```

Ignore TCN events.

```
configure ip igmp snooping tcn-query-solicit
```

Solicit an IGMP general query.

```
configure ip mroute IN-IF SRC-ADDRESS[/MASK] MC-ADDRESS[/MASK] OUT-IFS
```

Add a multicast route.

```
configure ip nameserver ADDRESS
```

Set the IPv4 address of the default name server.

```
configure ip nat NAME
```

Manage the definition of the NAT rule called *NAME*.

```
configure ip nat NAME description TEXT
```

Set *TEXT* as description string of the given NAT rule.

```
configure ip nat NAME enable
```

Enable the given NAT rule.

```
configure ip nat NAME in IF-NAME|any
```

Set *IF-NAME* as input interface for the given NAT rule.

```
configure ip nat NAME match ADDRESS NETMASK
```

Set traffic matching *ADDRESS* and *NETMASK* for the given NAT rule.

```
configure ip nat NAME out IF-NAME|any
```

Set *IF-NAME* as output interface for the given NAT rule.

```
configure ip nat NAME translate-to ADDRESS
```

Set the translation *ADDRESS* for the given NAT rule.

```
configure ip nat NAME type TYPE
```

Set NAT type for the given NAT rule. *TYPE* can be *dnat* (destination NAT), *snat* (source NAT) or *r-nat* (railway NAT).

```
configure ip route DESTINATION NETMASK GATEWAY [INTERFACE]
```

Add a static route with gateway and optional interface.

```
configure ip route DESTINATION NETMASK INTERFACE
```

Add a static route without gateway.

```
configure ip routing enable
```

Enable Layer3 IPv4 routing.

```
configure lldp holdtime VALUE
```

Set the LLDP holdtime value.

```
configure lldp run
```

Enable LLDP protocol.

```
configure lldp timer VALUE
```

Set the LLDP timer value.

```
configure lldp ttcmp extensions enable
```

Enable TTCMP protocol.

```
configure lldp ttcmp passive-mode enable
```

Set TTCMP protocol in passive mode.

```
configure mac-address MAC interface IF vlan VID
```

Add static MAC address to the forwarding table.

```
configure ntp client polling-time TIME
```

Set NTP client polling time. *TIME*, in seconds, must be an integer in the range 1-8192.

```
configure port-mirror destination IF-NAME source IF-LIST
```

Enable packet mirroring from all the interfaces specified in the comma-separated list *IF-LIST* to the destination interface *IF-NAME*.

```
configure qos cos-map QUEUE VALUE
```

Map a QoS CoS value to an internal priority queue. *QUEUE* must be an integer value in the range 1-8, where 1 is the queue with the lowest priority and 4 the one with highest priority.

```
configure qos dscp-map QUEUE VALUE
```

Map a QoS DSCP value to an internal priority queue. *QUEUE* must be an integer value in the range 1-8, where 1 is the queue with the lowest priority and 4 the one with highest priority.

```
configure rate-limit profile PROFILE
```

Define a rate-limit profile called *PROFILE*.

```
configure rate-limit profile PROFILE broadcast VALUE [K|M|G]
```

Define a rate limit for broadcast packets. By default the rate *VALUE* is expressed in Mbps; optionally the value can be expressed in kbps.

```
configure rate-limit profile PROFILE  
                                broadcast VALUE K|M|G burst auto|  
BURST
```

Define a rate limit for broadcast packets with explicit committed burst size *BURST* expressed in Bytes. The keyword *auto* will trigger the automatic computation of an optimal burst size (but carrying out this operation requires up to some seconds).

```
configure rate-limit profile PROFILE inbound VALUE [K|M|G]
```

Define a rate limit for inbound packets. By default the rate *VALUE* is expressed in Mbps; optionally the value can be expressed in kbps.

```
configure rate-limit profile PROFILE inbound VALUE K|M|G burst auto|  
BURST
```

Define a rate limit for inbound packets with explicit committed burst size *BURST* expressed in Bytes. The keyword *auto* will trigger the automatic computation of an optimal burst size (but carrying out this operation requires up to some seconds).

```
configure rate-limit profile PROFILE
                               inbound VALUE K|M|G burst auto|VALUE prio
PRIO
```

Define a priority based rate limit for inbound packets with priority *PRIO* (in range 1-8) and automatic or explicit committed burst size *BURST* expressed in Bytes.

```
configure rate-limit profile PROFILE inbound VALUE K|M|G prio PRIO
```

Define a priority based rate limit for inbound packets with priority *PRIO* (in range 1-8).

```
configure rate-limit profile PROFILE outbound VALUE [K|M|G|PPS]
```

Define a rate limit for outbound packets. By default the rate *VALUE* is expressed in Mbps; optionally the value can be expressed in kbps or in PPS.

```
configure router ospf default-information originate
                               [always] [metric METRIC] [metric-type 1|
2]
```

Configure Open Shortest Path First (OSPF): control distribution of default information. If *METRIC* is specified, it must be in the range 0-16777214.

```
configure router ospf default-information originate
```

Configure Open Shortest Path First (OSPF): always advertise default route.

```
configure router ospf neighbor ADDRESS
```

Configure Open Shortest Path First (OSPF): add a neighbor router with IP address *ADDRESS*.

```
configure router ospf network ADDRESS NETMASK area AREA-ID
```

Configure Open Shortest Path First (OSPF): enable routing on the IP network identified by *ADDRESS* and *NETMASK*. *AREA-ID* can be in decimal form (range 0-4294967295) or an IP address.

```
configure router ospf passive-interface INTERFACE
```

Configure Open Shortest Path First (OSPF): set routing updates suppression for data entering the given *INTERFACE*.

```
configure router ospf redistribute connected|static
                               [metric METRIC] [metric-type 1|
2]
```

Configure Open Shortest Path First (OSPF): redistribute *connected* or *static* routes from another protocol. If *METRIC* is specified, it must be in the range 0-16777214.

```
configure router ospf router-id ADDRESS
```

Configure Open Shortest Path First (OSPF): router-id in IP address format.

```
configure router rip default-information originate
```

Configure Routing Information Protocol (RIP): distribute a default route.

```
configure router rip distance VALUE
```

Configure Routing Information Protocol (RIP): set RIP administrative distance. *VALUE* must be in the range 1-255.

```
configure router rip neighbor ADDRESS
```

Configure Routing Information Protocol (RIP): add a neighbor router with IP address *ADDRESS*.

```
configure router rip network ADDRESS [NETMASK]
```

Configure Routing Information Protocol (RIP): enable routing on the IP network identified by *ADDRESS* and, optionally, *NETMASK*.

```
configure router rip passive-interface INTERFACE
```

Configure Routing Information Protocol (RIP): set routing updates suppression for data entering the given *INTERFACE*.

```
configure router rip redistribute connected|static [metric METRIC]
```

Configure Routing Information Protocol (RIP): redistribute *connected* or *static* routes from another protocol. If *METRIC* is specified, it must be in the range 0-16.

```
configure router rip timers basic INTERVAL TIMEOUT FLUSH
```

Configure Routing Information Protocol (RIP): basic update timers. *INTERVAL* is the time between updates with the given *TIMEOUT* and *FLUSH* time. All are in seconds and in the range 5-2147483.

```
configure router rip version 1|2
```

Configure Routing Information Protocol (RIP): set the protocol version in use.

```
configure service auto-usb-server
```

Enable Auto-USB service.

```
configure service console-server
```

Enable L2 Console service.

```
configure service http-rest-server
```

Enable HTTP based REST-like service.

```
configure service https-rest-server
```

Enable HTTPS based REST-like service.

```
configure service ssh-server
```

Enable Secure Shell (SSH) service.

```
configure service telnet-server
```


Enable Telnet service.

```
configure snmp-server
```

Enter specific configuration level for SNMP server.

```
configure snmp-server community COMMUNITY
```

Defines an SNMP read-only community string for authentication purposes.

```
configure snmp-server community COMMUNITY ro
```

Defines an SNMP read-only community string for authentication purposes.

```
configure snmp-server community COMMUNITY ro access ADDRESS
```

Defines an SNMP read-only community string for authentication purposes, restricting access to the specified source IP address.

```
configure snmp-server community COMMUNITY ro oid OID
```

Defines an SNMP read-only community string for authentication purposes, restricting access to the specified MIB oid

```
configure snmp-server community COMMUNITY ro oid OID access ADDRESS
```

Defines an SNMP read-only community string for authentication purposes, restricting access to the specified MIB oid and to the specified source IP address.

```
configure snmp-server community COMMUNITY ro view NAME
```

Defines an SNMP read-only community string for authentication purposes, restricting access to the specified named view.

```
configure snmp-server community COMMUNITY ro view NAME access ADDRESS
```

Defines an SNMP read-only community string for authentication purposes, restricting access to the specified named view and to the specified source IP address.

```
configure snmp-server community COMMUNITY rw
```

Defines an SNMP read-write community string for authentication purposes.

```
configure snmp-server community COMMUNITY rw access ADDRESS
```

Defines an SNMP read-write community string for authentication purposes, restricting access to the specified source IP address.

```
configure snmp-server community COMMUNITY rw oid OID
```

Defines an SNMP read-write community string for authentication purposes, restricting access to the specified MIB oid.

```
configure snmp-server community COMMUNITY rw oid OID access ADDRESS
```

Defines an SNMP read-write community string for authentication purposes, restricting access to the specified MIB oid and to the specified source IP address.

```
configure snmp-server community COMMUNITY rw view NAME
```

Defines an SNMP read-write community string for authentication purposes, restricting access to the specified named view.

```
configure snmp-server community COMMUNITY rw view NAME access ADDRESS
```

Defines an SNMP read-write community string for authentication purposes, restricting access to the specified named view and to the specified source IP address.

```
configure snmp-server contact TEXT
```

Set SNMP server contact info.

```
configure snmp-server description TEXT
```

Set *SNMP* server description.

```
configure snmp-server enable
```

Enable *SNMP* server.

```
configure snmp-server engineid ENGINE-ID-STRING
```

Defines the engineID string to use during SNMPv3 AutoDiscovery process and subsequent SNMPv3 requests. If NOT set is automatically computed based on system MAC-Address.

```
configure snmp-server group-access GROUPNAME noauth  
                                readview VIEWNAME writeview VIEWNAME
```

Defines the capabilities of the named group specifying the required minimum security level as NoAuth-NoPriv, and the read/write views to apply to restrict the scope of the received SNMP get/set requests.

```
configure snmp-server group-access GROUPNAME auth  
                                readview VIEWNAME writeview VIEWNAME
```

Defines the capabilities of the named group specifying the required minimum security level as Auth-NoPriv, and the read/write views to apply to restrict the scope of the received SNMP get/set requests.

```
configure snmp-server group-access GROUPNAME priv  
                                readview VIEWNAME writeview VIEWNAME
```

Defines the capabilities of the named group specifying the required minimum security level as Auth-Priv, and the read/write views to apply to restrict the scope of the received SNMP get/set requests.

```
configure snmp-server host ADDRESS COMMUNITY
```

Set *SNMP* server target host IP address for outgoing *SNMPv2c* traps, and also a trap-community.

```
configure snmp-server host ADDRESS COMMUNITY v1 traps
```

Set *SNMP* server target host IP address for outgoing *SNMP v1* traps, and a trap-community.

```
configure snmp-server host ADDRESS COMMUNITY v1 traps port PORT
```

Set *SNMP* server target host IP address for outgoing *SNMP v1* traps, and trap-

community, while choosing the non standard destination port *PORT*.

```
configure snmp-server host ADDRESS COMMUNITY v2c traps
```

Set *SNMP* server target host IP address for outgoing *SNMP* v2c traps, and a trap-community.

```
configure snmp-server host ADDRESS COMMUNITY v2c traps port PORT
```

Set *SNMP* server target host IP address for outgoing *SNMP* v2c traps, and trap-community, while choosing the non standard destination port *PORT*.

```
configure snmp-server host ADDRESS COMMUNITY v2c informs
```

Set *SNMP* server target host IP address for outgoing *SNMP* v2c informs, and a trap-community.

```
configure snmp-server host ADDRESS COMMUNITY v2c informs port PORT
```

Set *SNMP* server target host IP address for outgoing *SNMP* v2c informs, and a trap-community, while choosing the non standard destination port *PORT*.

```
configure snmp-server host ADDRESS USER v3 noauth informs
```

Set *SNMP* server target host IP address for outgoing *SNMP* v3 informs, a v3-user, without using any authentication nor privacy protocol.

```
configure snmp-server host ADDRESS USER v3 noauth informs port PORT
```

Set *SNMP* server target host IP address for outgoing *SNMP* v3 informs, a v3-user, without using any authentication nor privacy protocol, while choosing the non standard destination port *PORT*.

```
configure snmp-server host ADDRESS USER v3 noauth traps
                           remote-engineid ENGINE-ID-
STRING
```

Set *SNMP* server target host IP address for outgoing *SNMP*v3 traps, a v3-user, without using any authentication nor privacy protocol and using the specified *ENGINE-ID-STRING* as *engineID* used for the outgoing *SNMP*v3 traps.

```
configure snmp-server host ADDRESS USER v3 noauth traps
                           remote-engineid ENGINE-ID-STRING port
PORT
```

Set *SNMP* server target host IP address for outgoing *SNMP*v3 traps, a v3-user, without using any authentication nor privacy protocol and using the specified *ENGINE-ID-STRING* as *engineID* used for the outgoing *SNMP*v3 traps, while choosing the non standard destination port *PORT*.

```
configure snmp-server host ADDRESS USER v3 auth md5|sha1 AUTHPASS
informs
```

Set *SNMP* server target host IP address for outgoing *SNMP*v3 informs, a v3-user, and using specified authentication algorithm (MD5 or SHA1) with the specified password *AUTHPASS*.

```
configure snmp-server host ADDRESS USER v3 auth
                                md5|sha1 AUTHPASS informs port
PORT
```

Set SNMP server target host IP address for outgoing SNMPv3 informs, a v3-user, and using specified authentication algorithm with the specified password *AUTHPASS*, and choosing the non standard destination port *PORT*.

```
configure snmp-server host ADDRESS USER v3 auth
                                md5|sha1 AUTHPASS traps remote-engineid ENGINE-ID-
STRING
```

Set SNMP server target host IP address for outgoing SNMPv3 traps, a v3-user, using the specified authentication algorithm with the specified password *AUTHPASS*, and the specified *ENGINE-ID-STRING* as *engineID* used for the outgoing SNMPv3 traps.

```
configure snmp-server host ADDRESS USER v3 auth
                                md5|sha1 AUTHPASS traps remote-engineid ENGINE-ID-STRING port
PORT
```

Set SNMP server target host IP address for outgoing SNMPv3 traps, a v3-user, using the specified authentication algorithm with the specified password *AUTHPASS*, and the specified *ENGINE-ID-STRING* as *engineID* used for the outgoing SNMPv3 traps, while choosing the non standard destination port *PORT*.

```
configure snmp-server host ADDRESS USER v3 priv
                                md5|sha1 AUTHPASS des|aes PRIVPASS
informs
```

Set SNMP server target host IP address for outgoing SNMPv3 informs, a v3-user, and using the specified authentication algorithm with password *AUTHPASS*, and the specified privacy algorithm(DES or AES) with password *PRIVPASS*.

```
configure snmp-server host ADDRESS USER v3 priv
                                md5|sha1 AUTHPASS des|aes PRIVPASS informs port
PORT
```

Set SNMP server target host IP address for outgoing SNMPv3 informs, a v3-user, and using specified authentication algorithm with password *AUTHPASS*, and the specified privacy algorithm with password *PRIVPASS*, while choosing the non standard destination port *PORT*.

```
configure snmp-server host ADDRESS USER v3 priv md5|sha1 AUTHPASS
                                des|aes PRIVPASS traps remote-engineid ENGINE-ID-
STRING
```

Set SNMP server target host IP address for outgoing SNMPv3 traps, a v3-user, using the specified authentication algorithm with the specified password *AUTHPASS*, and specified privacy algorithm with password *PRIVPASS*, and the specified *ENGINE-ID-STRING* as *engineID* used for the outgoing SNMPv3 traps.

```
configure snmp-server host ADDRESS USER v3 priv md5|sha1 AUTHPASS  
des|aes PRIVPASS traps remote-engineid ENGINE-ID-STRING port  
PORT
```

Set SNMP server target host IP address for outgoing SNMPv3 traps, a v3-user, using specified authentication algorithm with the specified password *AUTHPASS*, and specified privacy algorithm with password *PRIVPASS*, and the specified *ENGINE-ID-STRING* as *engineID* used for the outgoing SNMPv3, while choosing the non standard destination port *PORT*.

```
configure snmp-server location TEXT
```

Set SNMP server location.

```
configure snmp-server name NAME
```

Set SNMP server name.

```
configure snmp-server traps-all-if TYPES
```

Defines a comma-separated list of the active SNMP notifications among the ones available in the *all-if* family. Set **ALL** as *TYPES* value to activate all the traps available for this family or use a comma-separated list of the following values: *link-status*, *link-flap* and, on ITBN, *link-poe-status*.

```
configure snmp-server traps-global TYPES
```

Defines a comma-separated list of the active SNMP notifications among the ones available in the *global* family. Set **ALL** as *TYPES* to activate all the traps available for this family or use a comma-separated list of the following values: *overtemp*, *high-cpu-load*, *configuration-change*, *sysadmin-change*, *new-root* and *topology-change*.

```
configure snmp-server traps-ttcmp TYPES
```

Defines a comma-separated list of the active SNMP notifications among the ones available in the *ttcmp* family. Specify **ALL** as *TYPES* to activate all the traps available for this family or use a comma-separated list of the following values: *ttcmp-anomaly*.

```
configure snmp-server user USERNAME group GROUPNAME  
md5 AUTHPASS des  
PRIVPASS
```

Defines an SNMPV3 user, belonging to the specified group and using MD5/DES as authentication/privacy protocols and the chosen passwords.

```
configure snmp-server user USERNAME group GROUPNAME  
sha1 AUTHPASS des PRIVPASS
```

Defines an SNMPV3 user, belonging to the specified group and using SHA1/DES as authentication/privacy protocols and the chosen passwords.

```
configure snmp-server user USERNAME group GROUPNAME  
md5 AUTHPASS aes PRIVPASS
```

Defines an SNMPV3 user, belonging to the specified group and using MD5/AES as authentication/privacy protocols and the chosen passwords.

```
configure snmp-server user USERNAME group GROUPNAME  
                                sha1 AUTHPASS aes PRIVPASS
```

Defines an SNMPV3 user, belonging to the specified group and using SHA1/AES as authentication/privacy protocols and the chosen passwords.

```
configure snmp-server view NAME OIDTREE excluded
```

Exclude the specified MIB OID-tree from the named view.

```
configure snmp-server view NAME OIDTREE included
```

Include the specified MIB OID-tree into the named view.

```
configure spanning-tree enable
```

Enable the Spanning Tree Protocol.

```
configure spanning-tree forward-delay VALUE
```

Set spanning-tree forward-delay timer in seconds. *VALUE* must be a value in range 4-30.

```
configure spanning-tree hello-time VALUE
```

Set spanning-tree hello-time timer in seconds. *VALUE* must be a value in range 1-10.

```
configure spanning-tree max-age VALUE
```

Set spanning-tree max-age timer in seconds. *VALUE* must be a value in range 6-40.

```
configure spanning-tree mode MODE
```

Set *MODE* to *stp* in order to use the old Spanning Tree algorithm; use *rapid-stp* (default) for the modern Rapid Spanning Tree algorithm.

```
configure spanning-tree priority VALUE
```

Set spanning-tree priority value. *VALUE* must be a value in range 0-61440, in steps of 4096.

```
configure spanning-tree tx-hold-count VALUE
```

Set spanning-tree tx-hold-count value. *VALUE* must be a value in range 0-10. Setting this to 0, disables any check on BPDU sending frequency.

```
configure svi-interfaces status dynamic
```

Enable dynamic status management of SVI interfaces.

```
configure system USER password SECRET
```

Change the password for the given user. *USER* can be *admin*, *operator* or *user*.

```
configure system authentication USER local
```

Use credentials from local database for authentication of the given user. *USER* can be *admin*, *operator* or *user*.

```
configure system authentication USER radius
```

Use credentials from configured RADIUS for authentication of the given user. *USER* can be *admin*, *operator* or *user*.

```
configure system authentication USER radius local
```

Use credentials at first from configured RADIUS and then fallback to local database for authentication of the given user. *USER* can be *admin*, *operator* or *user*.

```
configure system clock source PROTOCOL ADDRESS [ADDRESS2]
```

Set time auto-synchronization sources for given protocol. *PROTOCOL* must be one of *ntp* and *timep*. *ADDRESS* must be the valid IPv4 address of a network resource which provides synchronization service with the given protocol. The additional *ADDRESS2* is optional.

```
configure system hostname NAME
```

Set the system host name.

```
configure system logging debug SUBSYSTEMS
```

Enable recording of the log messages in the applog. *SUBSYSTEMS* is the comma-separated list of subsystems for which the recording is enabled.

```
configure system logging host ADDRESS
```

Enable recording of the log messages in the remote syslog server at the given IP *ADDRESS*.

```
configure system login banner TEXT
```

Set the given *TEXT* as login banner for the system shell.

```
configure system login failed-attempts VALUE
```

Enable failed logins counting. Lock the account after the login failed the given number of times.

```
configure system password complexity enable
```

Enable password complexity checking.

```
configure system radius-server primary ADDRESS SECRET
```

Configure a *primary* RADIUS server for use by local user authentication.

```
configure system radius-server primary options opt1=val1[,opt2=val2...]
```

Configure options to use for the *primary* RADIUS server. Option *optX* could be: *auth_port*, *acct_port*, *retry_cnt*, *timeout*, *src_ipaddr*

```
configure system radius-server backup ADDRESS SECRET
```

Configure a backup RADIUS server for use by local user authentication.

```
configure system radius-server backup options opt1=val1[,opt2=val2...]
```

Configure options to use for the *backup* RADIUS server. Option *optX* could be: *auth_port*, *acct_port*, *retry_cnt*, *timeout*, *src_ipaddr*

```
configure system timezone ZONE SITE
```

Set the current timezone. For valid values of *ZONE* and *SITE* arguments, see online help.

```
configure tcds address composition mask MASK
```

MASK is a special netmask used by TCDS in the computation of the position-dependent addresses; its bits are set in the portion of the IP address to be replaced by the position value.

```
configure tcds backbone-vlan VID
```

Set the Virtual Backbone VLAN identifier.

```
configure tcds enable
```

Enable the Train Composition Discovery Service.

```
configure tcds lock reboot after VALUE minutes
```

Configure timeout for rebooting after a composition lock; *VALUE* is expressed in minutes and 0 means *do not reboot*.

```
configure tcds mode plain|routed
```

Select the TCDS mode of operations: *plain* if the same IP network is used for all the train, *routed* if a different IP network is used on each consist.

```
configure tcds ttcmp forwarding
```

Enable the forwarding of TTCMP packets entering the consist boundaries.

```
configure tcds consist-network ADDRESS/MASK
```

Virtual TBN configuration: consist network IP *ADDRESS* in use.

```
configure tcds enable [no-nat]
```

Enable Virtual TBN role. Optionally the backbone NAT for consist networks can be disabled specifying the *no-nat* argument.

```
configure tcds ip-address-template TEMPLATE
```

Template used for building VTBN addresses.

```
configure tcds mroute IN-IF SRC-ADDR MC-ADDR
```

Add a IPv4 multicast route through the virtual backbone for traffic entering *IN-IF* and having multicast address *MC-ADDR*. *SRC-ADDR* can be the IPv4 address of the sender or 0.0.0.0 for any address.

```
configure tcds train-network-template TEMPLATE
```

Template used for building consist backbone addresses.

```
configure ttcmp auto confirm enable
```

Enable automatic confirmation of TTCMP project when all other devices are confirmed.

```
configure ttcmp auto cst-id enable
```

Enable automatic Consist-ID detection by TTCMP.


```
configure vlan ID [name NAME]
```

Create a VLAN with the given ID. *ID* must be an integer value in the range 2-4096. If *NAME* argument is not given, the VLAN will be assigned a name generated by the system.

```
debug [SUBSYSTEMS]
```

Enable debug messages. *SUBSYSTEMS* is the optional comma-separated list of subsystems for which the messages are enabled.

```
ping ADDRESS
```

Send some ICMP requests to the specified IP *ADDRESS*.

```
save running-config [force]
```

Save running configuration on persistent storage. The *force* option is required to save configurations generated by TTCMP projects.

```
show clock
```

Show clock properties, including current date/time, active auto-synchronization protocols and source, current timezone.

```
show counters protocol lldp|ttcmp
```

Show the packet counters recorded for the given protocol (LLDP or TTCMP).

```
show coupled-mode
```

Show the status of the coupled switch.

```
show diagnostics
```

Show the configuration of the diagnostics subsystem.

```
show dot1x supplicants [interface INTERFACE]
```

Report 802.1X data about all currently authenticated supplicants. The report can be restricted to the supplicant related to the optionally given *INTERFACE*.

```
show interface
```

Show the list of network interfaces.

```
show interface INTERFACE [details]
```

Show information about the given network interface. The details keyword adds more information.

```
show interface summary
```

Show a short list of network interfaces.

```
show interface trunks
```

Show the list of network interfaces configured in trunk mode and their characteristics.

```
show ip dhcp bindings [interface IF-NAME]
```

Show all the active DHCP bindings or the ones related to the specified

interface.

```
show ip dhcp services
```

Show the status of the DHCP services.

```
show ip filter
```

Show the status of the IP filtering services.

```
show ip igmp
```

Show the status of the IPv4 multicast services.

```
show ip interface summary
```

Summary report of Layer3 IP interfaces.

```
show ip mroutes [active]
```

Display active multicast routes.

```
show ip mroutes configured
```

Display all the configured multicast routes.

```
show ip nat
```

Show the status of the NAT service.

```
show ip protocols ospf
```

Display OSPF routing protocol status.

```
show ip protocols ospf database
```

Display OSPF routing protocol data-base.

```
show ip protocols ospf interface INTERFACE
```

Display information about the given *INTERFACE* as stored by the OSPF process.

```
show ip protocols ospf neighbor [INTERFACE] [detail]
```

Display OSPF neighbor list (optionally detailed), eventually restricted to the neighbors facing the given *INTERFACE*.

```
show ip protocols ospf neighbor ADDRESS
```

Display details about the OSPF neighbor identified by *ADDRESS*..

```
show ip protocols rip
```

Display RIP routing protocol status.

```
show ip routes
```

Show registered static IPv4 routes.

```
show link-aggregation interface grpNN
```

Show link aggregation status information of the given group interface.

```
show lldp
```

Show the current status of LLDP service.

```
show lldp neighbours [detail]
```

Show a list of discovered LLDP neighbours. The detail keyword adds detailed information to the output.

```
show log
```

Show log messages.

```
show log persistent
```

Show persistently recorded log messages.

```
show log update
```

Show log messages related to firmware updates.

```
show mac-address
```

Show MAC addresses collected from system network interfaces.

```
show mac-address interface INTERFACE
```

Show MAC addresses collected from the given network interface.

```
show mac-address vlan ID
```

Show collected MAC addresses which belong to the given VLAN.

```
show ntp status
```

Report information about the NTP subsystem.

```
show poe
```

Report information about the Power-over-Ethernet on ports.

```
show poe
```

Report information about the Power-over-Ethernet on ports.

```
show qos interfaces
```

Show current QoS per-interface configuration.

```
show qos prio-map
```

Show current QoS priority map assignments.

```
show running-config
```

Show current system configuration.

```
show spanning-tree
```

Show spanning-tree protocol status information.

```
show system
```

Display a summary of system specific data.

```
show system flash
```

Show flash memory usage info.

```
show system leds
```

Show current LEDs status.

```
show system memory
```

Show memory usage info.

```
show system temperature
```

Show system internal temperature.

```
show system time
```

Show current system date/time.

```
show startup-config
```

Show system configuration saved on persistent storage.

```
show tcds composition
```

Display active train composition discovered by the Train Composition Discovery Service.

```
show tcds local
```

Display TCDS local consist information.

```
show tech-info
```

Display data for technical analysis.

```
show topology
```

Display the list of network devices collected by TTCMP®.

```
show topology detail [device DEV-NAME]
```

Display the details about one or all the network devices collected by TTCMP®.

```
show ttcmp
```

Show the status of the TTCMP protocol and subsystem.

```
show ttcmp generated commands
```

Show the configuration commands generated by the current TTCMP project.

```
show version
```

Show system version info.

```
show vlans
```

Show defined VLANs.

```
show vlans <xx[,yy][,ww-zz]>
```

Show only the specified VLANs, if defined.

```
show vrrp
```

Show the status of the VRRP protocol and subsystem.

```
sysadmin delete image NAME
```

Remove the given firmware image from persistent storage.

```
sysadmin delete log persistent
```

Clear the persistent log.

```
sysadmin delete log update
```

Clear the firmware updates log.

```
sysadmin delete project ID
```

Delete the TTCMP© project identified by *ID*.

```
sysadmin delete startup-config
```

Remove the startup configuration from persistent storage.

```
sysadmin load URL
```

Load update data from the given URL.

```
sysadmin port-mirror destination IF-NAME source IF-LIST
```

Enable packet mirroring from all the interfaces specified in the comma-separated list *IF-LIST* to the destination interface *IF-NAME*.

```
sysadmin reboot
```

Reboot the system.

```
sysadmin reset coach-id
```

Reset the coach identifier value and description.

```
sysadmin reset composition head
```

Reset the Composition Head flag.

```
sysadmin reset composition lock
```

Reset composition discovery lock status.

```
sysadmin reset cst-id
```

Reset the current consist id value.

```
sysadmin reset global project
```

Reset the TTCMP© project on all the devices of the network.

```
sysadmin reset local project
```

Reset the TTCMP© project on the local device.

```
sysadmin reset poe cache
```

Clear the PoE cache.

```
sysadmin save configfile URL [external|plain]
```

Save current system configuration on the given network resource specified by *URL*. If *external* argument is present, the format will be suitable for external configuration mode. If *plain* argument is present, the configuration

will not be encapsulated in a KAD container.

```
sysadmin save image URL
```

Save current system image on the given network resource specified by *URL*.

```
sysadmin save log URL [plain]
```

Save current system logs on the given network resource specified by *URL*. If *plain* argument is present, the configuration will not be encapsulated in a KAD container.

```
sysadmin save tech-info URL [plain]
```

Save technical data on the given network resource specified by *URL*. If *plain* argument is present, the configuration will not be encapsulated in a KAD container.

```
sysadmin set coach-id VALUE [description DESCRIPTION]
```

Set the coach numeric identifier and an optional coach description.

```
sysadmin set composition head
```

Set the Composition Head flag.

```
sysadmin set composition lock [direct|forced|reversed]
```

Lock the composition discovery. If no argument is present, the lock direction takes into account the Composition head flag. If *direct* argument is present, the lock direction is the same of the discovered composition. If *reversed* argument is present, the lock direction is opposite to the discovered composition. If *forced* argument is present, the lock numbering starts from the local consist.

```
sysadmin set composition unlock
```

Unlock the composition discovery.

```
sysadmin set cst-id VALUE
```

Set the consist identifier as string (not exceeding 15 characters).

```
sysadmin set cst-id uuid/VALUE
```

Set the consist identifier as UUID; the value must be a valid UUID parsable using the `uuid_parse` system function.

```
sysadmin set default image NAME
```

Set the firmware image the system will load on next boot.

```
sysadmin set global project ID
```

Set the given TTCMP© project to be used on all the devices of the network.

```
sysadmin set time DAY MONTH YEAR HOURS MINUTES [SECONDS]
```

Set current system time. All arguments must be integer values. *YEAR* must be a value between 2010 and 2050.

```
sysadmin set topology match-mode default|default-fast|diffusion|none
```

Select a TTCMP© project matching algorithm.

```
sysadmin set topology status confirmed
```

Confirm the current TTCMP© project.

```
sysadmin set watchdog VALUE
```

Set the configuration watchdog feature activation time in minutes. If *VALUE* is 0, the watchdog is deactivated.

```
sysadmin show images
```

Show firmware images currently stored in persistent storage.

```
sysadmin show port-mirror
```

Report active port mirror setup.

```
sysadmin show projects
```

List the TTCMP© projects currently stored in the device.

```
sysadmin show watchdog
```

Show the configuration watchdog status.

NOTE

Please refer to VDS Rail representatives for further information about Konuendo Consist Network Concept (KCNC) and the Train Topology and Configuration Management Protocol (TTCMP©).

Chapter 32 - Copyright credits

32.1 Licenses

List of software packages and their licenses:

Package	License	Package	License
bluez-libs	GPLv2	libnl3	LGPLv2.1
bridge-utils	GPLv2	libol	GPLv2
busybox	GPLv2	libpam	GPLv2
chrony	GPLv2	libpcap	BSD
confscripts	GPLv2	libpcre	BSD
curl	MIT	libssh2	BSD
cracklib	LGPLv2.1	libtool	GPLv2
dbus	GPLv2	libtool-cross	GPLv2
dbus-daemon-proxy	GPLv2	libxml2	MIT
dbus-glib	GPLv2	lrzsz	GPLv2
dhclient	ISC	lzma	LGPLv2.1
dprocs	GPLv2	lzo	GPLv2
e2fsprogs	GPLv2	mgetty	GPLv2
ethtool	GPLv2	minicom	GPLv2
eventlog	BSD	mtd-utils	GPLv2
expat	MIT	mtools	GPLv2
flex	Flex	ncurses	MIT
gettext	GPLv3	net-snmp	BSD
glib-2.0	LGPLv2.1	net-tools	GPLv2
gnutls	GPLv3	ntpclient	GPLv2
gtk-doc	LGPLv2.1	openssh	BSD
iproute2	GPLv2	openssl	OpenSSL
iptables	GPLv2	ppp	BSD, GPLv2
iputils	BSD	quagga	GPLv2
libgcrypt	GPLv2, LGPL	radius	GPLv3
libgpg-error	GPLv2, LGPL	strace	GPLv2
libiptcom	lptcom	sysfsutils	GPLv2

Package	License		
tar	GPLv3	util-linux-ng	GPLv2
TCNOpen	MPLv2	xz	LGPLv2.1
tcpdump	BSD	zip	Info-ZIP
timezones	GPLv2	zlib	zlib
udev	GPLv2		

32.2 References

Software licenses:

License	URL
BSD	http://www.opensource.org/licenses/bsd-license.php
Flex	http://github.com/westes/flex
GPLv2	http://www.opensource.org/licenses/gpl-2.0.php
GPLv3	http://www.opensource.org/licenses/gpl-3.0.php
LGPLv2.1	http://www.opensource.org/licenses/lgpl-2.1.php
MIT	http://www.opensource.org/licenses/mit-license.html
MPLv2	http://opensource.org/licenses/MPL-2.0
Info-ZIP	http://www.info-zip.org/license.html
OpenSSL	http://www.openssl.org/source/license.html
zlib	http://www.opensource.org/licenses/zlib-license.php

License	Note
Iptcom	Copyright Bombardier © 2021, BT and others. All Rights Reserved.

Annex A - **Unavailable services in 7.2.x firmware series**

A.1 Consist Switch product models equipped with 19 or 28 Ethernet ports

The Consist Switch devices equipped with 19 or 28 ports work only with a firmware version 7.0.0 or greater.

The services listed below are, at present time, **not** available on firmware series 7.2.x:

- Port configuration:
 - Fast link detection
 - MDI/MDI-X and flow control
 - 802.1x
 - PoE mode 'as-link'
- Coupled-Switch[©] mode
- Link aggregation:
 - LACP
 - distributed
- Diagnostics data over IPTCom protocol

These services and functionalities will be added gradually to the next releases.

NOTE

The commands for the services or configurations listed above, even if not effective, may be available anyway in firmware series 7.2.x.